DEPARTMENT OF WAR
Small Business Technology Transfer (STTR) Program

2026 STTR Broad Agency Announcement (BAA)

IMPORTANT DATES

	Pre-Release	Open	Close
Release 1	Apr 13, 2026	May 06, 2026	Jun 03, 2026
Release 2	May 06, 2026	May 27, 2026	Jun 24, 2026
Release 3	Jun 03, 2026	Jun 24, 2026	Jul 22, 2026
Release 4	Jul 01, 2026	Jul 22, 2026	Aug 19, 2026
Release 5	Aug 05, 2026	Aug 26, 2026	Sep 23, 2026
Release 6	Sep 02, 2026	Sep 23, 2026	Oct 21, 2026

NOTE: DoW Services and Components may elect to release topics under this BAA once a month throughout fiscal year 2026. New research topics will pre-release on the first Wednesday of each month, based on the schedule above. Each release will have corresponding open and close dates applicable only to those topics within the release. If a topic in a release deviates from the open and close dates listed above, it will be noted within the Component-specific instructions and on DSIP.

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1.0 PROGRAM DESCRIPTION

1.1 Introduction

The Department of War (DoW) Small Business Technology Transfer (STTR) Program's objectives include stimulating technological innovation by strengthening the role of small businesses in meeting DoW Research and Development (R&D) needs, delivering innovative technology to the Warfighter, and growing the defense industrial base through commercialization. The STTR Program, while modeled after the Small Business Innovation Research (SBIR) Program, is a distinct and separately financed program. Subject to availability of funds, DoW Services/Components will support high quality cooperative research and development proposals of innovative concepts to solve the listed defense-related scientific or engineering problems, especially those concepts that also have high potential for commercialization in the private sector.

The Small Business Administration (SBA), through its SBIR/STTR Policy Directive, purposely departs from traditional government solicitation formats and requirements, simplifying the SBIR/STTR award process and minimizing the regulatory burden on small businesses. Consistent with the SBA SBIR/STTR Policy Directive, DoW is soliciting proposals as a Broad Agency Announcement (BAA). DoW invites Small Business Concerns (SBCs) with the capability to conduct R&D and commercialize the results in any of the defense-related topic areas described in this STTR Program BAA to submit proposals.

1.1.1 General BAA Procedures

Federal Acquisition Regulation (FAR) 2.101 defines a BAA as a general announcement of an agency's research interest including criteria for selecting proposals and soliciting the participation of all offerors capable of satisfying the Government's needs. FAR 35.016 prescribes the procedures for the use of BAAs as a competitive method for acquiring basic and applied research, and in some cases, advanced technology development. BAAs are intended to solicit scientific study and experimentation directed toward advancing knowledge, understanding and capability in areas of interest to the Government, rather than to acquire specific systems or hardware.

This BAA provides the framework for DoW to pursue innovative research concepts that address defense-related scientific and engineering challenges. By leveraging the flexibility of the SBA SBIR/STTR Policy Directive, this BAA encourages scientific and technical proposals most likely to yield significant results for DoW and the private sector.

This BAA may include broadly defined topics as well as more specific, individual program requirements. Topics are intentionally non-prescriptive and structured to accommodate varied technical and scientific approaches, encouraging novel solutions that align with DoW modernization goals. Proposals should be responsive to the specific problem statement(s) and area(s) of interest outlined in the release topics.

1.1.2 Acquisition Framework

This BAA is for research topics accepting Phase I proposals. A separate BAA will not be issued requesting Phase II proposals. All proposing SBCs that receive a Phase I award from this BAA will be eligible to participate in Phase II competitions and potential Phase III awards. DoW Services/Components will notify Phase I awardees of the Phase II proposal submission requirements.

Depending on the nature of the proposed effort, and the statutory authority available, the Government retains the flexibility to award a FAR-based procurement contract, purchase order, or Other Transaction (OT) under this solicitation, as appropriate. FAR-based contract types, including, Firm-Fixed-Price, Firm-Fixed-Price Level of Effort, Labor Hour, Time & Material, or Cost-Plus-Fixed-Fee type contracts may be negotiated at the discretion of the Component Contracting Officer. Additionally, the Government may award OTs pursuant to 10 U.S.C. 4021 (Research Projects) or 10 U.S.C. 4022 (Prototype Projects).

Follow-on production OTs may also be awarded in accordance with 10 U.S.C. 4022(f) if the statutory conditions are met. Separately, awards may also be made under 10 U.S.C. 4023 (Procurement for Experimental Purposes), when appropriate.

Multiple procurements are planned and anticipated to be awarded for each topic. Each proposal will be evaluated independently, and funding decisions will be made without regard to other submissions. DoW Services/Components may elect to fund several or none of the proposals received in response to a given topic. All awards are subject to both a risk-based due diligence security review and availability of funds, and the DoW is under no obligation to make any Phase I awards under this BAA. Offerors are solely responsible for all costs incurred in the preparation and submission of proposals. The DoW will not reimburse any expenses related to proposal development or pre-award activities. This BAA is not a mechanism for submitting unsolicited proposals, and unsolicited proposals will not be considered nor receive a response. Only proposals that are responsive to a posted topic release will be considered.

The instructions contained in this BAA are applicable to all scheduled topic releases. Each release will have corresponding open and close dates applicable only to those topics contained within the release. Additional instructions specific to an individual topic release will be contained within the component-specific instructions. All proposals submitted in response to any topic posted under this solicitation will be governed by this BAA. Proposals must conform to this announcement's terms.

1.2 A Three Phased Program

The STTR Program has three phases, Phases I, II, and III. Phase I determines, to the extent possible, an idea's scientific, technical, and commercial merit and feasibility within the STTR program. Phase I and II awards are made adhering to current SBA Policy Directive guidelines.

The Phase I period of performance is generally between six to twelve months. Proposals should focus on Research or Research & Development (R/R&D)to prove the proposed effort's scientific and technical feasibility, and commercialization potential, the successful completion of which is a prerequisite for further DoW support in Phase II. Proposing SBCs are encouraged to consider whether the research or R&D being proposed to DoW Services/Components also has private sector potential, either for the proposed application or as a base for other applications.

Phase II awards will be made to proposing SBCs based on results of Phase I awards and the Phase II proposal's scientific merit, technical merit, and commercialization potential. The period of performance is generally 24 months. The objective of Phase II is to continue and further develop the R/R&D effort from the completed Phase I award.

Phase III refers to work that derives from, extends, or completes an effort made under prior STTR funding agreements, but is funded by sources other than the STTR Program. Under Phase III, the SBC should focus on commercializing previously STTR-funded technology and is required to obtain funding from either the private sector, a non-STTR federal source, or both, to develop the prototype into a viable product or non-R&D service for sale in military or private sector markets.

1.3 OUSW(R&E) Critical Technology Areas

Although each DoW Service/Component develops SBIR and STTR topics tailored to their mission needs, topics may also align with the Office of the Under Secretary of Defense, Research & Engineering (OUSW(R&E)) Critical Technology Areas (CTAs). While many technologies may cross between these categories, these areas represent the broad and different approaches required to advance technologies crucial to the Department, with a focus on accelerating key capabilities' transitions to the Military Services and Combatant Commands.

OUSW(R&E) CTAs include:

- Applied Artificial Intelligence (AAI)
- Biomanufacturing (BIO)
- Contested Logistics Technologies (LOG)
- Quantum & Battlefield Information Dominance (QBID)
- Scaled Hypersonics (SHY)
- Scaled Directed Energy (SCADE)

Full descriptions of the OUSW (R&E) CTAs can be viewed here.

Below are additional Component Technology Priority Areas (TPAs) supporting DoW Services/Component-specific mission-critical areas:

- FutureG
- Trusted AI and Autonomy
- Biotechnology
- Advanced Computing and Software
- Integrated Sensing and Cyber
- Directed Energy (DE)
- Hypersonics
- Microelectronics
- Integrated Network Systems-of-Systems
- Quantum Science
- Space Technology
- Renewable Energy Generation and Storage
- Advanced Materials
- Human-Machine Interfaces
- Advanced Infrastructure & Advanced Manufacturing
- Combat Casualty Care
- Emerging Threat Reduction
- Military Infectious Diseases
- Military Operational Medicine
- Mission Readiness & Disaster Preparedness
- Nuclear
- Sustainment & Logistics

Full descriptions of the above TPAs can be reviewed here.

1.4 Eligibility and Performance Requirements

Each proposing entity must qualify as an SBC as defined in 13 C.F.R §§ 701-705 at time of award and certify to this in the proposal's cover sheet. The eligibility requirements for the SBIR/STTR programs are unique and do not correspond to those of other small business programs.

a. Proposing SBC must meet eligibility requirements for Small Business Ownership and Control (see 13 CFR § 121.702).
b. A minimum of 40% of each STTR project must be conducted by the small business concern and a minimum of 30% of the effort performed by the single research institution, as defined in Appendix B. The percentage of work is measured by both direct and indirect costs. Deviations from these STTR requirements are not allowed, as the performance of work requirements are specified in statute at 15 USC 638(e). For more information on the percentage of work calculation during proposal submission, refer to section 3.7.
c. For both Phase I and II, the principal investigator's primary employment must be with the SBC or the research institution at the time of award and during the conduct of the proposed effort. Primary employment means that more than one-half of the principal investigator's time is spent with the small business (based on a 40-hour work week). At the time of award of a Phase I or Phase II contract, the SBC must have at least one employee in a management position whose primary employment is with the small business and who is not also employed by the research institution. Primary employment with an SBC precludes full-time employment at another organization. Deviations from this requirement or changes to the principal investigator are subject to the Funding Agreement officer approval.

d. For both Phase I and Phase II, the SBC and its subcontractors must perform all research or R&D work in the United States.
e. Joint ventures and limited partnerships are permitted, provided that the entity qualifies as small business in accordance with the ownership requirements in 13 CFR 121.702(a)(1)(iii) and the size requirements in 13 CFR 121.702(c)(6). Proposing SBC must disclose joint ventures with existing (or planned) relationships/partnerships with any foreign entity or any foreign government-controlled companies. See sections 2.6 and 3.7 for more detail.
f. A small business concern must negotiate a written agreement between the small business and the research institution allocating intellectual property rights and rights to carry out follow-on research, development, or commercialization (see Model Agreement for the Allocation of Rights). In accordance with paragraph §8(d) of the SBA's SBIR/STTR Policy Directive (current version) and DFARS 252.227-7040, the proposing SBC must submit this agreement to the awarding agency with the proposal, under Volume 5: Supporting Documents.

1.5 Majority Ownership in Part by Multiple Venture Capital, Hedge Fund, and Private Equity Firms

Proposing SBCs that are multiple venture capital operating companies (VCOCs), hedge funds, or private equity funds majority owned are **ineligible** to submit applications to or receive awards under this solicitation.

1.6 Performance Benchmark Requirements/Increased Minimum Performance Standards for Experienced Firms

Proposing SBCs with multiple prior SBIR/STTR awards must meet minimum performance requirements to be eligible to apply for a new Phase I or Direct-to-Phase II award. The Phase I to Phase II Transition Rate addresses the extent to which an awardee progresses a project from Phase I to Phase II. The Commercialization Benchmark addresses the extent to which an awardee has moved past Phase II work towards commercialization.

The SBIR and STTR Extension Act of 2022 (Pub. L. 117-183) amended the benchmarks' application for more experienced firms. Find detailed information on benchmark calculations, increased performance standards for more experienced firms and consequence of failure to meet benchmarks here. SBA will notify companies failing either benchmark and the relevant officials at the participating agencies.

The SBIR/STTR Policy Directive defines the Departments of the Army, Navy, and Air Force each as its own federal agency, and the remaining DoW Components as an executive agency of the Department of War. Therefore, companies that fail to meet either of the benchmarks under the Increased Performance Standards for More Experienced Firms may not receive more than an overall total of 80 awards from DoW, as detailed in the breakdown below:

Army – 20 total Phase I and Direct to Phase II awards
Navy – 20 total Phase I and Direct to Phase II awards
Air Force – 20 total Phase I and Direct to Phase II awards
All other DoW Components combined – 20 total Phase I and Direct to Phase II awards

1.7 Program on Innovation Open Topics

15 U.S.C. §638 (ww) requires DoW establish innovation open topic activities to:

a. increase the transition of commercial technology to the DoW;
b. expand the small business nontraditional industrial base;
c. increase commercialization derived from DoW investments; and

d. expand the ability for qualifying SBCs to propose technology solutions to meet DoW needs.

Unlike conventional topics, which specify the desired technical objective and output, open topics use generalized mission requirements or specific technology areas to adapt commercial products or solutions to close capability gaps, improve performance, or provide technological advancements in existing capabilities.

Open topics released under this solicitation will be clearly identified as such in the title and topic objective. Proposal preparation instructions for open topics may vary significantly across DoW Services/Components. Proposing SBCs must carefully read and follow all instructions from the DoW Service/Component for the open topic of interest. Unless specifically noted in the Service/Component instructions, all requirements outlined in this solicitation remain in effect for open topics.

An SBC may only submit one proposal to each open topic. If an SBC submits more than one proposal for a single open topic, only the most recent certified proposal submitted prior to the submission deadline will receive an evaluation. All previously submitted proposals for the same open topic will be marked nonresponsive and will not receive an evaluation. Refer to the Service/Component-specific instructions for any additional requirements specific to open topics.

1.8 Discretionary Technical and Business Assistance (TABA)

DoW has not mandated the use of discretionary technical and business assistance (TABA). The proposing SBCs should review individual Service/Component-specific instructions to determine if TABA is offered by the Service/Component and follow instructions for requesting TABA funding.

1.9 Phase II Enhancement Policy

To further encourage the transition of STTR research into both DoW acquisition programs and the private sector, certain DoW Services/Components developed their own Phase II Enhancement policies. Under this policy, the Service/Component will provide a Phase II awardee with additional Phase II STTR funding if the proposing SBC can match the additional STTR funds with non-STTR funds from DoW acquisition programs or the private sector.

See Service/Component instructions for more details on Phase II Enhancement opportunities.

1.10 Commercialization Readiness Program (CRP)

The SBIR/STTR Reauthorization Act of 2011 established the Commercialization Pilot Program (CPP) as a long-term program called the Commercialization Readiness Program (CRP). Each Military Department (Army, Navy, and Air Force) has a CRP; refer to the Service/Component instructions for further information.

The DoW SBIR/STTR Program has established the OSW Accelerated Research for Transition (ART) for purposes of technology maturation and transfer. For more information contact osd.ncr.ousd-r-e.mbx.sbir-sttr-tech-transition@mail.mil.

1.11 State and Other Available Assistance

Many states have established programs to provide services to those proposing SBCs and individuals wishing to participate in the Federal STTR Program. These services vary from state to state, but may include:

a. Information and technical assistance;
b. Matching funds to STTR recipients;
c. Assistance in obtaining Phase III funding.

Contact your State SBIR/STTR Support office for further information. SBCs may seek general administrative guidance from small and disadvantaged business utilization specialists located in various defense contract management activities throughout the continental United States.

1.12 Fraud and Fraud Reporting

Knowingly and willfully making any false, fictitious, or fraudulent statements or representations may be a felony under the Federal Criminal False Statement Act (18 U.S.C. Sec 1001), punishable by a fine of up to $10,000, up to five years in prison, or both.

The DoW Office of Inspector General Hotline ("Defense Hotline") is an important avenue for reporting fraud, waste, abuse, and mismanagement within the DoW. The Office of Inspector General operates this hotline to receive and investigate complaints or information from contractor employees, DoW civilians, Service members, and the public. Individuals who wish to report fraud, waste or abuse may contact the Defense Hotline at (800) 424-9098 between 8:00 a.m. and 5:00 p.m. Eastern Time or visit their website to submit a complaint. Mailed correspondence should be addressed to the Defense Hotline, The Pentagon, Washington, DC 20301-1900, or email addressed to hotline@dodig.mil.

2.0 CERTIFICATIONS AND REGISTRATIONS

2.1 System for Award Management (SAM) Registration

The System for Award Management (SAM) allows proposing SBCs to provide basic information on business structure, capabilities, and financial and payment information with the Federal Government. Proposing SBCs must register in SAM here. Registration in SAM will generate the Unique Entity ID (UEI) number and the Commercial and Government Entry (CAGE) code. The UEI is required for registration in the U.S. Small Business Administration's (SBA) Company Registry. A proposing SBC that is already registered in SAM should verify the registration is active, and its representations and certifications are current to avoid award delay.

2.2 SBA Company Registry

Proposing SBCs must be registered in the SBA Company Registry. SBCs will be required to verify registration by providing the SBC Control ID and Proof of Registration/Certification during proposal submission.

2.3 Defense SBIR/STTR Innovation Portal (DSIP) Registration

Individuals from proposing SBCs must be registered in the DSIP to prepare and submit proposals. Proposing SBCs submitting through this site for the first time will be asked to register. All users are required to have an individual user account to access DSIP. It is recommended proposing SBCs register as soon as possible upon identification of a proposal opportunity to avoid delays in the proposal submission process.

DSIP user accounts are authenticated by Login.gov. Users who do not already have a Login.gov account will be required to create one. Users who already have a Login.gov account can link their existing Login.gov account with their DSIP account. Job Aids and Help Videos to walk you through the process are in the Learning & Support section of DSIP.

Be advised the sharing of accounts and passwords is a violation of the Terms of Use for Login.gov and DoW policy.

Registered SBCs will have a designated DSIP Firm Admin responsible for creating the Firm PIN, controlling access for other users in the SBC and completing and maintaining the Firm-level forms, which must be completed before any proposals can be submitted.

Users should complete their account registrations as soon as possible to avoid any delays in proposal submissions.

NOTE: The DSIP application is only accessible from within the United States, which is defined as the fifty states, the territories and possessions of the Federal Government, the Commonwealth of Puerto Rico, the Republic of the Marshall Islands, the Federated States of Micronesia, the Republic of Palau, and the District of Columbia.

2.3.1 DSIP Assistance and Support

For assistance with the DSIP application, please visit the Learning & Support section of DSIP. Email DSIP Support at DoDSBIRSupport@reisystems.com only for further assistance with issues pertaining directly to the DSIP application. Questions submitted to DSIP Support will be addressed in the order received during normal operating hours (Monday through Friday, 9:00 a.m. to 5:00 p.m. ET). Please include information on your SBC, a proposal number (if applicable), and screenshots of any pertinent errors or issues encountered.

DSIP Support cannot provide updates to proposal status after submission, such as proposal selection/non-selection status or contract award status. Contact the DoW Service/Component that originated the topic following the Service/Component-specific instructions given at the beginning of that Service/Component-specific instructions.

2.4 Required Certifications

At the time of proposal submission, each SBC must certify via the Cover Sheet of the proposal that it meets the size, ownership, and other requirements of the STTR Program. In addition, the Policy Directive includes certification requirements set forth in the SBIR and STTR Extension Act of 2022 (Public Law 117-183). SBCs are required to certify that they are meeting the Program's requirements during the life cycle of the funding agreement: at the time of the Phase I and Phase II award, prior to final payment on the Phase I award, prior to receiving 50 percent of the total Phase II award amount, and prior to final payment on the Phase II award.

2.5 Due Diligence Program to Assess Security Risks

15 U.S.C. §638(vv) requires the DoW, in coordination with the SBA, to establish and implement a due diligence program to assess and, when possible, mitigate national security risks from SBCs seeking a federally funded award. The Department will use the proposal and information in response to the "Disclosures of Foreign Affiliations or Relationships to Foreign Countries" (proposal submission Volume 7) to conduct a risk-based due diligence review of the following areas: cybersecurity practices; patent analysis; employee analysis and foreign ownership, including the SBC's financial ties and obligations (which shall include surety, equity, and debt obligations); and SBC employees' ties to a foreign country, foreign person, or foreign entity. Disclosure of foreign affiliations or relationships on this form does not mean denial. The Department will also assess proposals utilizing open-source analysis and analytical tools, for the purpose of confirming the accuracy of the information provided as well as determining if the proposing SBC failed to disclose the information set forth in 15 U.S.C. 638(g)(13).

After reviewing the proposing SBC's responses to the Disclosures of Foreign Affiliations or Relationships to Foreign Countries, if DoW determines it appropriate the Department may ask the SBC to provide true copies of any contractual or financial obligation or other agreement specific to a business arrangement or joint-venture like arrangement with an enterprise owned by a foreign state or any foreign entity in effect during the five-year period preceding the SBC's proposal submission.

The DoW may not make awards that pose unacceptable security risks pursuant to 15 U.S.C. §638(vv). If

the risk-based due diligence review concludes that an SBC should not be eligible for the specific SBIR or STTR award due to a national security risk that cannot be adequately mitigated, the proposal will not receive consideration for possible award regardless of the results of the technical review of the proposal. Attachment 2: Defense SBIR and STTR Due Diligence Program Common Risk Matrix of the memo titled "Defense Small Business Innovation Research and Small Business Technology Transfer Due Diligence Program", dated 13 May 2024, provides details on the factors for assessing SBC risk during the due diligence review.

2.5.1 Training for Understanding FOCI

DoW has partnered with Project Spectrum to provide an online course on Understanding Foreign Ownership, Control, or Influence (FOCI). This course defines FOCI, explains what it means to be under FOCI, and details FOCI's effect on a company seeking initial or continued eligibility for access to a federally funded award. Small business concerns can register and access this course by following the instructions below:

1. Go to projectspectrum.io
2. Click "Profile/Dashboard" in the top right and then click "Sign Up" from the dropdown menu.
3. Follow the instructions to sign up for an account. Descriptions of the account types are provided below each option.
4. Verify your email by entering the code sent to the email address you provided when signing up.
5. Log in to Project Spectrum by clicking "Profile/Dashboard > Login" in the top right.
6. Hover over "Courses" in the Navigation Bar, and then select "FOCI" from the dropdown listing.
7. Copy the provided password.
8. Click on the "Understanding Foreign Ownership, Control, or Influence (FOCI)" course, which will open a new browser tab.
9. From the new tab, log in to Encite.io using your email address and the copied password.
10. Enroll in the course and click "Enter" to begin.

For Project Spectrum registration or access assistance, please email support@projectspectrum.io.

2.6 Cybersecurity Maturity Model Certification (CMMC) Program

DoW has established the Cybersecurity Maturity Model Certification (CMMC) Program in order to verify contractors have implemented required security measures necessary to safeguard Federal Contract Information (FCI) and Controlled Unclassified Information (CUI). CMMC requirements will apply to all DoW solicitations and contracts for which a defense contractor or subcontractor will process, store, or transmit FCI or CUI on its unclassified contractor information systems. It is important to note that subcontractors are subject to flow-down requirements if the subcontractor will process, store, or transmit FCI or CUI on its unclassified contractor information systems.

The CMMC Program provides assessments at three levels, each incorporating security requirements from existing regulations and guidelines.

Level 1: Basic Safeguarding of FCI

- Requirements: Annual self-assessment and annual affirmation of compliance with the 15 security requirements in FAR clause 52.204-21.

Level 2: Broad Protection of CUI

- Requirements:
    1. Either a self-assessment or an independent assessment by an authorized CMMC Third-Party Assessment Organization (C3PAO) every three years, as specified in the solicitation.
        - Determined by the type of information processed, transmitted, or stored on the contractor or subcontractor information systems.
    2. Annual affirmation, verify compliance with the 110 security requirements in NIST SP 800-171 Revision 2.

Level 3: Higher-Level Protection of CUI Against Advanced Persistent Threats

- Requirements:
    1. Achieve CMMC Status of Final Level 2.
    2. Undergo an assessment every three years by the Defense Contract Management Agency's (DCMA) Defense Industrial Base Cybersecurity Assessment Center (DIBCAC).
    3. Provide an annual affirmation verifying compliance with the 24 identified requirements from NIST SP 800-172.

Projected CMMC level requirements will be identified within the respective topic. Proposers should carefully review and consider these updated requirements when preparing their proposals, as compliance may impact proposed costs and technical approaches. For additional CMMC information and resources visit https://dodcio.defense.gov/cmmc/About/.

2.6.1 Resources for Understanding FOCI & Cybersecurity Readiness

DoW has partnered with Project Spectrum to provide online resources and courses on Understanding Foreign Ownership, Control, or Influence (FOCI) and improving cybersecurity readiness, resiliency, and compliance as it relates to CMMC. SBCs can register and access these courses and resources at projectspectrum.io.

2.7 Joint Ventures

A small business joint venture entity must submit, with its proposal, the representation required in paragraph (c) of FAR solicitation provision 52.212-3, Offeror Representations and Certifications-Commercial Products and Commercial Services, and paragraph (c) of FAR provision 52.219-1, Small Business Program Representations, in accordance with 52.204-8(d) and 52.212-3(b) for the following categories:

a. Small business;
b. Service-disabled veteran-owned small business;
c. Women-owned small business (WOSB) under the WOSB Program;
d. Economically disadvantaged women-owned small business under the WOSB Program; or
e. Historically underutilized business zone small business.

These representations can be found here and must be uploaded to Volume 5, Supporting Documents of the proposal submission in DSIP, if applicable.

2.8 Conflicts of Interest

Contract awards to an SBC owned by or employing current, or previous, Federal Government employees could create conflicts of interest for those employees, which may be a violation of federal law.

2.9 Organizational Conflicts of Interest (OCI)

FAR 9.5 Requirements

In accordance with FAR 9.5, proposing SBCs are required to identify and disclose all facts relevant to potential organizational conflicts of interest (OCIs) involving the proposing SBC's organization and any

proposed team member (sub-awardee, consultant). The proposing SBC is responsible for providing this disclosure with each submitted proposal. The disclosure must include the proposing SBC's, and as applicable, proposed team member's OCI mitigation plan. The OCI mitigation plan must include a description of the actions the proposing SBC has taken, or intends to take, to prevent the existence of conflicting roles that might bias the proposing SBC's judgment, and to prevent the proposing SBC from having unfair competitive advantage. The OCI mitigation plan will specifically discuss the disclosed OCI in the context of each of the OCI limitations outlined in FAR 9.505-1 through FAR 9.505-4.

Agency Supplemental OCI Policy

DoW Services/Components also may have a supplemental OCI policy prohibiting contractors/performers from concurrently providing scientific engineering technical assistance (SETA), advisory and assistance services (A&AS), or similar support services, and being a technical performer. As part of the FAR 9.5 disclosure requirement, a proposing SBC must affirm whether the proposing SBC or any proposed team member (sub-awardee, consultant) is providing SETA, A&AS, or similar support to any DoW Service/Component office(s) under: (a) a current award or sub-award; or (b) a past award or sub-award that ended within one calendar year prior to the proposal's submission date. If SETA, A&AS, or similar support is or was provided to any DoW Service/Component office(s), the proposal must include:

a. The name of the DoW Service/Component office receiving the support;
b. The prime contract number;
c. Identification of proposed team member (sub-awardee, consultant) providing the support; and
d. An OCI mitigation plan in accordance with FAR 9.5.

Government Procedures

In accordance with FAR 9.503, 9.504 and 9.506, the Government will evaluate OCI mitigation plans to avoid, neutralize, or mitigate potential OCI issues before award and determine whether it is in the government's interest to grant a waiver. The U.S. Government will only evaluate OCI mitigation plans for proposals determined selectable under the solicitation evaluation criteria and funding availability.

The government may require proposing SBCs provide additional information to support evaluation of the proposing SBC's OCI mitigation plan.

If the government determines a proposer failed to fully disclose an OCI; or failed to provide a government waiver as described above; or failed to reasonably provide additional information the government requested when evaluating the proposer's OCI mitigation plan, the government may reject the proposal and withdraw it from consideration for award.

2.10 Research Involving Human Subjects/Human Subject Research (RIHS/HSR)

All research involving human subjects, to include use of human biological specimens and human data, shall comply with the applicable federal and state laws, and agency policy/guidelines for human subject protection (see Section 5.2 and Appendix B).

Institutions receiving funding for research involving human subjects must provide documentation of a current federal assurance of compliance with federal regulations for human subject protection; for example a Department of Health and Human Services, Office for Human Research Protections federal-wide assurance (http://www.hhs.gov/ohrp). The awarding DoW Service/Component may also request additional federal assurance documentation. All institutions engaged in human subject research, to include subcontractors, must also have a valid assurance. In addition, personnel involved in human subject research must provide documentation of completed appropriate training for the protection of human subjects. Institutions proposing to conduct human subject research that meets one of the exemption criteria in 32 CFR 219.101 are not required to have a federal assurance of compliance.

Proposing SBCs should clearly segregate research activities involving human subjects from other R&D activities in their proposal.

If selected, institutions must also provide documentation of institutional review board (IRB) approval, or a determination from an appropriate official in the institution, that the work meets one of the exemption criteria with 32 CFR 219. As part of the IRB review process, evidence that all investigators are appropriately trained should accompany the protocol. The protocol, separate from the proposal, must include a detailed description of the research plan, study population, risks and benefits of study participation, recruitment and consent process, and data collection and analysis.

The amount of time required for the IRB to review and approve the protocol will vary based on the IRB's procedures, the complexity of the research, the level of risk to study participants and the responsiveness of the investigator. The average IRB approval process can last between one and three months. Once the IRB has approved the research, the awarding DoW Service/Component will review the protocol and the IRB's determination to ensure that the research will be conducted in compliance with both DoW and Service/Component policies. The DoW review process can last between three to six months. Ample time should be allotted to complete both the IRB and DoW approval processes prior to recruiting subjects. **No funding can be used towards human subject research until ALL approvals are granted. Submitters proposing research involving human and/or animal use are encouraged to separate these tasks in the technical proposal and cost proposal to avoid potential contract award delays.**

2.11 Research Involving Animal Subjects

All research, development, testing, experimentation, education, or training involving the use of animals shall comply with the applicable federal and agency rules on animal acquisition, transport, care, handling, and use (see Section 5.2 and Appendix B).

For submissions containing animal use, proposals should briefly describe plans for their institutional animal care and use committee (IACUC) review and approval.

All recipients must receive their IACUC's approval, as well as secondary or headquarters-level approval from a DoW veterinarian trained or experienced in laboratory animal medicine and science. **No animal research may be conducted using DoW funding until all appropriate DoW office(s) grant approval. Submitters proposing research involving human and/or animal use are encouraged to separate these tasks in the technical proposal and cost proposal to avoid potential contract award delays.**

2.12 Research Involving Recombinant DNA Molecules

All research involving recombinant DNA molecules shall comply with the applicable federal and state law, regulation, and additional agency guidance. An institutional biosafety committee must approve the research.

In addition to the standard federal and DoW procurement certifications, the SBA SBIR/STTR Policy Directive requires the proposing business concerns provide certain information at time of award and during the award life cycle. Each proposing SBC must provide this additional information at the time of the Phase I and Phase II award, prior to final payment on the Phase I award, prior to receiving 50 percent of the Phase II total award amount, and prior to final payment on the Phase II award.

2.13 Federal Acquisition Supply Chain Security Act Orders

FAR 52.204-29 Federal Acquisition Supply Chain Security Act (FASCA) Orders—Representation and Disclosures and FAR 52.204-30 FASCA Orders—Prohibition are included in this solicitation. In accordance with FAR 52.204-29 and FAR 52.204-30, proposing SBCs must review FASCSA orders here for covered articles, or any products or services produced or provided by a source, that an applicable

FASCSA order prohibits.

During contract performance, the contractor shall review SAM.gov at least once every three months, or as the contracting officer advises, to check for covered articles, or products or services produced subject as part of any new FASCSA order(s) that could impact their supply chain, and report to the contracting officer any covered article, or product or service produced or provided by a source provided to the government or used during the contract performance.
The proposing SBC represents that, via proposal submission under this solicitation, it conducted a reasonable inquiry, and it does not propose to provide or use any covered article, or any products or services produced or provided by a source, if an applicable FASCSA prohibited the covered article or the source effective this solicitation's issue date.

2.14 BlueUAS Certification

SecWar *Unleashing U.S. Military Drone Dominance* memorandum dated 10 July 2025 provides a goal to eliminate barriers to purchase and train with emerging UAS technology. As of December 2025, the Department's BlueUAS Certification process has transitioned from the Defense Innovation Unit (DIU) to the Defense Contract Management Agency (DCMA). Companies who are seeking BlueUAS certification can review the process and submit their product(s) for assessment via DCMA's Blue List portal: https://bluelist.appsplatformportals.us/Blue-UAS-Portal/.

IMPORTANT: Unless specifically required per the topic information, companies are not required to have BlueUAS certification to compete for awards as included in this solicitation, nor are DoW SBIR/STTR program offices responsible for reimbursement of costs incurred by companies pursuing BlueUAS certification.

3.0 PROPOSAL PREPARATION INSTRUCTIONS AND REQUIREMENTS

3.1 Introduction

The proposal must provide sufficient information to demonstrate to the evaluator(s) that the proposed work represents an innovative approach to an important scientific or engineering problem and is worthy of support under the stated criteria. The proposed research or R&D must be responsive to the chosen topic, although it does not need to use the exact approach specified in the topic, if applicable. SBCs should consider the following:

a. Does the technical approach have a reasonable chance of meeting the topic objective (if applicable)?
b. Is this approach innovative, not routine, with potential for commercialization?
c. Does the proposing SBC have the capability to implement the technical approach or can it obtain the appropriate people and equipment for the task?

DSIP provides a structure for providing the following proposal volumes:

a. Volume 1: Proposal Cover Sheet
b. Volume 2: Technical Volume
c. Volume 3: Cost Volume
d. Volume 4: Company Commercialization Report
e. Volume 5: Supporting Documents
f. Volume 6: Fraud, Waste and Abuse Training
g. Volume 7: Disclosures of Foreign Affiliations or Relationships to Foreign Countries

Each Service/Component guidance on allowable proposal content may vary. A completed proposal submission in DSIP does NOT indicate that each proposal volume has been completed in accordance with the Service/Component-specific instructions. Accordingly, it is the proposing SBC's responsibility to consult the Service/Component-specific instructions for detailed guidance, including required proposal documentation and structure, cost and duration limitations, budget structure, TABA allowance and proposal page limits.

3.2 Export-Controlled Topic Requirements

Export of all unclassified technical data with military or space application in the possession of, or under the control of, a DoW Component information, which includes, in some circumstances, release to foreign nationals within the United States, without first obtaining approval, authorization, or license from the Department of State for items controlled by the International Traffic in Arms Regulations (ITAR), or the Department of Commerce for items controlled by the Export Administration Regulations (EAR), may constitute a violation of law.

Pursuant to Defense Federal Acquisition Regulation Supplement (DFARS) Procedures Guidance and Information 225.7901-2, your firm should direct its attention to the clause at DFARS 252.225-7048, Export-Controlled Items for questions concerning compliance with ITAR/EAR.

Further, in accordance with Department of War Directive 5230.25, Withholding of Unclassified Technical Data from Public Disclosure, contractors, or subcontractors that will handle technical data that might have military or space applications, must certify that they will comply with all applicable U.S. laws that control the export of sensitive data, as follows:

> If any portion of the proposed STTR effort is subject to ITAR an SBC must complete a fully certified DD Form 2345, Military Critical Technical Data Agreement, and provide a copy or evidence of application submission within Volume 5. The DD Form 2345, Military Critical Technical Data Agreement, instructions, and Frequently Asked Questions (FAQs) may be found at the United States/Canada Joint Certification Program (JCP) website, JCP Portal.
>
> If any portion of the proposed STTR effort is subject to EAR, your firm must submit for and obtain the proper export licenses through the Department of Commerce's Bureau of Industry and Security on-line system, SNAP-R. Failure to obtain the proper export licenses in a timely manner will significantly delay contract award, become grounds for cancellation of the contract action, or become grounds for termination of an existing contract.

Topics under this announcement may be subject to ITAR/EAR and may be identified as such. However, export control compliance statements found in this document are not meant to be all inclusive. They do not remove any liability from the applicant to comply with applicable ITAR or EAR export control restrictions. For additional guidance on export control considerations, SBCS are encouraged to review the SBIR|STTR ITAR tutorial, which provides an overview of ITAR and EAR and their applicability to federally funded research.

3.3 Classified Proposals

Classified proposals will not be accepted under the DoW STTR Program. If topics require classified work during Phase II, the proposing SBC must have a facility clearance to perform the work. For more information on facility and personnel clearance procedures and requirements, please visit the Defense Counterintelligence and Security Agency (DCSA) website at: https://www.dcsa.mil/mc/ctp/fc/.

3.4 Promotional Materials

Promotional and non-project related discussion is discouraged, and additional information provided via website links or on computer disks, CDs, DVDs, video tapes or any other medium will not be accepted or considered in the proposal evaluation.

3.5 Prior, Current, or Pending Support of Similar Proposals or Awards

While it is permissible, with proposal notification, to submit identical proposals or proposals containing a significant amount of essentially equivalent work for consideration under numerous federal program solicitations, it is unlawful to enter negotiation for contracts requiring essentially equivalent effort. If there is any question concerning prior, current, or pending support of similar proposals or awards, it must be disclosed to the soliciting agency or agencies as early as possible and declared on the proposal cover sheet.

3.6 Marking Proprietary Proposal Information

Proposing SBCs that include data in their proposals they do not want disclosed to the public for any purpose, or only used for government evaluation purposes, shall:

a. Mark the first page of each volume of the proposal submission with the following legend:

   "This proposal includes data that shall not be disclosed outside the Government and shall not be duplicated, used, or disclosed-in whole or in part-for any purpose other than to evaluate this proposal. If, however, a contract is awarded to this proposing SBC as a result of-or in connection with-the submission of this data, the Government shall have the right to duplicate, use, or disclose the data to the extent provided in the resulting contract. This restriction does not limit the Government's right to use information contained in this data if it is obtained from another source without restriction. The data subject to this restriction are contained in pages [insert numbers or other identification of sheets]"; and

b. Mark each sheet of data it wishes to restrict with the following legend:

   "Use or disclosure of data contained on this page is subject to the restriction on the first page of this volume."

The DoW assumes no liability for disclosure or use of unmarked data and may use or disclose such data for any purpose.

Restrictive notices notwithstanding, support contractors may handle proposals and final reports submitted through DSIP for administrative purposes only; they are required to adhere to appropriate non-disclosure agreements.

3.7 Phase I Proposal Instructions

a. **Proposal Cover Sheet (Volume 1)**

   The proposal cover sheet is prepared on DSIP. The cover sheet must include a brief technical abstract that describes the proposed R&D project and an anticipated benefits and potential commercial applications discussion. Each section should be no more than 3,000 characters. **Do not include proprietary or classified information in the proposal cover sheet**. If your proposal is selected for negotiation and possible award, the technical abstract and anticipated benefits discussion may be publicly released online. DSIP will assign a proposal number once the cover sheet is saved. You may modify the cover sheet as needed until the topic closes.

NOTE: the amounts listed in the percentage of work (POW) certification question on the proposal cover sheet are derived from SBC-entered information in the cost volume (Volume 3). Details on the calculation can be viewed in DSIP during proposal submission.

If the POW calculations fall below eligibility requirements, the small business concern will not be able to proceed with proposal submission. Deviations from the POW minimum requirements for STTR proposals are not allowed.

b. **Technical Volume Format (Volume 2)**

1. **File Type.** The Technical Volume must be a single PDF file, including graphics. Perform a virus check before uploading the technical volume file. If a virus is detected, the proposal may be rejected. **Do not lock, password protect or encrypt the uploaded file. Do not include or embed active graphics, such as videos, moving pictures, or other similar media, in the document.**

2. **Length.** It is the proposing SBC's responsibility to verify that the technical volume does not exceed the page limit after upload to DSIP. Please refer to Service/Component-specific instructions for how a technical volume is handled if the stated page count is exceeded. Some Services/Components will reject the entire technical proposal if the proposal exceeds the stated page count.

3. **Layout.** Number all proposal pages consecutively. Submit a direct, concise, and informative research or R&D proposal (no type smaller than 10-point on standard 8-1/2" x 11" paper with one-inch margins, including the header). Each header on each page in the technical volume should contain the proposing SBC's name, topic number, and the DSIP-assigned proposal number from the cover sheet.

c. **Technical Volume Content (Volume 2)**

The Technical Volume should cover the following items <u>in the order given below</u>:

1. Identification and Significance of the Problem or Opportunity
2. Phase I Technical Objectives
3. Phase I Statement of Work
4. Related Work
5. Relationship with Future Research or Research and Development
6. Commercialization Strategy
7. Key Personnel
8. Foreign Citizens
9. Facilities/Equipment
10. Subcontractors/Consultants
11. Prior, Current, or Pending Support of Similar Proposals or Awards
12. Identification and Assertion of Restrictions on the Government's Use, Release, or Disclosure of Technical Data or Computer Software

A Phase I technical volume template is available in Appendix A to provide details and helpful guidelines for completing each section of your Phase I technical proposal.

d. **Cost Volume Content (Volume 3)**

Complete the cost volume using the DSIP cost volume form. Some items in the cost breakdown may not apply to the proposed project. There is no need to provide information on each individual item; make sure to provide enough information for evaluators to understand the requested funds' planned use if a contract is awarded.

1. List all key personnels' names and include their individual hours dedicated to the project as direct labor.
2. While special tooling and test equipment and material cost may be included under Phase I, equipment and material inclusion will be carefully reviewed relative to need and appropriateness for the work proposed. Special tooling and test equipment purchases must, in the Service/Component contracting officer's opinion, be advantageous to the U.S. Government and should relate directly to the specific topic. These may include such items as innovative instrumentation or automatic test equipment. Title to property the U.S. Government furnished or acquired with government funds will be vested with the DoW Service/Component, unless it is determined that title transfer to the contractor would be more cost effective than the DoW Service/Component equipment recovery.
3. Cost for travel funds must be justified and related to the project needs.
4. Cost sharing is permitted for proposals under this solicitation; cost sharing is not required, nor will it be an evaluation factor in the Phase I proposal consideration.
5. A Phase I option (if applicable) should be fully costed separately from the Phase I (base) approach.
6. All subcontractor costs and consultant costs, such as labor, travel, equipment, materials, must be detailed at the same level as prime contractor costs. Provide detailed subcontractor costs substantiation in your cost proposal. Supporting Documents (Volume 5) may be used if additional space is needed.

If a proposal is selected for negotiation and possible award, you must be prepared to submit further documentation to the Service/Component contracting officer to substantiate costs (e.g., a cost estimates explanation for equipment, materials, and consultants or subcontractors). For more information about cost proposals and accounting standards, see visit DCAA's website.

e. **Company Commercialization Report (Volume 4)**

The company commercialization report (CCR) allows companies to report funding outcomes resulting from prior SBIR and STTR awards. SBA requires SBIR and STTR awardees to update and maintain their organization's CCR on SBIR.gov. Commercialization information is required upon the last deliverable's completion under the funding agreement. Thereafter, SBIR and STTR awardees are requested to voluntarily update the information in the database annually for a minimum 5-year period.

If the proposing SBC has prior DoW and/or non-DoW Phase I and/or Phase II SBIR/STTR awards, regardless of whether the project has any commercialization to date, the firm admin must download the CCR's PDF copy from SBIR.gov and upload it to DSIP's "Firm Forms" section. The DSIP firm admin completes the firm forms are completed and are applies them to all proposals the proposing SBC submits. To fulfill the DSIP CCR requirement complete the following:

1. Log into the firm account at https://www.sbir.gov/.
2. Navigate to My Dashboard > My Documents to view or print the information currently contained in the "Company Registry Commercialization Report."

3. Create or update the commercialization record, from the company dashboard, scroll to the "My Commercialization" section and click the "Create/Update Commercialization" tab under "Current Report Version." Please refer to the "Instructions" and "Guide" documents contained on dashboard for more detail on completing and updating the CCR. **Ensure the report is certified and submitted.**
4. Click the "Company Commercialization Report" PDF under the dashboard's "My Documents" section to download the CCR PDF.
5. Upload the CCR PDF (downloaded from SBIR.gov in previous step) to the "Company Commercialization Report" in DSIP's "Firm Forms" section. The firm admin must complete this upload action.

In Volume 4 of the DSIP proposal submission, the proposing SBC will be prompted to answer: "Do you have a new or revised Company Commercialization Report to upload?" There are three possible courses of action:

a. If the proposing SBC has prior DoW and/or non-DoW Phase I and/or Phase II SBIR/STTR awards and **has a new or revised CCR from SBIR.gov to upload to DSIP**, select YES.
   1. If the user is the firm admin, they can upload the CCR PDF from SBIR.gov directly on this page. It will also be updated in the "Firm Forms" and be associated with all new or in-progress proposals the proposing SBC submitted. If the user is not the firm admin, they will receive a message that they do not have access and must contact the firm admin to complete this action.
   2. **WARNING:** Uploading a new CCR under the DSIP "Firm Forms" section or clicking "Save" or "Submit" in one proposal's Volume 4 is considered a change for ALL proposals under any open solicitations. If a proposing SBC has previously certified and submitted any Phase I or Direct to Phase II proposals under ***any*** solicitation ***still open***, those proposals will be automatically reopened. Proposing SBCs will have to recertify and resubmit affected proposals. If a proposing SBC does not recertify or resubmit affected proposals, they will not be considered fully submitted and will not be evaluated.

b. If the proposing SBC has prior DoW and/or non-DoW Phase I and/or Phase II SBIR/STTR awards, and **no new or revised CCR from SBIR.gov to upload to DSIP**, select NO.
   1. If a prior CCR was uploaded to the "Firm Forms", the proposing SBC will see a file dialog box at the bottom of the page and can view the previously uploaded CCR. This read-only access allows the proposing SBC to confirm the firm admin uploaded the CCR.
   2. If no file dialog box appears at the bottom of the page **there is no previously uploaded CCR in the DSIP "Firm Forms**." To fulfill the DSIP CCR requirement the firm admin must follow steps 1-5 listed above to download a CCR PDF from SBIR.gov and upload it to the DSIP "Firm Forms" to be included with all proposal submissions.

c. If the proposing SBC has **NO** prior DoW and/or non-DoW Phase I and/or Phase II SBIR/STTR awards, the CCR upload from SBIR.gov is not required and SBC will select NO. The proposal's CCR section will be marked complete.

Please refer to the Service/Component-specific instructions for details on how the CCR information will be considered during proposal evaluations.

f. **Supporting Documents (Volume 5)**

Volume 5 is provided for proposing SBCs to submit additional documentation to support the Coversheet (Volume 1), Technical Volume (Volume 2), and the Cost Volume (Volume 3).

Proposing SBCs **MUST** submit the negotiated Allocation of Rights agreement between the SBC and research institution at the time of proposal submission. This agreement must be uploaded to Volume 5: Supporting Documents. Refer to section 1.4.f for more detail.

The following documents may also be included in Volume 5, if applicable to the proposal. Refer to Service/Component-specific instructions for additional Volume 5 requirements. **Reminder: A completed proposal submission in DSIP does NOT indicate the mandatory supporting documents have been uploaded in accordance with the Service/Component-specific instructions.**

1. Letters of support
2. Additional cost information
3. Funding agreement certification
4. Technical data rights (assertions)
5. Lifecycle certification
6. Verification of Eligibility of Small Business Joint Ventures, if applicable
7. DD Form 2345, Militarily Critical Technical Data Agreement, if applicable (see section 3.2)

g. **Fraud, Waste, and Abuse Training (Volume 6)**

The fraud, waste, and abuse (FWA) training is **required** for DoW SBIR/STTR proposals. FWA training provides information on what represents FWA in the SBIR/STTR program, the most common mistakes that lead to FWA, as well as the penalties and ways to prevent FWA. The training currently consists of a 3-page PDF, consistent with the tutorial provided by the SBA. This training material must be thoroughly reviewed once per year and can be found here and in the DSIP proposal submission module for Volume 6. Plan time to review the tutorial during completion of Volume 6, prior to the proposal submission deadline. The individual designated as the Proposal Owner must complete this training on behalf of the proposing small business.

h. **Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)**

In accordance with 15 U.S.C. §638(vv) and the SBA SBIR/STTR Policy Directive, the DoW will review all proposals submitted in response to this solicitation to assess security risks of SBCs seeking a federally funded award. SBCs must complete the DSIP Volume 7 webform "Disclosures of Foreign Affiliations or Relationships to Foreign Countries" (NOTE: PDF uploads are no longer accepted). The corporate official cannot certify and submit the full proposal until the Volume 7 webform is fully completed and submitted.

Please be aware that the "Disclosures of Foreign Affiliations or Relationships to Foreign Countries" form WILL NOT be accepted as a supporting document in DSIP's Volume 5 proposal submission. Do not upload any previous versions of this form to Volume 5.

For additional details, please refer to Section 2.5. The disclosure questions are below:

1. Is any owner or covered individual of the applicant or awardee party to any malign foreign talent recruitment program? If yes, disclose the first and last name of each owner or covered individual, identify their role (i.e., owner or covered individual), and the malign foreign talent recruitment program.
2. Is there a parent company, joint venture, or subsidiary, of the applicant or awardee that is based in or receives funding from, any foreign country of concern? If yes, disclose the name, full address, applicant or awardee relationships (i.e., parent company, joint venture, or subsidiary) of each entity based in, or funded by, any foreign country of concern.
3. Does the applicant or awardee have any current or pending contractual or financial obligation or other agreement specific to a business arrangement, or joint venture-like arrangement with an enterprise owned by a foreign state or any foreign entity? If yes, disclose the name of each enterprise or foreign entity, type of obligation, agreement, or arrangement (*i.e.,* contractual, financial, or other), description of obligation, agreement, or arrangement, and the foreign state(s) and/or the country of the foreign entity (or entities).
4. Is the applicant or awardee wholly owned in a foreign country? If yes, disclose the foreign country.
5. Does the applicant or awardee have any venture capital or institutional investment? If yes, proceed to question 5a. If no, proceed to question 6.

   5a. Does the investing entity have a general partner or any other individual holding a leadership role who has a foreign affiliation with any foreign country of concern? If yes or unable to determine, disclose the venture capital or institutional investing entity's name, the percentage of ownership obtained by the investing entity, and the type of investment (i.e., equity, debt, or combination of equity and debt).
6. During the previous 5-year period, did the applicant or awardee have any technology licensing or intellectual property sales or transfers, to a foreign country of concern? If yes, disclose the name, address, and country, of the institution or entity that licensed, purchased, or received the technology or intellectual property.
7. Is there any foreign business entity, offshore entity, or entity outside the United States related to the applicant or awardee? If yes, disclose the entity name, relationship type (i.e., foreign business entity, offshore entity, entity outside the United States), description of the relationship to the applicant or awardee, and entity address and country.
8. Does the applicant or awardee have an owner, officer, or covered individual that has a foreign affiliation with a research institution located in a foreign country of concern? If yes, disclose the first and last name of each owner, officer, or covered individual that has a foreign affiliation with a foreign country of concern, identify their role (i.e., owner, officer, or covered individual), and the name of the foreign research institution and the foreign country of concern where it is located.

3.8 Phase II Proposal Information

Only Phase I awardees may submit Phase II proposals. Phase II proposals submission must follow individual Service/Component instructions. Awarding DoW Services/Component, either in the Phase I award or via subsequent notification, will provide details on Phase II proposal due date, content, and submission requirements. If a proposing SBC submits their Phase II proposal prior to the individual Service/Component's dates, it may be rejected without evaluation.

Due to specific limitations on the amount of funding and number of awards awarded to a particular proposing SBC per topic using SBIR/STTR program funds, head of agency determinations are required

before a different agency may make an award using another agency's topic. This limitation does not apply to Phase III funding. Please contact your original sponsoring agency before submitting a Phase II proposal to an agency other than the one who sponsored the original topic.

SBIR/STTR Policy Directive Section 4(b)(1)(i) allows that, at the agency's discretion, projects awarded a Phase I under a solicitation for SBIR may transition in Phase II to STTR and vice versa. A proposing SBC wishing to transfer from one program to another must contact their designated technical monitor to discuss the reasons for the request and the agency's ability to it. The transition may be proposed prior to award or during the Phase II effort performance. Agency disapproval of a request to change programs shall not be grounds for granting relief from any contractual performance requirement. All approved transitions between programs must be noted in the contracting officer-signed Phase II award or award modification that indicates the removal or addition of the research institution and the revised percentage of work requirements.

3.8.1 Phase II Commercialization Strategy

At a minimum, the commercialization strategy must address the following five questions:

a. What will be the first product to use this technology?
b. Who will be the customers, and what is the estimated market size?
c. How much money is needed to bring the technology to market, and how will that money be raised?
d. Does the proposing SBC contain marketing expertise and, if not, how will the SBC acquire that expertise?
e. Who are the proposing SBC's competitors, and what is the price and/or quality advantage over those competitors?

The commercialization strategy must also include a schedule showing the anticipated quantitative commercialization results at 1) one year after the Phase II project starts, 2) at Phase II completion, and 3) after Phase II completion (i.e., additional investment amount, sales revenue, etc.). After Phase II award, the proposing SBC is required to report actual sales and investment data in its SBA company commercialization report via SBIR.gov's "My Dashboard" on a minimum annual basis. Please refer to the Service/Component-specific instructions for guidance on formatting, page count and other details.

3.8.2 Phase II Adequate Accounting System

To reduce the small business's risk and avoid potential contracting delays, companies interested in pursuing Phase II STTR contracts and other contracts of similar size with the DoW, may require an adequate accounting system in place per General Accepted Accounting Principles, Generally Accepted Government Auditing Standards, Federal Acquisition Regulation (FAR) and Cost Accounting Standards. The Defense Contract Audit Agency (DCAA) will audit the accounting system. See DCAA's website for requirements and standards, the audit process overview, and a pre-award system adequacy checklist.

4.0 METHOD OF SELECTION AND EVALUATION CRITERIA

4.1 Evaluation Process

All proposals will be evaluated and judged on a competitive basis in terms of technical capability and technical value. Proposals will be initially screened to determine responsiveness to the topic objective. Proposals passing this initial screening will be technically evaluated by engineers, scientists, or subject matter experts to determine the most promising technical and scientific approaches. As a common

statement of work does not exist, each proposal will be assessed on the merit of the approach in achieving the technical objectives established in the topic.

4.2 Evaluation Criteria

Proposals will be evaluated based on the criteria outlined below, unless otherwise specified in the Service/Component-specific instructions. Selections will be based on a determination of the overall technical value of each proposal and an evaluation of the cost volume, with the appropriate method of analysis given the contract type to be awarded, for selection of the proposal(s) most advantageous to the Government, considering the following factors which are listed in descending order of importance:

a. The soundness, technical merit, and innovation of the proposed approach and its incremental progress toward topic or subtopic solution.
b. The qualifications of the proposed principal/key investigators, supporting staff, and consultants. Qualifications include not only the ability to perform the R&D but also the ability to commercialize the results.
c. The potential for commercial (government or private sector) application and the benefits expected to accrue from this commercialization.

Cost or budget data submitted with the proposals will be considered during evaluation.

Technical reviewers will base their conclusions only on information contained in the proposal. Do not assume reviewers are acquainted with the proposing SBC, key individuals, or any referenced experiments. Relevant supporting data such as journal articles, literature, including Government publications, etc., should be included based on requirements listed in Service/Component-specific instructions.

4.3 Proposal Status & Feedback

Proposing SBCs will be notified of selection or non-selection status for an award by the DoW Service/Component that originated the topic <u>no later than 90 days</u> of the closing date for this solicitation. Please refer to the Service/Component-specific instructions for details.

After final selection decisions have been announced, the SBC may be provided proposal feedback in the form of a written debrief. This debriefing process varies across the DoW Services/Components. Please refer to the Service/Component-specific instructions for details on the debriefing processes.

4.4 Award Denials

The DoW will not make an award under the SBIR program if it determines:

a. The SBC submitting the proposal
    i. has an owner or covered individual that is party to a malign foreign talent recruitment program;
    ii. has a business entity, parent company, or subsidiary located in the People's Republic of China or another foreign country of concern; or
    iii. has an owner or covered individual that has a foreign affiliation with a foreign entity located in the People's Republic of China or another foreign country of concern; and
b. The relationships and commitments described in clauses (i) through (iii) of subparagraph (A)
    i. interfere with the capacity for DoW-supported activities to be carried out;
    ii. create duplication with DoW-supported activities;
    iii. present concerns about conflicts of interest;
    iv. were not appropriately disclosed to the DoW;
    v. violate Federal law or terms and conditions of DoW-awarded contracts or other agreements; or
    vi. pose a risk to national security.

4.5 Pre-Award and Post Award Solicitation Protests

Interested parties have the right to protest via procedures in FAR Subpart 33.1; protests exclusively related to this solicitation's terms must be served to: osd.ncr.ousd-r-e.mbx.SBIR-STTR-Protest@mail.mil

For protests filed with the Government Accountability Office (GAO), a copy of the protest shall be submitted to the email address listed above (pre-award ONLY) or DoW Service/Component POC (post-selection/award decision ONLY) within one day of filing with the GAO. Protests of small business status of a selected proposing SBC may also be made to the SBA via the procedures in FAR § 19.302.

For the purposes of a protest related to a particular topic selection, non-selection or award decision, protests should be served to the point-of-contact (POC) listed in the instructions of the DoW Component that authored the topic.

5.0 ADDITIONAL CONSIDERATIONS

5.1 Award Information

The number of awards will be consistent with the Component's RDT&E budget. No awards will be made until evaluation of all qualified proposals for a specific topic is complete.

Each proposal selected for negotiation and possible award will be funded under a negotiated instrument and may include a reasonable fee or profit consistent with normal profit margins consistent with normal profit margins provided to profit-making proposing SBCs for R/R&D work.

Contract value varies among the DoW Services/Components; it is important for proposing SBCs to review Service/Component-specific instructions regarding award size.

The SBA SBIR/STTR Policy Directive, Section 7(c)(1)(ii), states agencies should issue the award no more than 180 days after the closing date of the topic.

5.2 Contract Requirements

Upon contract award, the contractor will be required to make certain legal commitments through acceptance of U.S. Government contract clauses in the Phase I contract. The examples below are illustrative of the types of provisions the Federal Acquisition Regulation requires in the Phase I contract. This is not an exhaustive provisions list that could be included in Phase I contracts, nor does it contain specific clause wording. Appendix C of this solicitation contains additional potential required Federal Acquisition Regulation (FAR) and Defense Federal Acquisition Regulation Supplement (DFARS) clauses. Copies of complete general provisions will be made available prior to award.

While this solicitation does not address all potential clauses for resultant funding agreements, proposers are specifically directed to the Defense Federal Acquisition Regulation Supplement (DFARS) clause 252.204-7021, Cybersecurity Maturity Model Certification (CMMC) Requirements. This clause update reflects significant changes to cybersecurity requirements. Additional information provided in section 2.13 of this solicitation.

Examples of general provisions:

a. **Standards of Work**. Work performed under the Funding Agreement must conform to high professional standards.
b. **Inspection.** Work performed under the Funding Agreement is subject to Government inspection and evaluation at all times.

c. **Examination of Records**. The Comptroller General (or a duly authorized representative) must have the right to examine any pertinent records of the Awardee involving transactions related to this Funding Agreement.
d. **Default**. The Federal Government may terminate the Funding Agreement if the contractor fails to perform the work contracted.
e. **Termination for Convenience**. The Funding Agreement may be terminated at any time by the Federal Government if it deems termination to be in its best interest, in which case the Awardee will be compensated for work performed and for reasonable termination costs.
f. **Disputes.** Any dispute concerning the Funding Agreement that cannot be resolved by agreement must be decided by the contracting officer with right of appeal.
g. **Contract Work Hours**. The Awardee may not require an employee to work more than 8 hours a day or 40 hours a week unless the employee is compensated accordingly (for example, overtime pay).
h. **Equal Opportunity**. The Awardee will not discriminate against any employee or applicant for employment because of race, color, religion, sex, or national origin.
i. **Equal Opportunity for Veterans**. The Awardee will not discriminate against any employee or application for employment because he or she is a disabled veteran or veteran of the Vietnam era.
j. **Equal Opportunity for People with Disabilities**. The Awardee will not discriminate against any employee or applicant for employment because he or she is physically or intellectually disabled.
k. **Officials Not to Benefit**. No Federal Government official may benefit personally from the SBIR/STTR Funding Agreement.
l. **Covenant Against Contingent Fees**. No person or agency has been employed to solicit or secure the Funding Agreement upon an understanding for compensation except bona fide employees or commercial agencies maintained by the Awardee for the purpose of securing business.
m. **Gratuities**. The Funding Agreement may be terminated by the Federal Government if any gratuities have been offered to any representative of the Government to secure the award.
n. **Patent Infringement**. The Awardee must report each notice or claim of patent infringement based on the performance of the Funding Agreement.
o. **American Made Equipment and Products**. When purchasing equipment or a product under the SBIR/STTR Funding Agreement, purchase only American-made items whenever possible.

5.3 Agency Recovery Authority and Ongoing Reporting

In accordance with Section 5 of the SBIR and STTR Extension Act of 2022, the DoW will:

a. require an SBC receiving an award under its STTR program to repay all amounts received from the federal agency under the award if,
   1. the SBC makes a material misstatement that the federal agency determines poses a risk to national security; or
   2. there is a change in the SBC's ownership, entity structure, or other substantial change in circumstances that the federal agency determines poses a risk to national security; and

b. require an SBC receiving an award under its STTR program to regularly report to the federal agency and the administration throughout the duration of the award on
   1. any change to a disclosure required under the Disclosures of Foreign Affiliations or Relationships to Foreign Countries form;
   2. any material misstatement made under paragraph (A) above; and
   3. any change described in paragraph (B) above.

5.4 Copyrights

With prior written permission of the contracting officer, the awardee may copyright (consistent with any appropriate national security considerations) material developed with DoW support. DoW receives a royalty-free license for the Federal Government and requires that each publication contain an appropriate acknowledgment and disclaimer statement.

5.5 Patents

SBCs normally may retain the principal worldwide patent rights to any invention developed with U.S. Government support. The government receives a royalty-free license for its use, reserves the right to require the patent holder to license others in certain limited circumstances, and requires that anyone exclusively licensed to sell the invention in the United States must normally manufacture it domestically. 35 U.S.C. § 205 authorizes that the government will not make public any information disclosing a government-supported invention for a period of five years to allow the awardee to pursue a patent. See also Section 6.8, Invention Reporting.

5.6 Invention Reporting

STTR awardees must report inventions to the Service/Component within two months of the inventor's report to the awardee, via either paper documentation submission, including fax, or through the Edison Invention Reporting System at www.iedison.gov for participating agencies.

5.7 Technical Data Rights

Rights in technical data, including software, developed under the terms of any contract resulting from proposals submitted in response to this solicitation generally remain with the contractor, except that the U.S. Government obtains a royalty-free license to use such technical data only for government purposes during the period commencing with contract award and ending not less than twenty years after that date. This data should be marked with the restrictive legend specified in DFARS 252.227-7018 (JAN 2025). Upon expiration of the twenty-year restrictive license, the government has government purpose rights in the STTR data. During the license period, the U.S. Government may not release or disclose STTR data to any person other than its support services contractors except: (1) for evaluation purposes; (2) as expressly permitted by the contractor; or (3) a use, release, or disclosure that is necessary for emergency repair or overhaul government-operated items.

If a proposing SBC plans to submit assertions in accordance with DFARS 252.227-7017 (JAN 2025), those assertions must be identified and assertion of use, release, or disclosure restriction must be included with your proposal submission, at the end of the technical volume. The contract cannot be awarded until assertions are approved.

5.8 Final Technical Reports - Phase I through Phase III

a. **Content**: A final report is required for each project phase. The reports must contain in detail the project objectives, work performed, results obtained, and estimates of technical feasibility. A completed SF 298, "Report Documentation Page," will be used as the first page of the report. The DoW Service/Component also may require monthly status and/or progress reports.

b. **SF 298 Form "Report Documentation Page" Preparation**:
   a. If desirable, the proposing SBC may also use language from its Phase II proposal to cover Phase I progress in the final report.

   b. For each unclassified report, the proposing SBC submitting the report should fill in Block 12 (Distribution/Availability Statement) of the SF 298, "Report Documentation Page," with the following statement: "Distribution authorized to U.S. Government only; Proprietary Information, (Date of Determination). Other requests for this document shall be referred to the Service/Component STTR Program Office."

Note: Data developed under a STTR contract is subject to STTR Data Rights, under which DFARS 252.227-7018 (JAN 2025) (see Section 5.7, Technical Data Rights) provides protection. The sponsoring DoW activity, after reviewing the proposing SBC's entry in Block 12, has final responsibility for assigning a distribution statement.

For additional information on distribution statements see the following Defense Technical Information Center (DTIC) website.

c. Block 14 (Abstract) of the SF 298, "Report Documentation Page" must include as the first sentence, "Report developed under STTR contract for topic [insert solicitation topic number. [Follow with the topic title, if possible.]" The abstract must identify the work's purpose and briefly describe the work conducted, the findings or results, and the effort's potential applications. **Since DoW will publish the abstract, it must not contain any proprietary or classified data, and type "UU" in Block 17.**

d. Block 15 (Subject Terms) of the SF 298 must include the term "STTR Report".

c. **Submission**: In accordance with DFARS 252.235-7011, submit an electronic copy of the approved final scientific or technical report, not a summary, delivered under the contract to the Defense Technical Information Center (DTIC) through the web-based input system at https://discover.dtic.mil/submit-documents/ as required by DoD Instruction 3200.12, DoD Scientific and Technical Information Program (STIP). Include a completed Standard Form (SF) 298, Report Documentation Page, in the document, or complete the web-based SF 298. Additional submission resources are available here.

Delivery will normally be within 30 days after completion of the Phase I technical effort.

Other requirements regarding reports and/or other deliverables submission will be defined in each contract's contract data requirements list (CDRL). Special instructions for submitting CLASSIFIED reports will be defined in the contract's delivery schedule.

DO NOT email classified or controlled unclassified reports, or reports containing STTR Data Rights protected under DFARS 252.227-7018 (JAN 2025).

6.0 PROPOSAL SUBMISSION

6.1 Submission Details

DSIP is the official portal for DoW SBIR/STTR proposal submission. Proposers are required to submit proposals via DSIP; proposals submitted by any other means will be disregarded. Detailed guidance on DSIP proposal submission is found here.

Deadline for Receipt: Complete proposals must be certified and submitted in DSIP no later than the close date of the release listed on the cover page of this solicitation. Proposals cannot be submitted in DSIP after the deadline is reached and will not be accepted or evaluated.

The final proposal submission includes successful completion of all firm level forms, all required proposal volumes, and electronic corporate official certification. Although signatures are not required on the electronic forms at the time of submission, the proposal must be certified electronically by the corporate official for it to be considered submitted. If the proposal is selected for negotiation and possible award, the DoW Component program will contact the proposing SBC for signatures prior to award.

Please plan to submit proposals as early as possible to allow time for troubleshooting any possible issues before the topic close. DSIP Support is unable to assist with submission issues once a deadline has passed and cannot provide submission extensions. DoW is not responsible for missed proposal submission due to system latency.

If the proposal status is "In Progress" or "Ready to Certify" it will NOT be considered submitted, even if all volumes are added prior to the topic close date. The proposing SBC may modify all proposal volumes prior to the topic close date.

6.2 Technical Questions

a. **Direct Contact with Topic Authors**. During the pre-release period, the names of the topic authors, their phone numbers and/or email addresses are published with the topic on the DSIP Topics and Topic Q&A page. During this time, proposing SBCs can contact topic authors via telephone or email to ask technical questions about specific solicitation topics. Questions must be limited to specific information related to understanding a particular topic's requirements. Proposing SBCs may not ask for advice or guidance on solution approach and may not submit additional material to the topic author.
If information provided during an exchange with the topic author is deemed necessary for proposal preparation, that information will be made available to all parties through DSIP Topic Q&A.
After the pre-release period, questions must be asked through DSIP Topic Q&A as described below. No further direct contact is allowed between proposing SBCs and topic authors, unless the topic author is responding to a question submitted during the pre-release period.

b. **DSIP Topic Q&A.** Proposing SBCs may submit written questions through DSIP Topic Q&A here, where all questions and answers are posted on a non-attribution basis for public viewing. DSIP Topic Q&A opens on the pre-release date and closes two weeks prior to the topic close date.

   Proposing SBCs may use the topic search feature on DSIP to locate a topic of interest. Use the form at the bottom of the topic description, enter and submit the question. Answers are generally posted within seven business days of question submission and also e-mailed directly to the inquirer.

   Questions submitted through the DSIP Topic Q&A are limited to technical information focused on understanding a topic's requirements. Any other questions, such as asking for advice or guidance on solution approach, or administrative questions, such as SBIR or STTR program eligibility, technical proposal/cost proposal structure and page count, budget and duration limitations, or proposal due date are not appropriate and will not receive a response; for administrative questions, refer to a topic's Service/Component-specific instructions.

   Once the proposal submission closes, no communication of any kind is allowed either with the topic author or through topic Q&A regarding submitted proposals.

Throughout the topic solicitation period, proposing SBCs should frequently monitor DSIP for updates and amendments to the topics and DSIP Topic Q&A for questions and answers.

7.0 Participating Component Instructions & Research Topics

The following section contains all Component-specific proposal preparation instructions and research topics this BAA.

DoW 2026 SBIR CSO
Release 1

April 13, 2026: Topics Pre-Release
May 06, 2026: Topics Open; DoW begins accepting proposals in DSIP
May 20, 2026: DSIP Topic Q&A closes to new questions at **12:00 p.m. ET**
June 03, 2026: Topics Close; Deadline for receipt of proposals is **12:00 p.m. ET**

Participating Services/Components:

- Department of the Navy (Navy)
- Department of the Air Force (DAF)
- Defense Advanced Research Projects Agency (DARPA)

DEPARTMENT OF THE NAVY (DON)
Small Business Technology Transfer (STTR)
DoW 2026 STTR BAA Release 1
Proposal Submission Instructions

IMPORTANT

- **The following instructions apply to STTR topics only:**
  - **DON26TZ01-NV001 through DON26TZ01-NV019**

- Submitting small business concerns (SBCs) are encouraged to thoroughly review the DoW SBIR/STTR Program Broad Agency Announcement (BAA) and register for the DSIP Listserv to remain apprised of important programmatic changes.
  - The DoW Program BAA is located at: https://www.defensesbirsttr.mil/SBIR-STTR/Opportunities/#announcements. Select the tab for the appropriate BAA cycle.
  - Register for the DSIP Listserv at: https://www.dodsbirsttr.mil/submissions/login.

- The information provided in the DON Proposal Submission Instructions takes precedence over the DoW Instructions posted for this BAA.

- **DON Phase I Technical Volume (Volume 2) page limit is not to exceed 10 pages.**

- Phase I Technical Volume (Volume 2) and Supporting Documents (Volume 5) templates, specific to DON topics, are available at https://www.navysbir.com/links_forms.htm.

- The DON may consider the following FAR and Non-FAR contract strategies when issuing Phase I awards: Firm Fixed Price (FFP), Basic Ordering Agreement (BOA), or Prototype Other Transaction (OT). The DON may consider the following FAR and Non-FAR contracting strategies when issuing Phase II awards: Cost Plus Fixed Fee (CPFF), FFP, BOA, or Prototype OT.

- This BAA is issued under regulations set forth in Federal Acquisition Regulation (FAR) 35.016 and awards will be made under "other competitive procedures". The policies and procedures of FAR Subpart 15.3 shall not apply to this BAA, except as specifically referenced in it. All procedures are at the sole discretion of the Government as set forth in this BAA. Submission of a proposal in response to this BAA constitutes the express acknowledgement to that effect by the proposing SBC.

INTRODUCTION

The DON SBIR/STTR Programs are mission-oriented programs that integrate the needs and requirements of the DON's Fleet through research and development (R&D) topics that have dual-use potential, but primarily address the needs of the DON. More information on the programs can be found on the DON SBIR/STTR website at www.navysbir.com. Additional information on DON's mission can be found on the DON website at www.navy.mil.

For questions regarding this BAA, use the information in Table 1 to determine who to contact for what types of questions.

TABLE 1: POINTS OF CONTACT FOR QUESTIONS REGARDING THIS BAA

Type of Question	When	Contact Information
Program and administrative	Always	Navy SBIR/STTR Program Management Office usn.pentagon.cnr-arlington-va.mbx.navy-sbir-sttr@us.navy.mil or appropriate Program Manager listed in Table 2 (below)
Topic-specific technical questions	BAA Pre-release	Technical Point of Contact (TPOC) listed in each topic on the DoW SBIR/STTR Innovation Portal (DSIP). Refer to the Proposal Submission section of the DoW SBIR/STTR Program BAA for details.
	BAA Open	DoW SBIR/STTR Topic Q&A platform (https://www.dodsbirsttr.mil/submissions) Refer to the Proposal Submission section of the DoW SBIR/STTR Program BAA for details.
Electronic submission to the DoW SBIR/STTR Innovation Portal (DSIP)	Always	DSIP Support via email at dodsbirsupport@reisystems.com
Navy-specific BAA instructions and forms	Always	DON SBIR/STTR Program Management Office usn.pentagon.cnr-arlington-va.mbx.navy-sbir-sttr@us.navy.mil

TABLE 2: DON SYSTEMS COMMANDS (SYSCOM) STTR PROGRAM MANAGERS

Topic Numbers	Point of Contact	SYSCOM	Email
DON26TZ01-NV001 to DON26TZ01-NV004	Ms. Kristi DePriest	Naval Air Systems Command (NAVAIR)	navair-sbir@us.navy.mil
DON26TZ01-NV005 to DON26TZ01-NV006	Mr. Jason Schroepfer	Naval Sea Systems Command (NAVSEA)	NSSC_SBIR.fct@navy.mil
DON26TZ01-NV007 to DON26TZ01-NV019	Mr. Steve Sullivan	Office of Naval Research (ONR)	usn.pentagon.cnr-arlington-va.mbx.onr-sbir-sttr@us.navy.mil

PHASE I SUBMISSION INSTRUCTIONS

The following section details requirements for submitting a compliant Phase I Proposal to the DoW SBIR/STTR Programs.

(NOTE: Proposing SBCs are advised that support contract personnel will be used to carry out administrative functions and may have access to proposals, contract award documents, contract deliverables, and reports. All support contract personnel are bound by appropriate non-disclosure agreements.)

DoW SBIR/STTR Innovation Portal (DSIP). Proposing SBCs are required to submit proposals via the DoW SBIR/STTR Innovation Portal (DSIP); and follow proposal submission instructions in the DoW SBIR/STTR Program BAA on the DSIP at https://www.dodsbirsttr.mil/submissions. Proposals submitted by any other means will be disregarded. Proposing SBCs submitting through DSIP for the first time will be asked to register. It is recommended that SBCs register as soon as possible upon identification of a proposal opportunity to avoid delays in the proposal submission process. Proposals that are not successfully certified electronically in DSIP by the Corporate Official prior to BAA Close will NOT be considered submitted and will not be evaluated by DON. Proposals that are encrypted, password protected, or otherwise locked in any portion of the submission will be REJECTED unless specifically directed within the text of the topic to which you are submitting. Please refer to the DoW SBIR/STTR Program BAA for further information.

Proposal Volumes. The following seven volumes are required.

- **Proposal Cover Sheet (Volume 1).** As specified in DoW SBIR/STTR Program BAA.

- **Technical Proposal (Volume 2)**
  - Technical Proposal (Volume 2) must meet the following requirements or the proposal will be REJECTED:
    - Not to exceed 10 pages, regardless of page content
    - Single column format, single-spaced typed lines
    - Standard 8 ½" x 11" paper
    - Page margins one inch on all sides. A header and footer may be included in the one-inch margin.
    - No font size smaller than 10-point
    - Include, within the 10-page limit of Volume 2, an Option that furthers the effort in preparation for Phase II and will bridge the funding gap between the end of Phase I and the start of Phase II. Tasks for both the Phase I Base and the Phase I Option must be clearly identified. Phase I Options are exercised upon selection for Phase II.
    - Work proposed for the Phase I Base must be exactly six (6) months.
    - Work proposed for the Phase I Option must be exactly six (6) months.

  - Additional information:
    - A Phase I proposal template specific to DON to meet Phase I requirements is available at https://navysbir.com/links_forms.htm.
    - A font size smaller than 10-point is allowable for headers, footers, imbedded tables, figures, images, or graphics that include text. However, proposing SBCs are cautioned that if the text is too small to be legible it will not be evaluated.

- **Cost Volume (Volume 3).**
  - Cost Volume (Volume 3) must meet the following requirements or the proposal will be REJECTED:
    - The Phase I Base amount must not exceed $140,000.
    - Phase I Option amount must not exceed $100,000.
    - Costs for the Base and Option must be separated and clearly identified in Volume 3.

- For Phase I a minimum of 40% of the work is performed by the proposing SBC, and a minimum of 30% of the work is performed by the single research institution. The percentage of work requirement must be met in the Base costs as well as in the Option costs. The percentage of work is measured by both direct and indirect costs. To calculate the minimum percentage of effort for the proposing SBC the sum of all direct and indirect costs attributable to the proposing SBC represent the numerator and the total cost of the proposal (i.e., Total Cost before Profit Rate is applied) is the denominator. The single research institution percentage is calculated by taking the sum of all costs attributable to the single research institution (identified as Total Subcontractor Costs (TSC) 1 in DSIP Cost Volume) as the numerator and the total cost of the proposal (i.e., Total Cost before Profit Rate is applied) as the denominator.
    - Proposing SBC Costs (included in numerator for calculation of the SBC):
        - Total Direct Labor (TDL)
        - Total Direct Material Costs (TDM)
        - Total Direct Supplies Costs (TDS)
        - Total Direct Equipment Costs (TDE)
        - Total Direct Travel Costs (TDT)
        - Total Other Direct Costs (TODC)
        - General & Administrative Cost (G&A)

      NOTE: G&A, if proposed, will only be attributed to the proposing SBC.
    - Research Institution (numerator for Research Institution calculation):
        - Total Subcontractor Costs (TSC) 1
    - Total Cost (i.e., Total Cost before Profit Rate is applied, denominator for either calculation)
- **Cost Sharing: Cost sharing is not accepted on DON Phase I proposals. A value above or below $0.00 entered in the Cost Sharing field will not be considered in the Phase I contract award.**

- Additional information:
    - Provide sufficient detail for subcontractor, material, and travel costs. Subcontractor costs must be detailed to the same level as the prime contractor. Material costs must include a listing of items and cost per item. Travel costs must include the purpose of the trip, number of trips, location, length of trip, and number of personnel.
    - Inclusion of cost estimates for travel to the sponsoring SYSCOM's facility for one day of meetings is recommended for all proposals.
    - The "Additional Cost Information" of Supporting Documents (Volume 5) may be used to provide supporting cost details for Volume 3. When a proposal is selected for award, be prepared to submit further documentation to the SYSCOM Contracting Officer to substantiate costs (e.g., an explanation of cost estimates for equipment, materials, and consultants or subcontractors).

- **Company Commercialization Report (Volume 4)**. DoW collects and uses Volume 4 and DSIP requires Volume 4 for proposal submission. Please refer to the Proposal Preparation Instructions and Requirements section of the DoW SBIR/STTR Program BAA for details to ensure compliance with DSIP Volume 4 requirements.

- **Supporting Documents (Volume 5).** Volume 5 is for the submission of administrative material that DON may or will require to process a proposal, if selected, for contract award.

  All proposing SBCs must review and submit the following items, as applicable:

- **Allocation of Rights.** Required for all SBCs proposing to STTR. In accordance with the SBIR and STTR Policy Directive section 8(b) and DFARS 252.227-7040, the proposing SBC must submit this agreement as an upload as a separate PDF file in Volume 5, Supporting Documents. The STTR Allocation of Rights template must be used and is available on https://navysbir.com/links_forms.htm. Please refer to the Proposal Preparation Instructions and Requirements section of the DoW STTR Program BAA for information on this requirement.

- Proposing SBCs may include the following administrative materials in Supporting Documents (Volume 5); a template is available at https://navysbir.com/links_forms.htm to provide guidance on optional material the proposing SBC may want to include in Volume 5:
    - Additional Cost Information to support the Cost Volume (Volume 3)
    - SBIR/STTR Funding Agreement Certification
    - Data Rights Assertion
    - Disclosure of Information (DFARS 252.204-7000)
    - Prior, Current, or Pending Support of Similar Proposals or Awards
    - Foreign Citizens
- Details of Request for Discretionary Technical and Business Assistance (TABA), if proposed, is to be included under the Additional Cost Information section if using the DON Supporting Documents template.
- Do not include documents or information to substantiate the Technical Volume (Volume 2) (e.g., resumes, test data, technical reports, or publications). Such documents or information will not be considered.
- A font size smaller than 10-point is allowable for documents in Volume 5; however, proposing SBCs are cautioned that the text may be unreadable.

- **Fraud, Waste and Abuse Training Certification (Volume 6)**. DoW requires Volume 6 for submission. Please refer to the Proposal Preparation Instructions and Requirements section of the DoW SBIR/STTR Program BAA for details.

- **Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)**. In accordance with Section 4 of the SBIR and STTR Extension Act of 2022 and the SBA SBIR/STTR Policy Directive, the DoW will review all proposals submitted in response to this BAA to assess security risks presented by SBCs seeking a Federally funded award. SBCs must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission. Please refer to the Proposal Preparation Instructions and Requirements section of the DoW SBIR/STTR Program BAA for details.

PHASE I EVALUATION AND SELECTION

The following section details how the DON SBIR/STTR Programs will evaluate Phase I proposals.

Proposals meeting DSIP submission requirements will be forwarded to the DON SBIR/STTR Programs. Prior to evaluation, all proposals will undergo a compliance review to verify compliance with DoW and DON SBIR/STTR proposal eligibility requirements. Proposals not meeting submission requirements will be REJECTED and not evaluated.

- **Proposal Cover Sheet (Volume 1).** The Proposal Cover Sheet (Volume 1) will undergo a compliance review to verify the proposing SBC has met eligibility requirements and followed the instructions for the Proposal Cover Sheet as specified in the DoW SBIR/STTR Program BAA.

- **Technical Volume (Volume 2).** The DON will evaluate and select Phase I proposals using the evaluation criteria specified in the Method of Selection and Evaluation Criteria section of the DoW SBIR/STTR Program BAA, with technical merit being most important, followed by qualifications of key personnel and commercialization potential of equal importance. The information considered for this decision will come from Volume 2. This is not a FAR Part 15 evaluation and proposals will not be compared to one another. Cost is not an evaluation criterion and will not be considered during the evaluation process; the DON will only do a compliance review of Volume 3. Due to limited funding, the DON reserves the right to limit the number of awards under any topic.

  The Technical Volume (Volume 2) will undergo a compliance review (prior to evaluation) to verify the proposing SBC has met the following requirements or the proposal will be REJECTED:
  - Not to exceed 10 pages, regardless of page content
  - Single column format, single-spaced typed lines
  - Standard 8 ½" x 11" paper
  - Page margins one inch on all sides. A header and footer may be included in the one-inch margin.
  - No font size smaller than 10-point, except as permitted in the instructions above.
  - Include, within the 10-page limit of Volume 2, an Option that furthers the effort in preparation for Phase II and will bridge the funding gap between the end of Phase I and the start of Phase II. Tasks for both the Phase I Base and the Phase I Option must be clearly identified.
  - Work proposed for the Phase I Base must be exactly six (6) months.
  - Work proposed for the Phase I Option must be exactly six (6) months.

- **Cost Volume (Volume 3).** The Cost Volume (Volume 3) will not be considered in the selection process and will only undergo a compliance review to verify the proposing SBC has met the following requirements or the proposal will be REJECTED:
  - Must not exceed values for the Base ($140,000) and Option ($100,000).
  - Must meet minimum percentage of work; 40% of the work is performed by the proposing SBC, and a minimum of 30% of the work is performed by the single research institution. The percentage of work requirement must be met in the Base costs as well as in the Option costs.
  - **Cost Sharing: Cost sharing is not accepted on DON Phase I proposals. A value above or below $0.00 is entered in the Cost Sharing field will not be considered in the Phase I contract award.**

- **Company Commercialization Report (Volume 4).** The CCR (Volume 4) will not be evaluated by the DON nor will it be considered in the award decision. However, all proposing SBCs must refer to the DoW SBIR/STTR Program BAA to ensure compliance with DSIP Volume 4 requirements.

- **Supporting Documents (Volume 5).** Supporting Documents (Volume 5) will not be considered in the selection process and will only undergo a compliance review to ensure the proposing SBC has included items in accordance with the PHASE I SUBMISSION INSTRUCTIONS section above.

- **Fraud, Waste, and Abuse Training Certificate (Volume 6).** Not evaluated.

- **Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)**. Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7) will be assessed as part of the Due Diligence Program to Assess Security Risks. Refer to the DoW SBIR/STTR Program BAA to ensure compliance with Volume 7 requirements.

ADDITIONAL SUBMISSION CONSIDERATIONS

This section details additional items for proposing SBCs to consider during proposal preparation and submission process.

Due Diligence Program to Assess Security Risks. The SBIR and STTR Extension Act of 2022 (Pub. L. 117-183) requires the Department of War, in coordination with the Small Business Administration, to establish and implement a due diligence program to assess security risks presented by SBCs seeking a Federally funded award. Please review the Certifications and Registrations section of the DoW SBIR/STTR Program BAA for details on how DoW will assess security risks presented by SBCs. The Due Diligence Program to Assess Security Risks will be implemented for all Phases.

Discretionary Technical and Business Assistance (TABA). The SBIR and STTR Policy Directive section 9(b) allows the DON to provide TABA (formerly referred to as DTA) to its awardees. The purpose of TABA is to assist awardees in making better technical decisions on SBIR/STTR projects; solving technical problems that arise during SBIR/STTR projects; minimizing technical risks associated with SBIR/STTR projects; and commercializing the SBIR/STTR product or process, including intellectual property protections. Proposing SBCs may request, in their Phase I Cost Volume (Volume 3) and Phase II Cost Volume, to contract these services themselves through one or more TABA providers in an amount not to exceed the values specified below. The Phase I TABA amount is up to $6,500 and is in addition to the award amount. The Phase II TABA amount is up to $25,000 per award, is to be included as part of the award amount, and is limited by the established award values for Phase II by the SYSCOM (i.e., within the $2,000,000 or lower limit specified by the SYSCOM). As with Phase I, the amount proposed for TABA cannot include any profit/fee by the proposing SBC and must be inclusive of all applicable indirect costs. TABA cannot be used in the calculation of general and administrative expenses (G&A) for the STTR proposing SBC. A Phase II project may receive up to an additional $25,000 for TABA as part of one additional (sequential) Phase II award under the project for a total TABA award of up to $50,000 per project. An SBC receiving TABA will be required to submit a report detailing the results and benefits of the service received. This TABA report will be due at the time of submission of the final report.

Request for TABA funding will be reviewed by the DON SBIR/STTR Program Management Office.

If the TABA request does not include the following items the TABA request will be denied.

- TABA provider(s) (firm name)
- TABA provider(s) point of contact, email address, and phone number
- An explanation of why the TABA provider(s) is uniquely qualified to provide the service
- Tasks the TABA provider(s) will perform (to include the purpose and objective of the assistance)
- Total TABA provider(s) cost, number of hours, and labor rates (average/blended rate is acceptable)

TABA must <u>NOT</u>:

- be subject to any indirect costs, profit, or fee by the STTR proposing SBC
- propose a TABA provider that is the STTR proposing SBC
- propose a TABA provider that is an affiliate of the STTR proposing SBC
- propose a TABA provider that is an investor of the STTR proposing SBC
- propose a TABA provider that is a subcontractor or consultant of the requesting SBC otherwise required as part of the paid portion of the research effort (e.g., research partner, consultant, tester, or administrative service provider)

TABA requests must be included in the proposal as follows:

- Phase I:
    - Online DoW Cost Volume (Volume 3) – the value of the TABA request.
    - Supporting Documents (Volume 5) – a detailed request for TABA (as specified above) specifically identified as "TABA" in the section titled Additional Cost Information when using the DON Supporting Documents template.
- Phase II:
    - DON Phase II Cost Volume (provided by the DON SYSCOM) - the value of the TABA request.
    - Supporting Documents (Volume 5) – a detailed request for TABA (as specified above) specifically identified as "TABA" in the section titled Additional Cost Information when using the DON Supporting Documents template.

Proposed values for TABA must NOT exceed:
- Phase I: A total of $6,500
- Phase II: A total of $25,000 per award, not to exceed $50,000 per Phase II project

Disclosure of Information (DFARS 252.204-7000). In order to eliminate the requirements for prior approval of public disclosure of information (in accordance with DFARS 252.204-7000) under this award, the proposing SBC shall identify and describe all fundamental research to be performed under its proposal, including subcontracted work, with sufficient specificity to demonstrate that the work qualifies as fundamental research. Fundamental research means basic and applied research in science and engineering, the results of which ordinarily are published and shared broadly within the scientific community, as distinguished from proprietary research and from industrial development, design, production, and product utilization, the results of which ordinarily are restricted for proprietary or national security reasons (defined by National Security Decision Directive 189). An SBC whose proposed work will include fundamental research and requests to eliminate the requirement for prior approval of public disclosure of information must complete the DON Fundamental Research Disclosure and upload as a separate PDF file to the Supporting Documents (Volume 5) in DSIP as part of their proposal submission. The DON Fundamental Research Disclosure is available on https://navysbir.com/links_forms.htm and includes instructions on how to complete and upload the completed Disclosure. Simply identifying fundamental research in the Disclosure does **NOT** constitute acceptance of the exclusion. All exclusions will be reviewed and, if approved by the Government Contracting Officer, noted in the contract.

Partnering Research Institutions. The Naval Academy, the Naval Postgraduate School, and other military academies are Government organizations but qualify as partnering research institutions. However, DON laboratories DO NOT qualify as research partners. DON laboratories may be proposed only IN ADDITION TO the partnering research institution.

System for Award Management (SAM). It is strongly encouraged that proposing SBCs register in SAM, https://sam.gov, by the Close date of this BAA, or verify their registrations are still active and will not expire within 60 days of BAA Close. Additionally, proposing SBCs should confirm that they are registered to receive contracts (not just grants) and the address in SAM matches the address on the proposal. An SBC selected for an award MUST have an active SAM registration at the time of award or they will be considered ineligible.

Cybersecurity Maturity Model Certification (CMMC) Program. DoW has established the CMMC Program to verify that awardees have implemented required security measures necessary to safeguard Federal Contract Information (FCI) and Controlled Unclassified Information (CUI). CMMC Level requirements are identified within each topic. Proposing SBCs should anticipate that a Projected CMMC Level for Phase II award may be higher than the Projected CMMC Level advertised in the Phase I topic. Proposing SBCs should carefully review and consider the CMMC requirements as compliance may

impact proposed costs and technical approach. Please review the DoW SBIR/STTR Program BAA for additional information on the CMMC Program.

Notice of NIST SP 800-171 Assessment Database Requirement. The purpose of the National Institute of Standards and Technology (NIST) Special Publication (SP) 800-171 is to protect Controlled Unclassified Information (CUI) in Nonfederal Systems and Organizations. As prescribed by DFARS 252.240-7997, in order to be considered for award, an SBC is required to implement NIST SP 800-171 and shall have a current assessment uploaded to the Supplier Performance Risk System (SPRS) which provides storage and retrieval capabilities for this assessment. The platform Procurement Integrated Enterprise Environment (PIEE) will be used for secure login and verification to access SPRS. For brief instructions on NIST SP 800-171 assessment, SPRS, and PIEE, please visit https://www.sprs.csd.disa.mil/nistsp.htm. For in-depth tutorials on these items please visit https://www.sprs.csd.disa.mil/webtrain.htm.

Human Subjects, Animal Testing, and Recombinant DNA. Due to the short timeframe associated with Phase I of the SBIR/STTR process, the DON does **not** recommend the submission of Phase I proposals that require the use of Human Subjects, Animal Testing, or Recombinant DNA. For example, the ability to obtain Institutional Review Board (IRB) approval for proposals that involve human subjects can take 6-12 months, and that lengthy process can be at odds with the Phase I goal for time-to-award. Before the DON makes any award that involves an IRB or similar approval requirement, the proposing SBCs must demonstrate compliance with relevant regulatory approval requirements that pertain to proposals involving human, animal, or recombinant DNA protocols. It will not impact the DON's evaluation, but requiring IRB approval may delay the start time of the Phase I award and if approvals are not obtained within two months of notification of selection, the decision to award may be terminated. If the use of human, animal, and recombinant DNA is included under a Phase I or Phase II proposal, please carefully review the requirements at: https://www.nre.navy.mil/work-with-us/how-to-apply/compliance-and-protections/research-protections. This webpage provides guidance and lists approvals that may be required before contract/work can begin.

Government Furnished Equipment (GFE). Due to the typical lengthy time for approval to obtain GFE, it is recommended that GFE is not proposed as part of the Phase I proposal. If GFE is proposed, and it is determined during the proposal evaluation process to be unavailable, proposed GFE may be considered a weakness in the technical merit of the proposal.

International Traffic in Arms Regulation (ITAR). For topics indicating ITAR restrictions or the potential for classified work, limitations are generally placed on disclosure of information involving topics of a classified nature or those involving export control restrictions, which may curtail or preclude the involvement of universities and certain non-profit institutions beyond the basic research level. Small businesses must structure their proposals to clearly identify the work that will be performed that is of a basic research nature and how it can be segregated from work that falls under the classification and export control restrictions. As a result, information must also be provided on how efforts can be performed in later phases if the university/research institution is the source of critical knowledge, effort, or infrastructure (facilities and equipment).

SELECTION, AWARD, AND POST-AWARD INFORMATION

Notifications. Email notifications for proposal receipt (approximately one week after the Phase I BAA Close) and selection are sent based on the information received on the proposal Cover Sheet (Volume 1). Consequently, the e-mail address on the proposal Cover Sheet must be correct.

Debriefs. Requests for a debrief must be made within 15 calendar days of select/non-select notification via email as specified in the select/non-select notification. Please note debriefs are typically provided in writing via email to the Corporate Official identified in the proposal of the proposing SBC within 60 days of receipt of the request. Requests for oral debriefs may not be accommodated. If contact information for the Corporate Official has changed since proposal submission, a notice of the change on company letterhead signed by the Corporate Official must accompany the debrief request.

Protests. Interested parties have the right to protest in accordance with the procedures in FAR Subpart 33.1.

Pre-award agency protests related to the terms of the BAA must be served to: osd.ncr.ousd-r-e.mbx.SBIR-STTR-Protest@mail.mil. A copy of a pre-award Government Accountability Office (GAO) protest must also be filed with the aforementioned email address within one day of filing with the GAO.

Protests related to a selection or award decision should be filed with the appropriate Contracting Officer for an Agency Level Protest or with the GAO. Contracting Officer contact information for specific DON Topics may be obtained from the DON SYSCOM Program Managers listed in Table 2 above. For protests filed with the GAO, a copy of the protest must be submitted to the appropriate DON SYSCOM Program Manager and the appropriate Contracting Officer within one day of filing with the GAO.

Awards. Due to limited funding, the DON reserves the right to limit the number of awards under any topic. Any notification received from the DON that indicates the proposal has been selected does not ultimately guarantee an award will be made. This notification indicates that the proposal has been selected in accordance with the evaluation criteria and has been sent to the Contracting Officer to conduct compliance review of Volume 3, to confirm eligibility of the proposing SBC, and to take other relevant steps necessary prior to making an award.

Contract Types. A Firm Fixed Price (FFP), Basic Ordering Agreement (BOA), or Prototype Other Transaction (OT) may be used for Phase I awards.

Funding Limitations. In accordance with the SBIR and STTR Policy Directive section 4(b)(5), there is a limit of one sequential Phase II award per SBC per topic. Additionally, to adjust for inflation DON has raised Phase I and Phase II award amounts. The maximum Phase I proposal/award amount including all options (less TABA) is $240,000. The Phase I Base amount must not exceed $140,000 and the Phase I Option amount must not exceed $100,000. The maximum Phase II proposal/award amount including all options (including TABA) is $2,000,000 (unless non-SBIR/STTR funding is being added). Individual SYSCOMs may award amounts, including Base and all Options, of less than $2,000,000 based on available funding. The structure of the Phase II proposal/award, including maximum amounts as well as breakdown between Base and Option amounts will be provided to all Phase I awardees either in their Phase I award or a minimum of 30 days prior to the due date for submission of their Initial Phase II proposal.

Contract Deliverables. Contract deliverables for Phase I are typically a kick-off brief, progress reports, and a final report. Required contract deliverables (as stated in the contract) must be uploaded to https://www.navysbirprogram.com/navydeliverables/.

Payments. The DON makes three payments from the start of the Phase I Base period, and from the start of the Phase I Option period, if exercised. Payment amounts represent a set percentage of the Base or Option value as follows:

Days from Start of Base Award or Option	Payment Amount
15 Days	50% of Total Base or Option
90 Days	35% of Total Base or Option
180 Days	15% of Total Base or Option

Transfer Between SBIR and STTR Programs. Section 4(b)(1)(i) of the SBIR and STTR Policy Directive provides that, at the agency's discretion, projects awarded a Phase I under a BAA for SBIR may transition in Phase II to STTR and vice versa.

PHASE II GUIDELINES

Evaluation and Selection. All Phase I awardees may submit an **Initial** Phase II proposal for evaluation and selection. The evaluation criteria for Phase II is the same as Phase I (as stated in the BAA). The Phase I Final Report and Initial Phase II Proposal will be used to evaluate the SBC's potential to progress to a workable prototype in Phase II and transition the technology to Phase III. Details on the due date, content, and submission requirements of the Initial Phase II Proposal will be provided by the awarding SYSCOM either in the Phase I contract or by subsequent notification.

Awards. The DON will consider the following for Phase II award: Cost Plus Fixed Fee (CPFF), Firm Fixed Price (FFP), Basic Ordering Agreement (BOA), or Prototype Other Transaction (OT). Phase II awards can be structured in a way that allows for increased funding levels based on the project's transition potential. To accelerate the transition of SBIR/STTR-funded technologies to Phase III, especially those that lead to Programs of Record and fielded systems, the Commercialization Readiness Program was authorized and created as part of section 5122 of the National Defense Authorization Act of Fiscal Year 2012. The statute set-aside is 1% of the available SBIR/STTR funding to be used for administrative support to accelerate transition of SBIR/STTR-developed technologies and provide non-financial resources for the SBCs (e.g., the Navy SBIR Transition Program, STP).

Navy SBIR Transition Program (STP). All Phase II awardees have the opportunity to participate in the virtual Navy STP Kickoff during the first or second year of the Phase II contract. While there are no travel costs associated with this virtual event, Phase II awardees should budget time of up to a full day to participate. STP information can be obtained at: https://navystp.com. Phase II awardees will be contacted separately regarding this program.

PHASE III GUIDELINES

A Phase III SBIR/STTR award is any work that derives from, extends, or completes effort(s) performed under prior SBIR/STTR funding agreements, but is funded by sources other than the SBIR/STTR programs. This covers any contract, grant, or agreement issued as a follow-on Phase III award or any contract, grant, or agreement award issued as a result of a competitive process where the awardee was an SBIR/STTR firm that developed the technology as a result of a Phase I or Phase II award. The DON will give Phase III status to any award that falls within the above-mentioned description. Consequently, DON will assign SBIR/STTR Data Rights to any noncommercial technical data and noncommercial computer software delivered in Phase III that were developed under SBIR/STTR Phase I/II effort(s). Government prime contractors and their subcontractors must follow the same guidelines as above and ensure that companies operating on behalf of the DON protect the rights of the SBIR/STTR firm.

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DON26TZ01-NV001 TITLE: Thermally Tolerant Optical Fire Detectors

COMPONENT TECHNOLOGY PRIORITY AREA(S): Sustainment

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Development and demonstration of an optical fire detector capable of an artificial intelligence/machine learning (AI/ML)-enhanced Optical Fire Detector (OFD) capable of operating in temperatures up to 400°F, enabling deployment in high-performance engine nacelles without compromising responsiveness or coverage.

DESCRIPTION: All aircraft engine nacelles require reliable and rapid-fire detection systems to ensure airworthiness and flight safety. OFDs are preferred over other technologies due to their fast response times and comprehensive coverage. However, existing OFDs are typically limited to operating in temperatures below 200°F, rendering them unsuitable for certain high-temperature nacelle environments that exceed this threshold.
Current Limitations of the Alternative (Thermally Robust Temperature-sensing Lines):
• Slower detection response compared to optical methods
• Limited coverage due to sensor placement constraints
• Lack of non-destructive calibration, increasing maintenance complexity and downtime
AI/ML Integration for False Alarm Reduction:
• Real-time signal classification to distinguish between genuine fire signatures and benign stimuli such as sunlight, engine exhaust, or infrared (IR) reflection
• Adaptive filtering based on operational context, reducing nuisance alarms and increasing system confidence
Thermal Design Enhancements:
• Material and packaging innovations to withstand prolonged exposure to 400°F (204°C) environments
• Calibration methodologies resilient to the thermal cycling, vibration, and Electromagnetic Interference common to nacelle-mounted systems
• Integration compatibility for both retrofit of legacy platforms and new platforms.
Expected Benefits:
• Improved fire detection performance in thermally extreme zones
• Increased aircraft survivability and mission readiness
• Enhanced maintainability through non-invasive self-test and diagnostic capabilities
• Improved fleet sustainability

PHASE I: Develop an innovative approach for an OFD suitable for use in a hot (up to 400°F) aircraft environment. Demonstrate the OFD detection capability of a hydrocarbon flame in a 400°F environment, as well as the ability to avoid a false alarm from other light or heat sources.
The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Develop and demonstrate practical implementation of a production-scalable OFD developed in Phase I that can function as a “drop-in” replacement for current fire detectors (mounting and wiring). Evaluate performance of production scalable unit in accordance with MIL-F-23447 and MIL-STD-810H.

PHASE III DUAL USE APPLICATIONS: Produce a functional OFD that passes all environmental and fire detection qualification testing. Prepare the OFD for installation on a selected aircraft. Deliver all needed data to verify functionality for fire detection and any limitations due to false positives. Document any software needed for fire detection used in the OFD.
Thermally tolerant and robust OFD benefits commercial aircraft by enabling mounting in high temperature areas. The increased operational temperature would increase the life of the component. OFDs

can also be used in manufacturing, chemical process, and oil refinement. High temperature environments may be needed or result in these industries. Any improvements to software detection through AI/ML would reduce the number of false positives, which results in a reduction of expensive cleanup, extensive operation downtime, and potential environmental issues.

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3. “MIL-F-23447 (NOTICE-1), MILITARY SPECIFICATION: FIRE WARNING SYSTEMS, AIRCRAFT, RADIATION SENSING TYPE, TEST AND INSTALLATION OF (30-AUG-1999).” http://everyspec.com/MIL-SPECS/MIL-SPECS-MIL-F/MIL-F-23447_NOTICE-1_46780/
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KEYWORDS: Optical Fire Detector; OFD; Fire protection; High temperature; Electronics; Fire; Flame

DON26TZ01-NV002 TITLE: Modeling for Frontal Polymerization Curing Process

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop a multiphysics model or toolset to predict frontal polymerization phenomena and to optimize the resin additives (e,g., catalyst, inhibitor, etc.) for an optimized cure with less distortion or residual stress, while ensuring that the front does not self-extinguish.

DESCRIPTION: Frontal polymerization is the process of curing a resin monomer into a polymer with a localized self-sustaining and moving reaction zone. Frontal polymerization has many benefits over traditional resin cure methods, such as reduced cure time from many hours to seconds or minutes [Refs 1,-2], a significant reduction of the energy required to cure (in some cases over 99.5%) [Ref 3], and reduced cost associated with curing a resin [Ref 3].
Frontal polymerization has many potential applications such as increasing cure percentage for thermoset additive manufacturing processes without requiring a post cure, rapid manufacturing of composite structures, and rapid composite curing for accelerated repairs of composite structures.
Frontal polymerization is a very boundary condition dependent process. Changes in boundary conditions, initial conditions (including temperature and initiation methods), resin formulations, resin or composite thickness, as well as the addition of reinforced fibers or materials can drastically affect characteristics like front velocity, front temperature, and whether a front is sustained or terminated. This can make it challenging to predict and synthesize resin systems that can sustain a frontally polymerized cure with different initiation methods, environmental conditions, composite/resin thicknesses, and reinforcement materials.
Currently, phenomenological multiphysics modeling efforts for frontal polymerization are limited to 1D, 2D, or small 3D models, since they are very computationally demanding due to the highly nonlinear coupling of the governing equations and short timescales required for accurate solution convergence. Furthermore, many models do not predict the mechanical response resulting from the frontal polymerization process (i.e., warpage or residual stress of the polymer caused by the frontal polymerization process). Surrogate modeling can drastically reduce the time to simulate a front but often requires training to create the surrogate model in the form of many finite element analyses or experiments that can be very time consuming. Recently a mechanism-based approach has been created, allowing for prediction of frontal polymerization phenomena without requiring differential scanning calorimetry (DSC) testing to obtain properties for different resin formulations [Ref 4].
This STTR topic calls for development of a model or toolset to predict characteristics of the frontal polymerization process such as front temperature, front velocity, and cure percentage, as well as the resulting effects from the frontal polymerization process such as warpage, residual stress, or post cure mechanical strength. The model should work for multiple initiation methods (i.e., a point initiation of the front, line initiation, and planar initiation for the front (for simulating a point heat source, a line/wire heat source, and a planar heat source). The model should also be scalable, allowing for simulation of different/larger geometries without detrimental increases in computational time. This topic falls under the NAWCAD STTR focus area for in situ material detection and repair solutions.

PHASE I: Develop the framework for a model and determine if the model can predict a frontal cure of a resin with at least one experimentally cured front of a resin as the starting foundation for validation of the model. The model should predict characteristics such as cure percentage, thermal gradients, and front velocity.
The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Use a model or framework to optimize additive concentrations (e,g,, catalyst wt%, inhibitor wt%, etc.) to reduce distortion, front temperature, or residual stress for a frontally cured polymer, while ensuring that the front is sustained leading to a fully cured polymer. Use a model or have a framework for simulating or predicting the curing of a fiber reinforced resin. Optimize the additive concentration percentages to reduce distortion of a frontally cured composite patch or panel, while maximizing the percent fiber volume fraction, and ensuring a sustained front/cure. Experimentally validate the model via frontally cured resin and frontally cured composite samples. Optimize the frontal polymerization process for a successful composite cure for different boundary or initial conditions (i.e. ambient temperatures, thicknesses of composite, initiation methods, etc.).

PHASE III DUAL USE APPLICATIONS: Fully develop the model and transition the model and frontal polymerization technology targeting repair applications for NAVAIR. Create representative panels and/or repair patches that meet specifications required for Navy adoption of the technology.
This technology could benefit the private sector by reducing time and cost associated with curing composite structures or performing composite repairs. This could lead to a reduced cost for cured composite structures. Potential secondary applications include reduced cost automotive composite manufacturing, marine/boat composite manufacturing, and renewable energy composite manufacturing.

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KEYWORDS: Frontal Polymerization; Finite Element Analysis; Composites; Rapid Manufacturing; Rapid Repair; Multiphysics Modeling; Polymerization

DON26TZ01-NV003 TITLE: Advanced Liquid Hydrogen Storage and Employment Methodologies for Unmanned Aerial Systems

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop a cryogenic liquid hydrogen storage and delivery solution that can achieve high hydrogen mass fraction and a low boil off rate. Demonstrate that the cryogenic liquid hydrogen storage system improves endurance, range, and continuous payload power in an unmanned aerial system (UAS).

DESCRIPTION: Hydrogen fuel-cell-powered air systems are becoming more prevalent in aviation [Refs 1-4]. Although compressed gaseous hydrogen has traditionally been employed to power these systems, cryogenic liquid hydrogen has recently started gaining traction [Refs 5-8]. Overall, liquid hydrogen storage provides added benefits such as reduced weight and volume compared to gaseous hydrogen storage, but there are still challenges to air vehicle integration and long-term use due to the extreme low temperature and other properties of liquefied hydrogen [Refs 9-10].
This STTR topic is seeking a liquid hydrogen storage and delivery solution that achieves high performance metrics while also maintaining longevity, safety, and usability for US Navy and US Marine Corps UASs. The performance metrics of interest for the delivered solution include a gravimetric hydrogen storage efficiency = 40% and volumetric hydrogen storage density of > 40 g/L. The integrated solution must also maintain a hydrogen boil-off rate of < 10% mass per day, as well as a gaseous hydrogen delivery system that can meet the flow and temperature requirements of the fuel cell. A liquid hydrogen filling method and procedure shall be defined with an emphasis on minimizing loss of hydrogen. Additionally, the storage vessel shall be reusable and able to achieve > 100 fill cycles. The storage solution and filling procedure must also meet standard safety requirements such as those called out in DOC 06/02/E on the H2 Tools website [Ref 11].
Additionally, consideration should be made for integration into a range of UAS sizes from a Group 2 to Group 5. This shall include considerations for fuel level monitoring and sloshing effects during flights, as well as meeting necessary environmental (basic hot and basic cold), shock, and vibration requirements called out in MIL-STD-810-H [Ref 12]. Ability to demonstrate that the new cryogenic liquid hydrogen delivery system can manage and mitigate thermal loads of UAS mission systems is of particular interest. Finally, cryo-compressed hydrogen solutions will also be considered if it meets the key performance parameters outlined here.

PHASE I: Develop a design for a holistic liquid hydrogen storage and delivery solution that is validated through material analysis and/or modeling and simulation. The design should include a trade study that demonstrates how metrics such as size, weight, and volume affect the overall boil-off rate as well as gravimetric and volumetric storage efficiency. The analysis should assume normal hydrogen and include expected liquid hydrogen fill rate, precooling requirements, and storage vessel cycle life. UAS load profile and fuel cell requirements will be provided by the Government and/or UAS supplier and should be incorporated into the heat leak rate and hydrogen flow requirements to optimize design. Incorporation of a battery to meet peak loads can also be considered in the optimization trade study. Investigation should emphasize UAS integration, be performed over a range of liquid hydrogen storage amounts from 0.5 kg up to 100 kg and consider thermal management opportunities such as cooling UAS systems like payload and avionics. Overall, incorporation of test data to validate analysis is encouraged.
The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Build a prototype liquid hydrogen storage vessel and gaseous hydrogen delivery system and demonstrate a filling procedure. Complete an end-to-end bench test of the overall system to demonstrate

operational performance in a relevant environment. Collaborate with a UAS manufacturer chosen by the Government to integrate the storage and delivery system.

PHASE III DUAL USE APPLICATIONS: Demonstrate the manufacturing maturity of the integrated storage, delivery, and filling system. Develop the operation, maintenance, storage, and safety procedures for using the system in an unmanned aerial vehicle. Demonstrate that the system meets all UAS requirements while also improving its operational capabilities such as endurance, range, and continuous payload power.
This technology is highly relevant to the commercial urban air mobility market. A high-performance hydrogen system would have improved range and flight time compared to current platforms that rely on batteries.

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KEYWORDS: Hydrogen; Liquid Hydrogen; Cryogenics; Storage; Fuel Cell; Unmanned Aerial System; UAS

DON26TZ01-NV004 TITLE: Non-Proximate Chemical Analysis by Field Portable Mass Spectrometry and Robotics

COMPONENT TECHNOLOGY PRIORITY AREA(S): Integrated Sensing and Cyber

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Design, build, and operate a portable mass spectrometer outfitted for proximal detection with a flexible inlet on a land-based robot, to collect real time mass spectra and chemical data at the source.

DESCRIPTION: Mass spectrometers provide unparalleled chemical detection and identification, specifically by use of high-resolution system or tandem mass spectrometry (MS/MS). Field portable mass spectrometers have been commercialized for decades and have led to the ability to detect chemicals of concern at the source. Unlike traditional mass spectrometry where sample preparation is required to get analytes into a form factor amenable for analysis, ambient ionization mass spectrometry has demonstrated proximate detection, with no sample preparation, if the test subject can be placed in front of the mass spectrometer inlet. A plethora of ambient ionization sources for drug, chemical warfare, explosive, and environmental detections of bulk objects in their original form factors with no sample preparation.
Not every test subject however will fit in front of the mass spectrometer's inlet, nor can the ionization source be positioned in such a way to accommodate the test subject. Other ionization sources such as swabs and contact transfer touch sprays have been developed to sample an area and bring the sample to the mass spectrometer. This requires a trained user and sampling error can often be the largest challenge in these samplings.
Non-proximate methods, essentially changing the inlet of the mass spectrometer, have been developed and demonstrated. For example, sampling explosives and chemical warfare agents from ambient surfaces at distances of up to 3 meters from the mass spectrometer has been demonstrated [Ref 9]. However, this method was performed with a rigid inlet and while non-proximate distances were achieved, the flexibility of the sampling was limited, and it would be difficult to adapt to a robotic arm on a rover such as those used by Explosive Ordnance Disposal (EOD).
The objective of this STTR topic is to demonstrate a portable mass spectrometer that has a flexible inlet that could be brought to the test subject and manipulated by a robotic arm platform to collect chemical data at the source. The inlet and the subsequent ionization source combination must be ruggedized and manipulated by a robotic arm to move both to position for sampling. The mass spectrometer must provide remote red light / green light results to the operator from a standoff distance. It must be operational in varying levels of humidity, temperature, and ability to detect a wide array of chemicals.

PHASE I: Design a detailed model and parts inventory for a non-proximate mass spectrometry inlet or ion transport device that is flexible and positionable by a robotic arm. Detail ionization source that will be selected to interface with mass spectrometry inlet and demonstrate that it can be moved by a robotic arm. Design a detailed model and parts inventory for a field portable mass spectrometer (MS) that will couple to the flexible non-proximate inlet, taking into consideration Size, Weight and Power (SWaP) requirements and how it will interface and operate on a robotic arm on a rover. Adaptation of commercial field portable mass spectrometer to interface with a flexible inlet is also appropriate.
Provide a parts list with material type and weight for each component (required). The battery for the MS should last at least 3 hours before needing to be recharged. The casing of the flexible inlet, ionization source, and MS should be ruggedized, water resistant, and ensure that the instrument is not damaged upon movement. The MS must be operational during and after movement. Additionally, the MS must be operational at a temperature range of -25 °F to 120 °F (Threshold (T)), -35°F to 135°F (Objective (O)). Consult MIL-STD-810 Test Methods: 501.6 High Temperature, 502.6 Low Temperature, 507.6 Humidity, and 510.6 Sand and Dust when designing and selecting the material and layout of the system.

The overall layout of these components for the final system should be sketched and the overall size should be as compact as possible. The communication of the MS to the control system at the operator should be detailed. The MS should have an operating mass range at least from m/z 50 to m/z 500 to detect a wide range of chemical threats. Detection limits for chemical threats must be less than 100 ppb (T), in the ppt range (O).
The Phase I Option, if exercised, should be used to further develop and improve the design and, possibly, to demonstrate key components.
The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Construct the non-proximate inlet and interface with field portable MS with the aforementioned operational specifications from Phase . Interface the non-proximate inlet and ionization source with a commercial off-the-shelf (COTS) robotic arm that could be mounted to a rover for sampling remote and hard to reach locations. The MS should be able to be operated remotely and have a database capability of providing red light / green light results to the operator based upon both mass spectra and tandem mass spectra. The system must provide remote results and be fully operational while the robotic arm is in motion. Demonstrate the detection of three target chemical threat simulants from surfaces that cannot be moved to a traditional stationary inlet. Guidance will be provided to threat simulants at the start of Phase II, but they will be within the required mass range. Deliver one operational unit to the Navy.

PHASE III DUAL USE APPLICATIONS: Final testing and demonstration / evaluation would be conducted in theater and on forward operating bases. Scenarios could be standoff detection of unknown objects and suspected hazardous threats, gathering chemical information from object difficult to sample that may have moved through a hazardous environment (under side wing of an aircraft), or when human / physical interaction with a sample is not ideal.
Ample forensic applications for this technology exist in drug, chemical warfare agent, and explosives detection. There are also potential dual use applications in the pharmaceutical and industrial processes such as inspecting the inside of a chemical reactor / checking the cleanliness after a process or within quality control areas. There are also environmental dual use applications for traversing hard to sample locations and instead being able to send the rover and sampling apparatus into the space.

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KEYWORDS: Mass spectrometry; Non-proximate detection; Rover; Field portable; Automation; Chemical analysis

DON26TZ01-NV005 TITLE: Automatic Cable Tester

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Contested Logistics Technologies (LOG)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Integrated Sensing and Cyber

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop a low-cost and user-friendly automatic cable tester capable of universally testing continuity, resistance, and isolation of both Copper, Radio Frequency (RF) cables, and the impedance of Fiber Optic cables via Optical Time Domain Reflectometry (OTDR), while rapidly generating easily read quality assurance reports.

DESCRIPTION: While a DDG 51 Class Ship is undergoing modernization, significant time is spent testing continuity, resistance, and isolation on large numbers of Copper, RF cables, and impedance of Fiber Optic cables. For example, within the combat systems alone, there are over 2,900 interfaces that require such testing. With the numerous amounts of cables needed to be tested on board a ship, manual testing of each cable can take several hours compared to several minutes or less with utilization of an automatic tester.
While approximately 175 adapters and kits are available for automatic cable testing, there are no universal devices capable of testing Copper, RF, or Fiber cables. The automatic tester must be equipped with low-cost software and adapter kits for both the "local" and the "remote" sides of the varieties of copper cables and connectors under test. The Navy needs a cable analyzer that can perform a variety of multi-pin connections along with a Fiber Optical Loss Test Set (OLTS)/ OTDR tester capable of utilizing the existing and approved testing standards or featuring an innovative unconventional low-cost means of examining each cable and loopback.
The Navy seeks an automatic cable tester capable of testing the connectivity, continuity, and isolation of both Copper, RF and Fiber Optic cables. The development of an inexpensive, portable, universal cable tester system, able to portray data in real time is desired. The tester must be able to easily connect to the variety of connectors on the cables previously mentioned and reduce both the number of test-connectors necessary to operate as well as the overall cost of the prototype/production system. The software used by the tester should be able to be either Microsoft-based software or one easily convertible into an Excel format for recording test data. The solution should be easily transported by one sailor to allow for convenient movement through tight hatchways and spaces found within a DDG 51 Class Ship. The prototype developer should also document specifics of a life cycle management program both for the tester and all components. The developed solution should shorten the length of time required to test all connections on a ship undergoing modernization.

PHASE I: Develop a concept for an automatic cable tester and demonstrate that the concept meets all parameters in the Description. Demonstrate the feasibility of the concept in meeting Navy needs by component evaluation and analytical modeling. The Phase I Option, if exercised, should include the initial layout and capabilities to build the prototype in Phase II. Prepare a Phase II plan.

PHASE II: Develop and deliver for evaluation and testing a prototype accompanied by a complete connector kit for Navy testing. Design all system components to meet all standard Navy environmental testing. The prototype will be evaluated and tested to determine capability in meeting the performance goals defined in the Description. Develop device designs that can be efficiently fabricated/assembled and detailed plans for fabrication intended to reduce the number and/or cost of test-connectors/ devices across the spectrum of cables mentioned in the Description. Identify software concepts that can be used for testing at minimum continuity and insulation resistance testing of copper and fiber optic verification testing for performance, insertion and return loss for both single and multi-mode fiber cabling. Product

performance will be demonstrated through prototype evaluation, modeling, and demonstration over the required range of parameters. An extended laboratory test will be used to refine the prototypes into a design that will meet Navy requirements. Prepare a Phase III manufacturing and development plan to transition the system to Navy use.

PHASE III DUAL USE APPLICATIONS: Support the Navy in transitioning the Automatic Cable Tester to Navy use. Develop installation, maintenance, and operations manuals for the Automatic Cable Tester to support transition to the fleet.
The finished product has potential commercial applications for commercial communication maintenance personnel.

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KEYWORDS: Radio Frequency; Automatic Cable Testing; Fiber Optic Cable; Continuity Testing; Test Connectors; Resistance Testing

DON26TZ01-NV006 TITLE: Waste Heat Recovery

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Contested Logistics Technologies (LOG)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Sustainment

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop a low-cost waste heat recovery system capable of converting the heat energy within DDG 51 main engine exhaust into electrical power.

DESCRIPTION: LM 2500 gas turbine engines' maximum thermal efficiency is approximately 38%. This means at least 62% of the energy in every drop of fuel consumed by the process of propelling a DDG 51 Class ship is unused and available for harvesting as it is being expelled in the form of heat via engine exhaust. Significant energy that is currently "wasted" could be recovered from exhaust to save on fuel costs and increase the range of surface combatants. To effectively utilize all resources, the Navy seeks to capture this waste heat as usable energy source.
In the past, the Navy recovered this heat energy via the Rankin cycle to heat galley appliances with steam. However, there has never been a durable, effective, weight- and space-economizing system that utilizes waste heat to produce electrical power on a Navy ship. Within the context of enhancing the environmental record of the Navy, this initiative would productively tap an "alternative" energy source to reduce fuel consumption and subsequent emissions.
The Navy seeks a solution that provides an innovative system for waste heat collection and utilization that maximizes capture and use of thermal energy while minimizing impacts on any other ship system or prominent feature (especially the main engines). Also important to the Navy is an emphasis on moderating use of or impacts to the ship's profile and/or Radar Cross Section, available onboard space, and any serious impacts to weight and stability characteristics. Keeping these difficult limitations in mind, it is the Navy's goal to produce the greatest possible amount of electrical power from harvesting the abundant thermal energy from every ship's main engine exhaust. While the DDG 51 Class Gas Turbine Generators (GTGs) also have similar thermal efficiencies and the scope of this STTR topic may become inclusive of GTGs in the future, the immediate focus of the topic is on the waste heat from the LM 2500 main engines.
The proposer should quantify the level of stress the material can incur while in an operational environment, and provide a preliminary concept design and validation plan and an in-depth examination in scalability and the potential for miniaturizing any technologies highlighted within the feasibility study, as these proposed technologies will need to create a system able to fit and effectively/safely operate within the DDG 51 Class footprint(s) and meet weight and stability requirements.

PHASE I: Develop a concept for waste heat recovery of the LM 2500 engine that accomplishes the requirements listed in the Description. Demonstrate the feasibility of the concept with a development plan and proposed test plan that will include testing to failure and compliance with environmental standards. Accompany the feasibility study with a recommendation of how the technology could be best incorporated into DDG 51 Class ships. The Phase I Option, if exercised, will include the initial design specifications and capabilities description to build a prototype solution in Phase II. Prepare a Phase II plan.

PHASE II: Develop and deliver a prototype and/or a comparable simulation able to demonstrate the conformance with power-generation industry standards and according to actual operating specifications, conditions, and DDG 51 Class footprints. A high-fidelity industry-standard computerized predictive model/simulation of the system displaying all significant data points of the system while in operation is needed and/or a high-fidelity (to no less than 1/32 scale) working prototype of the system. The simulation

must validate the functionality/effectiveness of the system. A comprehensive installation plan, itemizing any required materials and their sources, recommending the safest and most cost and time-effective installation techniques will also accompany all Phase II documentation as a deliverable. Conduct a thorough examination and estimate of potential electrical output across the range of ship speeds and engine conditions to include idle.

PHASE III DUAL USE APPLICATIONS: Support the Navy in transitioning the technology to Navy use. The product will be validated, tested, qualified, and certified for Navy use.
There are any number of industries that utilize gas turbines, and this technology will likely be applicable to many of those industries where abundant spare electrical power can and would be fully utilized.

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KEYWORDS: Energy Recovery; Fuel Efficiency; Heat Recovery; Gas Turbine Generators; Electricity; LM 2500 Turbine Engine

DON26TZ01-NV007 TITLE: Novel Computing for Streaming Radio Frequency in Low Size, Weight and Power Environments

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Computing and Software;Integrated Sensing and Cyber;Trusted AI and Autonomy

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Create a small and computationally powerful Radio Frequency (RF) sensor that meets or exceeds the requirements of extremely limited low size, weight, and power (LOW SWaP) platforms. This computational engine should weigh < = 1.5 lbs., require < = 100 watts of power, and be able to ingest and process > = 2 GHz of instantaneous bandwidth of RF spectrum continuously.

DESCRIPTION: Today's electronic processing technology is not keeping pace with the DOW's computational demands. Growing networks of new higher resolution and higher fidelity sensors yield vast quantities of data, and deep neural networks are being deployed to reduce these data streams to actionable information for the warfighter. Concurrently, the constraints imposed by network capacity, latency, and data security are driving this sensor processing to the tactical and network edge where the data is collected. This transition is compounding processing throughput shortfalls because of edge platform challenges.

In today's battle space, the concept of putting payloads on smaller and smaller unmanned platforms is a huge need. This STTR topic focuses on extremely low SWaP RF sensors that can be less than 5 lbs. The critical piece is the computational engine that will turn streaming Intermediate Frequency (IF) (with an Instantaneous Band Width of > = to 2 GHz) and perform the detection and data product formation for RF analysis and eventual classification and localization of emissions in the extremely broad RF spectrum. For these payloads to go on Group 2 or smaller Unmanned l Aerial platforms, the entire sensor package must weigh less than 5 lbs. and the processing engine is allocated less than 1.5 lbs. of this total weight and consume less than 150 Watts. The critical component is a computational engine that is small enough yet computationally powerful enough to make these payloads a reality.

The Navy seeks a single chip and infrastructure that is less than 1 lb. and less than 3 mm on either side. This will include the input/output (I/O) to the sensor data sources, the memory, the computational devices, and the I/O to the operator or decision engine (preferably the decision engine would be part of this device).

These computational engines must meet an extremely high performance metric while being extremely lightweight and energy efficient. The products resulting from this STTR topic will be utilized in Group 1 and 2 UASs as well as buoys that are less than 3 inches in diameter. These devices must be able to be zeroized and support data at rest encryption standards.

Work produced in Phase II is expected to become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and ONR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Develop and provide a detailed schedule out through Phase II Option periods and a detailed technical description as to how they will achieve success.
Participate in a kickoff meeting during which details on how to get to the final briefing and its specifics will be presented.
If the Phase I Option is exercised, showcase software modules and fundamental breadboard designs.
The final briefing showing specifically how they will meet the following requirements:
- Less than or equal to 3 mm by 3 mm size
- Meeting class B shipboard installation Environmental Qualification Testing (EQT)
- Less than 100 watts energy consumption
- Demonstrate the ability to move 2 GHz of IBW continuously through the pipeline and show what higher order data products will come out the backend
- Data at rest security requirements
- Non-Volatility certification requirements
- Networking architecture demonstrating ability to configure to multiple types of networks.
- Interface descriptions on how external systems will interface and operate the prototype device remotely and locally
- Examples of the user interfaces and the schema for formatting, recording, librarying, and playback
- Cost and schedule program management plan

PHASE II: Participate in a kickoff meeting and present a detailed development plan including costing (recurring and non-recurring separated) development, security and testing plans.
These plans will include:
- detailed technical plans
- detailed security plans
- detailed EQT plans
- detailed lab testing plans (both at developers facility and at government labs) utilizing different types of RF sensors.
- detailed ship installation and at sea testing (Phase II Option will be integration and testing at sea)

After the kickoff meeting and with government concurrence of the plan, focus on developing the solution meeting all the security, environmental qualifications, and performance requirements agreed to.
The system to be developed shall meet the following requirements (as stated in the Phase I section):
- Less than or equal to 3 mm by 3 mm size
- Meeting class B shipboard installation Environmental Qualification Testing (EQT)
- Less than 100 watts energy consumption
- Demonstrate the ability to move 2 GHz of IBW continuously through the pipeline and show what higher order data products will come out the backend
- Data at rest security requirements
- Non-Volatility certification requirements
- Networking architecture demonstrating ability to configure to multiple types of networks.

The prototype system:
- will show compliance with shipboard installation environmental qual requirements
- shall show the ability to perform data at rest encryption and the ability to meet volatility requirements for system posture changes.
- shall show the ability to consume data from a defined sensor and parse and tag this data.
- shall demonstrate the ability to record and playback from both local and remote users
Perform a minimum two lab demonstrations at the developer's facility and one integration and demonstration at a government lab. The government lab will provide testing and validation of the

capabilities and provide immediate feedback to the developer for further refinement of the prototype.
Work with the government lab to develop a shipboard installation and testing plan.
The Phase II Option, if exercised, will focus on readying the prototype system to be installed and tested at sea during a government-defined testing event.
It is probable that the work under this effort will be classified under Phase II (see the Description section for details).

PHASE III DUAL USE APPLICATIONS: Clearly and in detail describe how this capability will transition to a Navy program of record (POR). This plan will also describe how it will be used in the POR and the initial concept of what data products will be recorded and how.
The government feels that any commercial industries needing real time video / data compression, or greater than 4K video streaming capabilities will benefit from this technology.

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KEYWORDS: RF Spectrum sensing, RF Payload processing, optical computing, Photonic Convolution, Adaptive processing, Signal detection

DON26TZ01-NV008 TITLE: Automated Software Test Generation and Augmentation for Improved Debloating

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Quantum and Battlefield Information Dominance (Q-BID)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Computing and Software;Integrated Network Systems-of-Systems;Integrated Sensing and Cyber

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop an automated solution for developing, enhancing, expanding, and augmenting software tests to more safely broaden the employment of proactive cyber techniques such as debloating and post-construction software refactoring. Technology is needed to refine a suite of tests to a level such that it may serve as a practical expression of a software transformation objective to drive other tools as well as validate their output. Technology should leverage multi-modal methods such as ingesting code and documentation as well as be compatible with DevOps processes.

DESCRIPTION: Modern software development practices such as industrialized code reuse and artificial intelligence (AI) assistance enable developers to produce increasingly complex and capable software more quickly and cheaply than ever before. The tools to ensure that all this software is well-tested and that all of the included code is well-tailored to the deployment scenario, however, have lagged by comparison.

Modern applications often include hundreds to thousands of libraries and other dependencies, with often only a small portion of the code in each being ever needed by users in each deployment scenario. The excess code that remains often tends to be less used in general, less well-scrutinized, and full of obscure features that will often be found (sometimes only years later) to contain vulnerabilities. To address this problem, numerous tools have been developed to identify bloat and then modify the software by removing unneeded code [Ref 1]. Configurations, usage logs, and tests that are fed as inputs to code transformation tools to tell them what to cut are referred to as the debloat specifications [Refs 1, 2].

Because the economics of code reuse will continue to drive library and package developers to maximize generality, debloating must happen through a separate process that begins after those components are built into a specific application. The fact that another process will be modifying code separate from the original one that designed, implemented, and tested those components adds risk—it is not uncommon to see flawed or incomplete transformations. Evaluation results in [Ref 2] showed that 37% of the debloated binaries they created failed to correctly execute the functionality they were intending to retain.

Many factors can contribute to a transformation yielding a broken application, but one of the biggest is a low-quality debloat specification. Developer-authored tests are often limited and the users of debloating tools rarely can specify in exact detail all the features they actually need for a given deployment scenario. These incomplete specifications can lead tools to be overly aggressive in things like security checks and exception handlers that are critical to application safety and robustness [Ref 3].

To better address the problem of low-quality and incomplete debloat specifications, new technology is needed to more fully incorporate and automate the capturing of desired software behaviors for input to a debloat tool. The technology should be able to take advantage of code analysis as well as analysis of related artifacts such as documentation, build configs, existing tests, and even user input, as long as it can be made practical and easy for a user to answer. Various works have explored methods and techniques for capturing exception handers [Ref 3], balancing reduction with a targeted amount of generality [Ref 4], and leveraging AI to incorporate new tests [Refs 5, 6, 7]. All may inform strategies for automated test generation and augmentation that can lead to higher quality debloat specifications.

PHASE I: Define and develop a concept for automated multi-modal processing of code and other DevOps repository artifacts such as user guides, etc. to generate and augment a suite of tests that can serve as the inputs to proactive cyber security tools, namely debloating. Work toward a design that can develop tests based on unstructured documents and interact with a user to refine the tests. The Phase I Option, if exercised, would develop the initial test augmentation capability to create the full prototype in Phase II.

PHASE II: Develop a prototype containerized test augmentation capability to validate the concept defined in Phase I. Demonstrate the automated multi-modal processing of code, DevOps repository artifacts, and, if necessary, user interview inputs, into developing, enhancing, expanding, and augmenting software tests by the prototype. Ensure that the prototype is deployable in a software factory environment and able to develop many tests to sufficiently, reliably, and robustly enable the debloat of (1) an application using only its existing limited test suite, (2) unstructured program documents like user guides, and (3) real-time user input at the non-expert level by the end of Phase II.

PHASE III DUAL USE APPLICATIONS: Integrate the Phase II developed test augmentation capability with Program of Record systems and their applications. Field containerized solutions that integrate with existing build pipelines.
Potential commercial applications include automated software testing and fuzzing harness generation, a growing need due to the proliferation of AI-generated code.

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KEYWORDS: Cyber; Software Testing; Automation; artificial intelligence; AI; machine learning; ML; large language model; LLM; Debloating; Feature Specification

DON26TZ01-NV009 TITLE: Robust Universal Adaptive Denoising Technology

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Computing and Software;Trusted AI and Autonomy

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop robust denoising approaches that are highly adaptive and effective.

DESCRIPTION: Signal denoising has shown to be highly effective in improving performance of signal processing radio frequency and acoustic sensing systems. The main hindering signal in these applications is noise as it degrades the ability to sense low level signals masked by ambient noise sources which may be external to the sensor or generated by the sensor itself. The main goal of this SBIR topic is to develop a denoising technology that suppresses noise while preserving the underlying signal features. Traditionally, denoising methods have struggled to maintain performance when presented with highly non-stationary or complex noise patterns. The traditional approaches typically require extensive and time-consuming tuning to achieve desired performance. On the other hand many of learning-based methods have demonstrated excellent denoising performance but suffer from limited robustness. Therefore, the method's performance will drop if the training conditions do not adequately reflect the characteristics of the operational environment. The Navy seeks improvements in denoising performance greater than 10 dB.
For such a system installed on an aircraft, it will experience both wind- and aircraft-generated noise. That noise has components that are narrow band (< 10 Hz wide) and broadband (10s to 100s of Hz wide). The spectrum of interest for sensing extends from approximately 10 Hz to 1000 Hz. When compared with more traditional active noise cancellation techniques, the denoising approach should be capable of providing 6 dB of additional cancellation and show potential to deliver 10 dB or more cancellation.

PHASE I: Develop concepts for a robust denoising approach requiring minimal training and are effective in highly non-stationary or complex noise environments. Modeling and simulation to include laboratory measurements to assess the efficacy of the approach based on an in-air or ground-based mobile acoustic sensing system is desired. Consider how the approach may be extended to a radio frequency (RF) system operating in the 1-10 GHz range. The Phase I effort will include prototype plans to be developed under Phase II.

PHASE II: Develop and demonstrate an end-to-end denoising approach on an acoustic frequency sensing system in a laboratory environment and ultimately in a representative operational environment. The prototype assessment should include narrow and broadband noise removal performance while preserving desired signal characteristics, robustness in the presence of non-stationary noise environments. At least 10 dB of noise cancellation is needed with 15 dB desired over traditional active noise cancellation techniques. Consideration for the ease of integration and fielding should be made. Demonstrating the efficacy of the denoising approach on a variety of host platforms is desired. Further refine the extension of the denoising technique to use by RF sensing systems.

PHASE III DUAL USE APPLICATIONS: Support the transition to Navy use.
A universal highly adaptive denoising approach could find applications across remote sensing, communication systems, biomedical signal processing, audio restoration, and image enhancement.

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KEYWORDS: Denoising; Signal Processing; Deep Learning; Non-Stationary; Acoustic; Radio Frequency

DON26TZ01-NV010 TITLE: Next Generation Tropical Cyclone Analysis, Forecasting, and Dissemination Tactical Decision Aid Software

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Computing and Software;Sustainment;Trusted AI and Autonomy

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Demonstrate an improved automated tropical cyclone forecasting, analysis, and dissemination tactical decision aid capability that uses a modern containerized software backend/frontend and is able to easily integrate legacy and novel component algorithms, models, databases, and Application Programming Interfaces (APIs).

DESCRIPTION: Domestic operational tropical cyclone forecasting at the Joint Typhoon Warning Center (DOW), Fleet Weather Centers (DOW), and National Hurricane Center (NOAA) have relied on the Automated Tropical Cyclone Forecast System (ATCF®) software suite for end-to-end tropical cyclone analysis, forecasting, and product dissemination for over three decades. This one-stop-shop for all data, modeling, post-processing, and user interaction for tropical cyclone information has endured due to its robust assured infrastructure, reliability, speed for executing actions, and long continuity even as forecasters and information have evolved. However, as compute environments and programming languages have changed, it has become more difficult to maintain and upgrade legacy software to take advantage of new capabilities.

This STTR topic seeks the development of a prototype software suite that can learn lessons from the success of ATCF®, but is architected in a modern software ecosystem to mitigate current workflow disadvantages. Fundamentally, the goal is a modular and containerized software application that can variously interact with legacy, current, and future software suites such as components of ATCF®, the Naval Integrated Tactical Environmental System Next Generation (NITES-Next) program, the NOAA Advanced Weather Interactive Processing System (AWIPS), and other back-end and front-end APIs. The software architecture must be designed from the outset to comply with DOW DevSecOps principles and prepare the system for the Risk Management Framework (RMF) process. Desired software requirements include design in a modern broadly supported and maintained open programming language that can run online or offline on premises or in a cloud compute environment; hardening against connectivity and bandwidth issues; separation of functionality between logic, database, analysis algorithms, forecast generation , user interface, and dissemination layers; modular component development where different parts can interoperate with other software; and the ability to quickly address software updates and functionality and revert on the client side.

A dual-pronged approach to improving workflow for the tropical cyclone forecast process is envisioned, with parallel development tracks for software architecture creation and decision support aid integration. While the focus of tropical cyclone tools in the 1980s through 2000s was on track and intensity prediction, the forecasting mission has increasingly expanded. This includes the ability to track before storms have formed; 3D storm structure and rainfall evolution; ocean and wave field information; storm surge and hazards; probabilistic uncertainty; and emerging machine learning tools. The software should be capable of generating and disseminating an automated, objectively optimal analysis and forecast product from available data — that is, without significant manual human effort. It is not expected that all these efforts are achievable within the scope of this STTR effort; however, the priority development schedule must be justified, and the software solution must be able to accommodate all of these components within a future strategic plan.

Back-end capability should include: (1) a modern development software framework that can easily include or remove proprietary and open source algorithms as desired and is built and deployed as a

containerized architecture; (2) a robust state management (e.g., database) for storm information and aids with backwards compatibility and export for current ATCF® "deck file" formatting; (3) concurrency for data fetching and processing to reduce data latency for time-critical forecast workflows; and (4) defined APIs to provide data access to clients, such as front-end graphical user interfaces (GUI) or downstream machine-to-machine programs. Further, the back-end capability should support backup capability, likely through a distributed system of servers.
Front-end capability should include: (1) thin client(s) for use on desktop workstations and possibly within web browsers, facilitating both on-site and remote operation with both client-side and server-side rendering tested for DOW network responsiveness; (2) flexible means of calling external scripts/functions/APIs with configurable input data, allowing forecasters to trigger different production pipelines and workflows that could be defined externally; (3) editable runtime configuration facilitating separate profiles for different operational systems or users; (4) means to integrate with internal and/or external AI systems or agents to facilitate future workflows; and (5) GUI and storm management capabilities consistent with ATCF®, with additional emphasis on performant map navigation and rendering, multi-product and format overlays (from gridded and sparse data, such as from kml, kmz, ShapeFiles, GeoTiff, HDF, netCDF, GRIB2, Zarr, GeoJSON, etc.), and dynamic filtering and alerts for data as they populate in real time. It is imperative that software developed be done so with an emphasis that support does not require skills beyond those currently required and normally used by support staff at forecast centers.

PHASE I: Design and develop an architecture concept for an improved tropical cyclone software capability, identifying the most challenging technical components. Present a demonstration for viable solutions to the technological problems. Capabilities should be contextualized and contrasted with current ATCF functionality and forecasting requirements. Integration of emerging technologies should be explicitly described, and include machine learning prediction methods, higher resolution or local forecasting, and new machine learning tools to develop and publish a product. Required Phase I deliverables include the reporting documents on progress and outcomes, the final proposed architectural solution with justifications and risk/mitigation analysis, and a description/demonstration of the unique methods to address the development challenges.

PHASE II: Develop, demonstrate, and validate a prototype software suite. Effort should focus on proving back-end and front-end capabilities from the above topic description that address greatest needs from the concept developed in the Phase I. It is expected that regular engagement with Naval users and frequent software demonstrations and side-by-side comparison with other operational software suites will be performed throughout software maturation. The outcome by the end of the Phase II should be an end-to-end prototype (not expected to be feature complete) that addresses substantial aspects of the ATCF mission requirements and is able to be run robustly for real-time operational usage.

PHASE III DUAL USE APPLICATIONS: Focus on continuing development of software capability, while ramping up integration with other decision support tools in the operational environment. Performance equivalence with, or superiority to, ATCF is expected and should be demonstrated on different compute platforms, cloud systems, and classified systems. Performance metrics include: graphical rendering latency (both locally and over remote display protocols) and the time required for a user to complete a standard set of storm analysis and forecast tasks within the software. Acceptance of new data, analysis, and model capabilities from internal and external partners are expected. Awardee should be prepared to deploy as a stand-alone system with thin client interface, as a component backend in a thick client front end, and/or as a module in a multi-software system.
Beyond DOW, interaction and potential transition with NOAA is expected. There is potential for supporting multiple commercializations, such as supporting academic and basic open-source software community, software sales to commercial forecasting and risk companies, and further government capability development.

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KEYWORDS: Tropical cyclone; typhoon; Automated Tropical Cyclone Forecasting; ATCF; user interface; algorithms; modular; refactor; cloud; database; container; Application Programming Interface; API; graphical user interface; GUI; software

DON26TZ01-NV011 TITLE: Field Deployable Welding Technologies for In-Situ Repair of Thermoplastic Composites Components on Naval Aviation Platforms

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Sustainment

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop and transition a portable induction welding system capable of in-situ repair of thermoplastic composite components on naval aviation platforms enabling rapid, field-ready maintenance capabilities for next-generation naval aircraft.

DESCRIPTION: Modern aviation platforms are increasingly using high-performance thermoplastic composites such as PAEK, PEEK, PPS, or PEI reinforced with Carbon Fiber for structural and semi-structural components. Their attractiveness is due to their superior damage tolerance, impact, and ability to be reworked for repair. Unlike traditional thermoset composites, which can only be repaired by bonded patches or bolted panels, thermoplastic composites can also be repaired by welding, which restores strength without the need to remove additional material. However, currently available welding systems have a large footprint and are available mostly with OEM and only suited for deployment at the Depots. Thus, without field deployable technofixes, repairs will result in long downtime for repair and likely higher scrap rates.
This STTR topic seeks to leverage the research expertise of academic or government labs in thermoplastic processing and electromagnetic heating to partner with a small business in designing a rugged, portable induction welding system that can be deployed shipboard and/or in Aircraft Intermediate Maintenance Detachments.
The proposed system should: (1) be capable of welding aerospace-grade thermoplastics (at temperatures up to 400°C); (2) be lightweight and field operable, including on aircraft carriers; (3) be electromechanically ruggedized and safe to operate near avionics and flight-critical systems; (4) have a closed-loop thermal control for temperature; (5) be able to repair skins, fairings, panels, and access doors; and (6) have a weld strength of at least 70% of the parent material.

PHASE I: Identify key thermoplastic components and repair scenarios relevant to Navy and Marine Corp Aircraft; develop preliminary induction welding system architecture; build and integrate a TRL 3-4 configuration in collaboration with the research institution; and conduct an initial weld strength test. The weld strength should be at least 70% of parent material. Additionally, the awardee should assess the electromagnetic compatibility and ergonomics of the system.
While this is not an allowable developments program, the awardee may propose a limited amount of testing for calibration and validation of the prototype. The awardee should note that the Phase II down select is based on the performance and final deliverables of the Phase I Base period – so plan accordingly. The Phase I deliverables should include at a minimum: (a) a Weld Feasibility report which should include results of weld trials meeting the key parameters stated above. It should also include assessment of the thermal profile, fusion quality, and repeatability; and (b) a preliminary system concept and architecture that will meet the topic's goals.

PHASE II: Build and validate a ruggedized TRL 6-7 prototype system. Integrate smart controls and thermal feedback into the prototype. Adapt tools and fixtures for real world geometry and repair location. Validate the tool on a sub-element level aircraft part. Demonstrate mechanical integrity of repairs by mechanical testing and nondestructive inspection (NDI). Demonstrate that it meets full EMC qualification per appropriate MIL STD [Ref 3].
Ensure that the prototype meets the Phase II goal (stated in the Description).
Provide a test report summarizing all tests done in Phase II, and an electronic user instruction manual for the prototype and a maintainability and support plan for the delivered prototype.

PHASE III DUAL USE APPLICATIONS: Support the transition to Navy use. Expected transition within government is to fleet resource centers servicing MQ25 and next gen platforms including Maritime Strike.
Additionally, the awardee will be encouraged to transition to commercial maintenance depots.

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KEYWORDS: induction welding; field deployable system; automated process control; forward deployed repair; shipboard repair; structural repair

DON26TZ01-NV012 TITLE: Sensing to Measure and Validate Corrosion in Naval Systems

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Integrated Sensing and Cyber;Sustainment

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop and deliver a sensory tool that can be used to monitor and assess several modes of corrosion activity as a function of time within Navy ship systems and subsystems. The sensory tool will incorporate artificial intelligence (AI) identify and estimate component life in a given platform/system for a given material selection, CAD geometry, and environment during the ship operations. AI can incorporate a set of mathematical models that will detect when the error happens and when to do maintenance. The main objectives of AI are to reduce maintenance time, production downtime, and the cost of component supplies.

DESCRIPTION: It is increasingly important for corrosion rate analysis to be performed on steel structures such as ships, offshore platforms and bridges to determine their safe operating life and for the development of effective and efficient maintenance practices. Optimal timeframes for asset availability and for planned redundancy also demand information about corrosion rates. Corrosion loss affects the effective load capacity of steel plating through causing plating thickness loss. The design of steel ships typically incorporates a corrosion allowance, i.e., an amount of corrosion loss that can be tolerated before the structural system is considered compromised. Corrosion protection measures include paint coatings and sacrificial anode systems for immersed areas. However, these methods are not always wholly effective, and continual maintenance usually is required but not always applied. In extreme cases, repair and replacement of structural details may be necessary, incurring very considerable cost penalties due to direct repair costs. It follows that the estimates of the expected rate of deterioration are important inputs for optimal maintenance and repair decisions for ships.

Naval ships are exposed to a range of corrosive environments and as a result the patterns of corrosion vary widely. The structural details and the orientation and position within the space within a given environment also will cause different corrosion patterns and rates. For immersion environments, influences on corrosion include chemical factors such as salinity, oxygen content, pH, and presence of pollutants; physical factors such as temperature and pressure; and biological factors such as bacteria and biomass. For ballast tanks the immersion environment usually is considered the most critical but in modelling the corrosion process attention might also need to be given to the occurrence of repeated wet/dry cycles as a result of the tanks being filled and emptied to adjust the freeboard trim of the ship. In addition, the presence of sacrificial anodes may have some influence, although they are effective only under immersed conditions and for uncoated areas. Thus, a de-ballasted tank will not be protected. It follows that the amount of corrosion in a ballast tank is a function of the environment, the type of corrosion protection, and the tank status. Apart from corrosion protection and operational practices, the main influence on the environmental parameters is the result of the conditions encountered during operations – what might be called the trading route, including geographical influences.

The number of hours a ship is generally in an operating or training status have decreased. Navy corrosion maintenance costs continue to escalate, reaching upwards to nearly $10B/year. Roughly 40% of those costs are caused by corrective maintenance that can be attributed to the improper selection of materials, usually from design process decisions that addressed system requirements without considering materials corrosion behavior in environments for which they are planned.

The application of a resistant coating on ships, offshore structures, and pipelines is the primary prevention method of corrosion wastage in the marine industries. To guarantee coating integrity and to be able to thoroughly survey for corrosion wastage on marine structures, new advanced nondestructive methods are being sought. The requirements of convenient and rapid determination of corrosion wastage on coated structures, even in the difficult spatial positions of the structure, will require advanced technologies which are being developed for other industries that also require very high structural integrity. Corrosion detection and monitoring are essential diagnostic and prognostic means for preserving material "health" and reducing life-cycle cost of industrial infrastructures, weapon systems, ships, aircraft, ground vehicles, pipelines, etc.

Sensor system attributes of small size, low weight, open plug-and-play interface architecture, self-diagnostics and validation make this a valuable interface and controller platform for other industrial and military monitoring applications. The system simplicity and low cost allows for wide area coverage by monitoring multiple sites on an individual structure and for fleet-wide vehicle condition monitoring. Other than Military vehicles, the smart sensor system has market potential in stationary structures, industrial processes, and civil and commercial transportation. By collecting and consolidating datasets into a fleet management system, DOW can better allocate maintenance resources and increase availability and service life objectives for these platforms. The collected data drive sustainment analytics and fleet management by increasing the accuracy of predictive maintenance schedules and decreasing inspection intervals and unnecessary preventative maintenance.

Artificial Intelligence (AI) plays a pivotal role in interpreting the vast amounts of data collected by drones. Machine learning (ML) algorithms analyze the images to identify patterns of corrosion, thereby enabling more accurate and timely maintenance decisions. This level of automation reduces human error and ensures that Navy vessels remain in optimal condition. AI is a machine's capability to impersonate human behavior, respond perceptively, solve problems, and make decisions automatically without human interference or with less human interference. The main objective of AI research involves general intelligence, automated planning, perception, natural language processing, knowledge representation, and robotics.

PHASE I: Explore the various non-destructive and electrochemical technologies through a literature search and downselect to the two or three most promising evaluation options that are capable of sensing the most corrosion degradation mechanisms. Non-contact technologies are preferred if degradation sensitivities are not lost. There is a critical need for the development of a real-time monitoring capability for U.S. Navy assets that has the potential to identify the onset of various corrosion modes like pitting as well as actively characterize stress corrosion (SCC) initiation and progression.

Optimization of quasi- and fully- distributed fiber optic sensing hardware for corrosion and SCC monitoring in Navy-relevant environments, including ultrasonic acoustics. Employ laboratory corrosion and SCC experiments on instrumented structural alloy coupons to develop a correlation between acoustic emission and corrosion/SCC signatures. Create physics-based modeling of both ultrasonic guided wave non-destructive examination (NDE) and acoustic emission to develop a training set for the AI-classification framework. Improve correlations by using training and validation of AI-classification framework and application for identification, localization, and classification of various corrosion modes and SCC in relevant alloys.

The NDE or electrochemical methods should be assessed as to the quality and accuracy of the objective measurements. The speed at which the requisite information can be obtained (ft2/minute) will also be an evaluation parameter. Offerors must show at least one technology that can reliably characterize the quality of the materials interface and bulk interior, assess the spatial resolution of the technique, and

assess the substrate surface conditions such as corrosion including sites with significant surface roughness.

PHASE II: The technology(ies) selected in Phase I should be further tested using larger uncoated and coated coupons of various alloys to better gauge what the speed (ft2/minute) of detection of decohesive sites, coating defects, and substrate corrosion, if present. Work with a Navy laboratory for collaborations in assisting the offeror in maturing and transitioning the technology(ies). Further modeling validation in select field Navy environments. This will be required to assert the reliability and sensitivity of the selected technology will be needed. Other acceptance testing as dictated by the Navy Laboratory should also be done and the evaluation/monitoring technology must be assessed as to its compatibility.

PHASE III DUAL USE APPLICATIONS: The ability of some NDE and electrochemical methods to penetrate most non-metallic materials allows non-contact examination of materials. The properties of interest across the industries may be broadly categorized into three areas—layer thickness, defects and contamination, susceptibility to SCC, and material characterization. Commercial and military ships both operate in a marine environment, and although they operate in different duty cycles, both are exposed to the aggressiveness of seawater and associated micro-marine environments. Both commercial and military ships suffer from similar corrosion failures.

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KEYWORDS: Electrochemical; monitoring; marine; corrosion; ship; mechanical failures; artificial intelligence, performance; protection

DON26TZ01-NV013 TITLE: Flexible Printed Thermoelectric Cooling Film

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Contested Logistics Technologies (LOG)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Microelectronics

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop a low-cost and lightweight thermoelectric cooling film that could be used to cool the warfighter (small scale) or surfaces on military platforms (larger scale) using printed organic semiconductors. The flexible cooling films should have a bending radius of less than one inch to easily wrap around pipes, wrists, and ankles, and be able to conform to complex curvatures on larger surfaces.

DESCRIPTION: Thermoelectric cooling devices based on narrow bandgap semiconductors such as bismuth telluride are commercially available. They are solid state devices and thus do not have the large footprint and moving parts associated with vapor compression refrigeration systems; however, they operate with lower efficiency. They are well-suited for cooling small flat surfaces where one is more concerned with the form factor than efficiency. For many practical applications, these square ceramic tile thermoelectric devices are heavy and too rigid, and do not offer conformal contact to curved surfaces. Over the past fifteen years, a lot of progress has been made on organic thermoelectric materials. Though the thermoelectric figure of merit (ZT) has not caught up to that of bismuth telluride and other inorganic materials, the potential to make low-cost, lightweight, and flexible devices has opened a new application space for thermoelectric cooling where flexibility and large-area conformal contact are prioritized over efficiency. For instance, lightweight headbands and wristbands only need to remove a small amount of heat to provide significant cooling sensation to the user. Likewise, there are diffuse, large surface area applications with similar cooling needs. Prior research was summarized in a recent review article by Segalman [Ref 1].

The conducting polymer Poly(3,4-ethylenedioxythiophene) [PEDOT] was identified as a strong candidate for the p-type leg in the p-n device, but device performance has been limited by the lack of suitable n-type materials. The organic electronics community has long wrestled with n-type materials due to potential oxidation of the electron carriers. A number of inherently stable and high performing n-type polymers have recently been developed [Ref 2] that should complement the available p-type materials and enable significantly improved thermoelectric cooling device performance. New device designs obtainable with simple fabrication must be developed to take advantage of the anisotropic thermal conductance and charge transport in these materials, which is typically maximized in-plane and along the polymer molecular backbones, such that measured thin film behaviors successfully translate into device performance. A number of design and fabrication strategies have been demonstrated but much more innovation is possible [Ref 1]. It is an appropriate time to develop lightweight, flexible thermoelectric cooling devices for these niche applications.

This STTR topic is for low-cost, lightweight, and flexible thermoelectrics for personal cooling as well as for large area applications.

The flexible cooling films should have a bending radius of less than one inch to easily wrap around pipes, wrists, and ankles, and be able to conform to complex curvatures on larger surfaces. The stated applications are near-ambient temperatures though the conjugated polymers should be able to handle temperatures up to 200°C. Composite approaches that are appropriate are welcome. This topic is not soliciting a fabric-based solution.

PHASE I: Select n and p type materials. Demonstrate that the selected n and p type materials can be processed (and doped) into films with reasonable Seebeck coefficient, thermal conductivity, and electrical conductivity in device relevant materials planes. Measure these properties in device relevant planes. Prepare a simple thermoelectric device and characterize performance. Model this simple device structure and compare with achieved performance. Model novel device geometries that could be manufactured with low cost processing approaches for both the personal cooling (small area, 4 kelvin gradient) and surface cooling (large area, 20 kelvin gradient) applications. Describe material and processing advances that would be accomplished in Phase II to enable these devices.

PHASE II: Optimize materials properties and device designs for personal and surface cooling applications. Model new designs as necessary. Make prototypes for both application s and start work towards commercially relevant device fabrication processes. In year two, prepare larger devices (10 square inches) by using commercially relevant processing methods and fully characterize device performance including achievable temperature gradients and efficiency. Model power requirements for both applications. Compare performance against commercial thermal electrics for ambient cooling applications. Prepare the cost analysis and business case for the two products.

PHASE III DUAL USE APPLICATIONS: Support transition to Navy use.
Both wearable thermoelectric coolers and large area films should have commercial and military applications. For wearable devices, performance metrics (power requirements) should be known for the Phase II prototype and there would be a lot of work to optimize the product for skin contact. In the longer term, some level of elasticity to the substrates would enable better contact and comfort. For the surface cooling film, a modestly stretchable substrate would also enable better contact with the large, curved surfaces of military platforms. An adhesive backing would be needed for large area applications.
In addition to directly developing products to keep warfighters cooler, industry that provides clothing and accessories to construction workers, law enforcement, and agricultural workers could develop appropriate products. Larger area devices could cool and heat automobile seats and outdoor surfaces.

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KEYWORDS: Thermoelectric cooling; Peltier effect; organic electronics; PEDOT; n-type polymer; p-type polymer

DON26TZ01-NV014 TITLE: Low Power Seawater Converter for Aircrew Survival

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Contested Logistics Technologies (LOG)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Biotechnology;Sustainment

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop a lightweight, compact, rugged, and reliable device that can convert seawater into safe, drinkable water. The device should minimize bulk and human energy expenditure, while maximizing output.

DESCRIPTION: Survival in a life raft on the open ocean depends greatly on the availability of potable water. Naval aircrew currently carry prepackaged water in soft packets placed within the ejection seat survival kit and aircrew survival vest sufficient to sustain life for less than one day. Reverse osmosis desalinators and forward osmosis nutrient packs are commercially available to the recreational seafarer. However, neither of these approaches are designed to maximize the amount of drinkable water while minimizing the amount of human energy expended, while constrained by limited space within a survival kit. Manual Reverse Osmosis Desalinator (MROD) devices are labor intensive, requiring more than 2500 pumps to produce one liter of water in one hour. Such human powered devices may require more energy expenditure than the calories available to stranded aircrew. Forward osmosis products available for the recreational sailor can produce potable beverages with little manual effort, but the total output capacity for aircrew is limited by the storage volume of the ejection seat survival kit. Current options for supplying sufficient drinking water to sustain life throughout extended rescue durations are inadequate. Innovative solutions will minimize or eliminate aircrew physical activity/exertion, while producing at least one gallon of drinkable water per day, with a minimum salt rejection of 95%. Concepts utilizing novel chemical processes or nanotechnology are preferred over simple refinements of current osmosis technology.

The device should:
a) fit within a Naval Aircraft Common Ejection Seat (NACES) survival kit (an envelope approximately 6½"x14½"x4½") along with an Emergency Oxygen System (EOS) and an LRU-38/P life raft, but not exceed 114 cubic inches.
b) operate in near freezing brine water/freshwater/saltwater.
c) operate in turbulent or calm water conditions.
d) operate reliably in cold and hot ambient air from -40° to +125°F (-40° to +51°C).
e) operate after exposure to temperature extremes from -65° to +160°F (-54° to +71°C).
f) operate after exposure to mold, mildew, flame, and salt fog.
g) not create hazards (i.e., burn, injury, Foreign Object Debris (FOD), snag/trip, and static discharge) inany mission or survival operations.
h) operate following a 600-knot seat ejection.
i) operate after repeated exposure to altitudes up to 70,000 ft (0.65 psi).
j) operate after exposure to typical fixed-wing ejection seat aircraft vibration levels, at frequencies from 5 Hz-2000 Hz).
k) provide resistance to environmental contaminants (i.e., sand, petroleum, oil, lubricants, and solar radiation).
l) not interfere with survival vest or mounted gear, armor/armor release, seat harnesses, helmets or head mounted gear.
m) be capable of operating after 15 months in a packed state (360-day inspection cycle plus 90 day shelf life) while exposed to temperature ranges of -65° to 160°F (-54° to +71°C).
n) weigh less than 2 lbs.

o) use Berry Amendment-compliant materials and manufacturing techniques.

PHASE I: Design and determine the feasibility of a concept seawater conversion device that meets the requirements provided in the Description. Demonstrate feasibility through analysis, modeling, simulation, and limited laboratory demonstrations. Provide performance, size, weight, cost and reliability estimates.

PHASE II: Develop, demonstrate, and validate a prototype seawater conversion device based on the design concept created in Phase I. Demonstrate device operation and capabilities in laboratory and simulated ocean environments. Provide draft design specifications, engineering drawings, and cost-benefit and life-cycle analyses.

PHASE III DUAL USE APPLICATIONS: Fabricate, validate, and deliver additional prototype devices for testing in open ocean environments. Provide support in transitioning the technology to Navy use. Provide a technical data package including a performance specification, an interface control document, and engineering drawings in accordance with military standards. Develop and assist with required qualification testing and training. Document the quality assurance test program in accordance with industry best practices.
The transfer and modification of commercial technology can benefit other military and recreational seafarers, as well as industrial, merchant, and marine operators and their crews or passengers.

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KEYWORDS: survival; osmosis; desalinate; portable; seawater; life raft

DON26TZ01-NV015 TITLE: Advancing Human Modeling Tools for Enhanced Performance and Survivability in Austere Environments

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Human-Machine Interfaces;Sustainment

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop an advanced suite of parametric human modeling tools incorporating USN/USMC aircrew anthropometric databases, empirical posture data, and 3D scans.

DESCRIPTION: The goal of this STTR topic is to leverage newly available data and advances in digital human modeling to improve modeling fidelity for USN/USMC and other DOW aircrew to improve acquisition outcomes. Resulting improvements to operational and environmentally appropriate protective clothing and equipment size, design, and tariffing (i.e., determination of how much of each size needs to be procured and distributed) will yield significant benefits to Fleet readiness and sustainment, safety, performance, protection, and affordability.

Digital Human Modeling (DHM) applications and tools are used to design and assess items for the DOW including protective clothing, footwear, body armor, flight equipment (e.g., helmets, oxygen masks, survival vests, G-suits, torso harnesses, etc.), seating, restraint systems, workstations, cockpits, controls, ground vehicles, and much more. Using this technology early in the product lifecycle is essential to reducing development cost and schedule and informing design tradeoff decisions. Historically, use of DHM has been subject to a variety of limitations that affect model fidelity, which is how well the model represents reality. These limitations result in reduced utility of the technology when the limitations are understood, but more concerning are the potential adverse outcomes where the limitations have either not been understood or have been ignored. This is concerning for all types of design applications, but especially problematic in aviation where safety of flight is crucial. There is an abundance of feedback from aircrew regarding poor fit or lack of availability of the sizes of protective clothing and operational equipment they need. They experience pain and injury, reducing performance and impacting readiness. There is now the potential to exponentially improve DHM capabilities due to a variety of advances in 3D scanning, model development, and availability of aircrew population specific anthropometric data and empirical posture data representing real-world conditions for military aircrew.

Limitations to current DHM capabilities related to the users include issues with intuitiveness of the tools, the degree of expertise required for effective use, and the significant amount of time it takes to develop expertise. There is a shortage of expert users in both the DOW and industry. Manikins used in DHM analysis are commonly selected from built-in software libraries with inappropriate anthropometric measurements for the population and/or design being evaluated. DHM users with a poor understanding of anthropometry often fail to consider the multivariate nature of anthropometric accommodation ignoring the need to consider more than one measurement at a time and neglecting the critical interactions of the measurements. Users positioning/posturing manikins routinely use guesswork in the absence of empirical data to account for clothing and flight equipment, restraint systems, cushion compression, flesh compression, and postural variation. They often have a limited understanding of aircrew operations and/or environment leading to incorrect assumptions when setting up their models.

For some DHMs the anthropometric measurements that can be adjusted are not the ones that matter for design application and the underlying anthropometric data used in the application may not represent the target population. Multivariate use cases have been developed and in use on DOW aircraft acquisition programs since the mid-90s [Ref 1], but manikins representing the use cases are often not included in DHM manikin libraries causing users to default to inappropriate use of the manikins that are available.

Until recently, the only USN/USMC aircrew anthropometric data available was from a 1960s database that did not include women. Currently, there are no DHM applications that include USN/USMC aircrew anthropometric data or associated multivariate use cases.

Another important consideration is that the commercially available DHM applications allow for analysis of one or more manikins, to include a family of multivariate use cases, but do not allow for parametric modeling of an entire population needed to accurately quantify the accommodation levels of a design. The NAWCAD Human Systems Engineering Department has recently completed an aircrew/aviator anthropometric survey and is also collaborating with the USAF on the Seat Specific Posture Model (SSPM) Project to collect empirical posture data to improve modeling fidelity. This project was initially intended for the purpose of developing an aviation specific postural analysis tool in the RAMSIS DHM but will be useful for other applications as well. One example that this STTR topic proposes is that this aircrew data be used in in the development of aviation-specific parametric accommodation models. The US Army has successfully developed this type of modeling tool for ground vehicles with a great many advantages to their acquisition programs and alleviation of many of the limitations documented above [Refs 2,3,4].

There have also been significant advances to head, hand, and body models that can be leveraged to greatly improve DHM state of the art and acquisition outcomes [Refs 5-11]. Integration of aircrew-specific anthropometric and 3D scan databases would ensure modeling efforts reflect the intended population. Aviators are a distinctly different population and appropriate representation of them in modeling applications is essential. Model input parameters can be adjusted to represent the goals of the modeling effort (i.e., desired accommodation levels and target population or subpopulation) with adjustable demographic variables such as sex, age, and race/ethnicity. Modeling tools can incorporate the ability to consider not only traditional 2D anthropometric measurements, but 3D shape and/or non-traditional measurements with the goal of improving size design and fit prediction [Refs 12, 13]. Through new and affordable 3D body scanning technologies [Refs 14,15], it is possible for an individual's specific anthropometry as well as their feedback on fit and preferred size to be run through an artificial intelligence (AI) algorithm to allow for ongoing improvements in size design, fit prediction, and tariffing. There have been advances in the development of head models that do not include hair artifacts [Ref 16], an important consideration in design. Improvements of head and hand models for dynamic or functional fit can improve the ability to digitally evaluate if masks maintain a seal when pilots talk or change facial expression and if gloves are designed appropriately for all pilot tasks, not just one static hand position. Posable manikins representing intended individuals or populations (multivariate use cases) can be easily customized and imported into any CAD environment or DHM software application for a variety of uses. It is important to note that the proposed tools are meant to be supplemental not duplicative of other modeling tools currently available or in development. Having these proposed modeling tools be interoperable or integrated with existing or emerging tools is highly desirable.
What makes these tools unique from existing/emerging modeling tools:

- Inclusion of USN/USMC aircrew anthropometric databases and 3D scans.
- Inclusion of SSPM project aircrew posture and reach data.
- Solution is not computationally and/or time prohibitive to use.
- Fills a gap in providing a solution that does not require an artisan modeler to make use of the models (easy to learn, simple user interface).
- Leveraging existing models/methods for expeditious transition.
- Models to be exported in common file formats to be interoperable with a broad range of CAD/DHM applications. No specific software applications are required.
- Not strictly PPE focused but also applicable to clothing design.
- Includes accommodation modeling tool for aircraft cockpits and workstations.

- Will represent digital twins of individuals like other modeling tools, but will also provide population virtual assessment of fit, size design, tariffing recommendations, and report population accommodation levels.
- Will allow for principal component analysis on a population and representation of boundary cases customized for specific applications.
- Includes ability to import anthropometric data for a group of participants and create bivariate plots for visual comparison to aircrew population data.
- Models will be web-hosted and freely/easily available to DOW civilians and contractors.
- Intention is to have web-hosted instructional materials, user forum, document library, and subject matter expert information to encourage best practices and collaboration.
- Framework will be built in to allow import of other population databases so other military populations including foreign military partners can be represented.

The proposed suite of tools would need to be easy to use, affordable, and easily accessed (e.g., hosted webapps and/or downloadable standalone applications) to facilitate practitioner usage and standardization. Accompanying guidance in the form of teaching materials, a user forum, links to relevant papers and reports, and a registry for subject matter experts and facilities wishing to be listed would be beneficial inclusions. The ability to create visualizations should also be considered. Allowing the import of anthropometry in a .CSV file for overlay with existing anthropometric databases in the form of bivariate plots of key anthropometric measurements is extremely helpful for population comparisons as well as confirming that human participants used for physical assessments adequately represent the target population. This proposed effort also seeks to put a framework in place that will allow incorporation of data from other populations and use of the models for other applications and users to include the entire DOW, foreign military partners, NASA, industry, and academia.

PHASE I: Identify, discuss, and demonstrate an approach to develop new or update existing models to create a suite of tools that will improve modeling fidelity for aviation applications. Ensure that the approach would seek to address limitations of the current state of the art as well as leverage recent improvements where feasible within the scope of this topic. Include plans for development and testing of prototypes to be developed during Phase II.

PHASE II: Develop and demonstrate prototype parametric aviator head, hand, and body shape models as well as an accommodation model tool. Provide access to the prototype for evaluation by end-user DOW subject matter experts (SMEs).

PHASE III DUAL USE APPLICATIONS: Upon completion of modeling tool development, the tools will be web hosted and made freely available to DOW users (civilian and contractors), vendors, and academia. They will have immediate benefit to numerous programs/platforms (e.g., PMA-202, Joint Strike Fighter), Human Systems SMEs, and DOW manufacturers/suppliers. Accompanying guidance in the form of teaching materials, a user forum, links to relevant papers and reports, and a registry for subject matter experts and facilities wishing to be listed are desired inclusions. The USN will not incur an ongoing webhosting cost. Model improvements (e.g., incorporation of new/additional scans and anthropometry, customization or new development of a tool for a specific application that was outside the scope of the STTR, etc.) may provide follow-on funding opportunities by Program Offices or other DOW entities. The STTR partners may choose whether they would like to make the tools freely available to the public or charge a fee for use for other organizations that may find the tools useful.
Other commercial opportunities include expanding the populations represented by the tools to include foreign military and civilian populations.

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KEYWORDS: Digital Human Modeling; Human Systems; Protective Clothing; Protective Equipment; Cold Weather Gear; Aircrew

DON26TZ01-NV016 TITLE: Nudging Behaviors for Better Sleep

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Human-Machine Interfaces;Trusted AI and Autonomy

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop software for personalized and adaptive behavioral interventions (i.e., nudges) using commercial off-the-shelf (COTS) wearable hardware devices to promote and improve sleep outcomes and human performance in dynamic environments.

DESCRIPTION: Despite extensive research on the mechanisms of sleep and behavioral modifications to improve sleep, relatively little is known about how context-sensitive behavioral nudging systems—those that dynamically suggest small, adaptive changes based on real-time data—can improve sleep quality and overall performance outcomes in complex, high-stakes settings. Fatigue caused by inadequate sleep negatively affects service members' performance and has contributed to accidents—resulting in deaths and hundreds of millions of dollars in damage to ships, vehicles, and aircraft [Ref 1]. "Nudging" refers to subtle interventions that steer behavior without restricting choices [Ref 2]. For example, non-obvious changes in how options are presented (e.g., ordering, timing, framing) have been shown to significantly affect sleep behaviors and dietary choices [Ref 3]. Recent advances in wearable sensor technology (e.g., smartwatches, rings, sleep trackers, etc.) allow for continuous collection of physiological and behavioral data. Many hardware devices are coupled with software that provide notifications, advice, and suggestions, but these are often canned, static statements that are simply pushed to the user (i.e., a one-way notification) and are not personalized to the user and/or their data.

Delivering adaptive behavioral nudges that learn and track the user's state and responses, evolve over time, and promote sustained positive behavior change is also critical for mitigating the impact of sleep on operations. The objective of this STTR topic is to develop personalized and adaptive behavioral interventions (i.e., nudges) using COTS wearable devices to promote and improve sleep outcomes and human performance in dynamic environments. Achieving this objective requires: (1) research into integrated theoretical frameworks for personalized behavior change, grounded in cognitive, physiological, and contextual variables, and informed by mathematical tools such as dynamical systems modeling; (2) the development of adaptive algorithms that leverage Machine Learning (ML) and Artificial Intelligence (AI) to integrate with existing wearable and embedded sensors to identify optimal timing, modality, and content for real-time, minimally-intrusive, adherence-supporting behavioral nudges across diverse user states and operational contexts; (3) the exploration of human-centered communication strategies for delivering behavioral insights and recommendations, ensuring interventions are not only well-timed but also subtle and capable of supporting an ongoing user-system relationship built on trust and voluntary engagement; and (4) empirical testing in ecologically valid environments, including experiments that collect sleep and performance metrics to evaluate effectiveness, generalizability, and long-term behavioral impact.

Equal emphasis will be placed on (1) advancing theoretical models of behavior change, sleep regulation, and performance adaptation and (2) developing AI/ML systems and communication strategies for delivering behavioral nudges.

This topic focuses on sleep behavior due to its broad applicability to the general population, its foundational role in human performance, and the relative ease and reliability of measurement. Proposed efforts should aim to develop generalizable algorithms that integrate complex mathematical modeling and

ML with cognitive-behavioral theory to drive adaptive behavioral interventions. These interventions must be compatible with existing wearable and embedded sensor ecosystems – this topic explicitly does not aim to develop new hardware, but instead to maximize the utility of currently available commercial sensors as inputs to a personalized, adaptive nudging system.

PHASE I: Develop early research plans, concepts/prototypes, and requirements for investigations into: (1) theoretical models of real-time behavioral change, intervention receptiveness, and nudging effectiveness that combines psychology (e.g., behavioral, cognitive, decision sciences), mathematics (e.g., dynamical systems), computer science (e.g., AI/ML, human-computer interaction), physiology (e.g., sleep science, chronobiology), and communications (e.g., persuasive communication, dialogic interaction systems; (2) develop a system that integrates AI/ML decision engines that dynamically adapts nudge timing, content, and delivery method based on physiological, cognitive, and contextual data and measures nudge compliance and effectiveness all in service of positively impact sleep behavior.

Tasks and environments should reflect the unique operational demands of the naval context, including irregular sleep schedules, sustained attention during extended operations, and decision-making under fatigue. Critical elements of the Phase I effort are to describe the research and engineering plans that would be executed during a potential Phase II and provide evidence of the feasibility of the approach, emphasizing the current state of the research and how the approach is both innovative and achievable. Prepare Phase II plans that should include key component technological milestones and plans for at least one operational test and evaluation to include user testing.
Due to the potential for long review times involved, formal human subject research is prohibited during Phase I.

PHASE II: Conduct and implement interdisciplinary research in the fields of psychology, physiology, mathematics, and computer science to develop and evaluate a prototype system with sailors (coordination aided by ONR) outlined during the Phase I. Include parallel but interrelated research and engineering tracks that will deliver iterative prototypes that will undergo design and testing reviews, to include usability assessment and effectiveness evaluations where appropriate. Collect both subjective and objective metrics regarding nudge usefulness, compliance, sleep quantity and quality through the development process. Perform all appropriate engineering tests and reviews, including a critical design review to finalize the system design. Produce the following deliverables: (1) additional research into development, adaptation, and delivery of behavioral nudges with a particular emphasis on sleep and fatigue; (2) a working prototype of the system that leverages existing COTS sensors, wearables, and hardware; (3) evaluation of system usability and compliance regarding effectiveness of nudges; (4) a system effectiveness evaluation of system capabilities to produce improved sleep quantity and/or quality. DON will provide Phase II awardees with the appropriate guidance required for human research protocols. Institutional Review Board (IRB) determination as well as processing, submission, and review of all paperwork required for human subject use can be a lengthy process. As such, no human research will be allowed until Phase II and work will not be authorized until approval has been obtained, typically as an Option to be exercised during Phase II.

PHASE III DUAL USE APPLICATIONS: Support the Navy in transitioning the technology for fielding. Develop the software for evaluation to determine its effectiveness in operational settings. As appropriate, focus on broadening capabilities and commercialization plans. Development of affordable, scalable, non-proprietary technologies are needed to take data generated by COTS hardware and sensors into actionable information that can be delivered as personalized nudges.

The commercial sector is developing some of these AI-enabled sleep technologies, but they often do not deal with critical issues regarding complex environments and dynamic contexts, do not address

encryption and classification requirements, and often come with prohibitive licensing and usage fees. This technology will have broad application in the commercial sector.

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2. Thaler, R. H. and Sunstein, C. R. “Nudge: Improving Decisions About Health, Wealth, and Happiness.” Penguin, Feb 24, 2009. https://books.google.com/books/about/Nudge.html?id=NGA9DwAAQBAJ&source=kp_book_description
3. Kahneman, D. “Thinking, Fast and Slow.” Farrar, Straus and Giroux, 2011. https://en.wikipedia.org/wiki/Thinking,_Fast_and_Slow

KEYWORDS: behavior change; adaptive systems; human performance; sleep and fatigue; dynamical systems modeling

DON26TZ01-NV017 TITLE: High Energy Laser Optically Rugged Maritime Beam Director Components & Subassemblies

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Scaled Directed Energy (SCADE)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Directed Energy (DE);Microelectronics

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop new, innovative processes and methods of reproduction, and deliver prototypical end item high precision optics suitable for use with high energy lasers in beam directors - as scalable components and/or subassemblies, through automated and additive manufacturing techniques for structures, optics, and mirrors (flat and parabolic) - including any required finishing processes, (e.g., coating and polishing processes) to develop, document, achieve and demonstrate "end item" durable, rugged, reliable, tested components and/or products.

DESCRIPTION: Highly precise, small to large diameter (10 to 50 to 100cm) high energy laser optics and mirrors have very long lead times often exceeding individual fiscal year funding, and experience a high rejection rate due to complex, multi-step processing between multiple dislocated facilities. Resulting optics have high defect rates and low ruggedness requiring depot supplies of spares and replacements, creating logistical shortages and non-availabilities which impact readiness and capacity.
Creating multiple kinds of components for a notional or specific beam director that offers a series of developmental components and elements toward a finalized ruggedized beam director, suitable for at-sea deployment for up to ten years without maintenance is the objective. Threshold shall be the development of an optic that provides initial research and development value that can be tested in multiple laser inducted damage tests (LiDT). Examination of capabilities for scale, with optics from 10cm to 50cm or 100cm diameters, is expected.

Specifically, there is a very high interest in creating components from bulk materials with finished or near finish high quality optical surfaces and properties, transmissive or reflective, at a greatly reduced cost compared to traditional optical components (e.g., an optical transformation lens, a simple transmissive optic, or a fast steering mirror) utilizing "on-demand" adaptive, additive 3-D printing, etching, and highly automated finishing techniques. High interest exist in optical elements from 40 to 50 centimeters in diameter (e.g., ceramic, metal or other optical materials), small lightweight optics (e.g., from plastics or ceramics), and items that are completed to form a fully finished component through "no touch" human intervention processes or via fully automated decision-based manufacturing and processing (e.g., including finished robust optical coatings suitable for sea water based atmospheric exposure – such as fog or sea water splash contamination).

The Navy seeks a capability to create custom optical components, potentially including required integrated subassemblies, from processes that result in highly precise end item optics for high energy laser beam directors and laser weapons systems, either as components, replacements and/or subassemblies, through automated and additive manufacturing techniques for structures, optics, mirrors both shorten timelines for availability, and also enable innovative laser architectures - including or beyond current state-of-the-art modular architecture designs. Especially those where limited lifetimes due to environmental exposure require unique materials and innovative generational designs that change based on emergent requirements and increased commercial capacity. These can potentially open new avenues that enable new, innovative laser architectures - including capabilities or beyond current state of the art modular architecture designs, such as "ball on gimbal", heliostats and celiostats – but the focus is on the processes and means to scale component designs, rapidly prototype multiple initial designs, and then

move to quickly produce production grade high quality optics for initial use or as replacement utility spares. Preference shall be given for use of existing, commercially available materials, starting feed stock, or machine tooling. Similarly, preference shall be given for use of existing or modified “open system, open software” code and manufacturing methods.

The Navy has special interest in those components where limited lifetimes are expected (e.g., exit apertures, rotating or moving optics) due to environmental exposure and require unique materials (e.g., hard coatings for dust resistance, hydrophobic water shedding or chemical resistance) and innovative designs (e.g., flexible substrates) that can adapt, be replaced quickly, or change based on when emergent requirements and increased commercial capacity are noted.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and ONR in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Demonstrate at small scale (threshold: 10-15cm, objective: > 25cm) ability to create, finish, produce and replicate a single custom highly precise optical component design or integrated subassembly from a proposed set of processes - that results in highly precise end item optic, that can be tested in a standardized high energy laser induced damage test (LIDT). The design should include characteristics that show the resulting optical component can serve in the development of a full-scale beam director for a high energy laser weapons system at scale (small [10cm] or large [50cm]) using a surrogate commercial laser of no less than 1 kilowatt (kW). The final Phase I test shall enable proof in the applicability of additive or subtractive machined optical components, using highly automated processes toward meeting replacements for components and/or subassemblies. Included in the proof shall be demonstration of the ability to incorporate a fully automated manufacturing technique for assembling structures, optics, and/or mirrors that both shorten timelines for availability, or potentially enable innovative laser architectures for an individual component or subsystem. As an objective, one specific optical design that utilizes or extends an existing/anticipated high energy laser beam director (e.g., the planned DOW Joint Beam Control System (JCBS) or other service lightweight beam director) availability or functionality beyond current state of the art is expected.

PHASE II: Demonstrate at larger scales [30 to 50cm (threshold) to 100cm (objective)] of custom optical components or integrated subassemblies developed from the Phase I design concepts that will support a proposed DOW service led high energy laser beam director effort. Include development of all required mechanical and electronic control considerations for a full scale beam director use of the developed component or subsystem, suitable for a prototypical level integration effort within a high energy laser weapons system using a surrogate commercial laser of no less than 10 kW (Threshold) to 50 kW (Objective), showing applicability of additive or subtractive machined components, using only highly automated processes toward meeting replacements for components and/or subassemblies. Complete one specific optical component design that replaces or extends a high energy laser beam director functionality beyond the current state of the art in modular architecture and manufacturable designs by identifying an accurate unit price or cost estimate against documented and identified performance requirements. Identify

potential means where limited lifetimes due to environmental exposure can be monitored and rapidly replaced through the use of unique materials and innovative generational designs that offer changes based on emergent requirements and increased commercial capacity with an objective being to demonstrate a simulated component failure and manufacture a complete replacement optic or component within 1 month.
It is probable that the work under this effort will be classified under Phase II (see the Description section for details).

PHASE III DUAL USE APPLICATIONS: Support transition for Navy use.
The processes and components developed shall transition directly into the DOW development of high energy laser weapons systems, including spares.

Additional commercial products may include sensing and measurements systems that require a rugged, highly accurate optical element for video or still imagery. In particular, multiple service high energy laser beam directors are under development. This also includes potential for near term transition, including but not limited to high energy laser precision optics for LADAR systems; optics that could transition into the Golden Dome 4 America missile defense initiative for a potential directed energy weapon based on high energy lasers; or for a high power optical sensing and tracking capability.

REFERENCES:

1. Perram, Glen P.; Cusumano, Salvatore J.; Hengehold, Robert L. and Fiorino, Steven T. "An Introduction to Laser Weapon Systems." Directed Energy Professional Society , 2010. https://www.deps.org/store/merchandise/TOCs/IntroLaserWeaponsDetails.html
2. Merritt. Paul H. and Albertine, John R. "Beam control for high-energy laser devices." , Optical Engineering 52(2), 021005, February 2013. https://www.spiedigitallibrary.org/journals/optical-engineering/volume-52/issue-2/021005/Beam-control-for-high-energy-laser-devices/10.1117/1.OE.52.2.021005.short
3. Berglund, Gregory; Wisniowiecki, Anna; Gawedzinski, John; Applegate, Brian and Tkaczyk, Tomasz S. "Additive manufacturing for the development of optical/photonic systems and components.", Optica 9, 623-638, 2022. https://opg.optica.org/optica/fulltext.cfm?uri=optica-9-6-623&id=476811
4. Xin, Chenxing; Li, Zheng; Hao, Liang and Li, Yan. "A comprehensive review on additive manufacturing of glass: Recent progress and future outlook." Materials & Design, Volume 227, 111736, 2023. ISSN 0264-1275, https://doi.org/10.1016/j.matdes.2023.111736

KEYWORDS: laser; optics; additive manufacturing; directed energy; weapons; beam director

DON26TZ01-NV018 TITLE: Production of Norbornadiene

COMPONENT TECHNOLOGY PRIORITY AREA(S): Advanced Materials;Hypersonics;Sustainment

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop efficient and scalable methods for the production of norbornadiene from abundant domestic feedstocks.

DESCRIPTION: Norbornadiene is a critical chemical used in the manufacture of fuels and cross-linked polymers. The conventional process for production of norbornadiene relies on a Diels-Alder coupling reaction between cyclopentadiene and acetylene. Acetylene air mixtures can be explosive, which has increased the cost of norbornadiene and reliance on foreign supply chains. The intent of this STTR topic is to establish a manufacturing process that will enable the safe and efficient domestic production of norbornadiene, which will in turn reduce acquisition costs.
Ultimately, this topic seeks to establish a process for the domestic production of norbornadiene at > 500 metric tons/year with target acquisition costs below $20/kg. The norbornadiene synthesized in this effort should have a purity > 97%. The utilization of advanced manufacturing techniques that generate acetylene on demand or incorporate novel methods for the safe storage of acetylene on-site are encouraged. Other approaches that generate norbornadiene via unique intermediates are also of interest. A preferred approach is to utilize domestic bio-feedstocks, including hemicellulose and furfural, as substrates for the production of norbornadiene.

PHASE I: Identify a chemical process for the synthesis of norbornadiene. Particular attention will be paid to the safe handling of acetylene and other reactants/reagents and solvents used in the manufacturing process. The use of domestic biomass feedstocks as the carbon source for norbornadiene production is preferred. A preliminary technoeconomic analysis will be conducted with a target selling price < $20/kg. Awardee(s) will conduct laboratory scale reactions and demonstrate the ability to generate small quantities of norbornadiene (10-100 mL) at purities comparable to commercial norbornadiene (> 97%). The purity of the product will be confirmed through standard analytical techniques including NMR spectroscopy and gas chromatography.

PHASE II: Demonstrate a prototype scalable approach for the synthesis of norbornadiene and provide 1 kg of material for evaluation by Navy researchers. Demonstrate production of norbornadiene at the 100 kg scale and develop plans for a facility capable of producing > 500 metric tons/year. An updated technoeconomic analysis will be performed utilizing information obtained during the pilot-scale production runs. The assessment will include a comparison of processes utilizing inputs from both conventional petrochemicals and biofeedstocks. At the completion of Phase II, 100 kg of norbornadiene with a purity > 97% will be delivered to the Government.

PHASE III DUAL USE APPLICATIONS: Assist the government in the establishment a CONUS facility capable of producing norbornadiene at the estimated scale of 10-20 metric tons/year, likely doing this in collaboration with a larger chemical supplier. Informed by the standup of this facility, update the plans and technoeconomic analysis for a facility capable of producing > 500 metric tons/year.

REFERENCES:

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2. “CN1580014A - Method for preparing 2,5-norbornadiene from dicyclopentadiene.”

Sinopec Shanghai Petrochemical Co Ltd, 14 August 2003. https://patents.google.com/patent/CN1580014A/en
3. "RU2228324C1 - Norbornadiene preparation method." https://patents.google.com/patent/RU2228324C1/en?oq=RU2228324C1
4. Zhang, X. et al. "Integration of Bio-JP-10 Synthetic Route from Furfuryl Alcohol." Catalysis Today, 443, 1 January 2025, 114987. https://www.sciencedirect.com/science/article/pii/S0920586124004814
5. Xie, J. et al. "Synthesis of JP-10 Analogues High-Density Fuels via One-Pot Diels-Alder/Hydrodeoxygenation Reaction." Fuel 2024, 361, 1 April 2024, 130738. https://www.sciencedirect.com/science/article/abs/pii/S0016236123033525
6. Li, G. et al. "Making JP-10 Superfuel Affordable with a Lignocellulosic Platform Compound" Chem. Int. Ed. 2019, 58, 12154

KEYWORDS: Chemical Synthesis; Chemical Engineering; Manufacturing; Norbornadiene; Technoeconomic Analysis; Bio-feedstocks

DON26TZ01-NV019 TITLE: High Voltage and Current Silicon-Carbide (SiC) Metal-Oxide Semiconductor Field-Effect Transistor (MOSFET) for Fast Turn-On Current Applications

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Scaled Directed Energy (SCADE)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Directed Energy (DE);Microelectronics;Renewable Energy Generation and Storage

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop state-of-the-art silicon carbide (SiC) Metal-Oxide Semiconductor Field-Effect Transistors (MOSFETs) packaged for improved size, weight, and power (SWaP) for applications where a high-blocking voltage of more than 10 kV, a high pulsed current density of greater than +/- 5 kA (10 kA ideal), and tens of nanoseconds turn-on time, with low-jitter, are needed for integration with high power microwave (HPM) systems.

DESCRIPTION: The DOW needs SWaP-favorable solutions for fast turn-on and low-jitter SiC MOSFETs to generate high current densities from high voltage capacitors. Current methods of high-current/voltage switching from SiC MOSFETS rely on an array created from series and parallel combinations of commercial off the shelf (COTS) devices [Ref 1]. However, these device arrays are limited in the voltage and amplitude they can switch, have complicated gate driving circuits, and can become size limited. To improve current state-of-the-art capability, the DOW has a need for the development of MOSFETs that have a blocking voltage greater than 10 kV for a single wafer, such that a low-side gate driver can be used to turn on the MOSFET, and a high pulsed-current capability. The requirements for a 5 kA peak current (10 kA ideal) may require multiple parallel combinations of MOSFET wafers, and if so, packaging is to be minimized and vertically stacked packaged arrays should be utilized. It is understood that at higher blocking voltages and current densities an additional diode may be necessary to accommodate the desired pulse current [Ref 2]. Minimizing gate charge and gate resistance for an array of MOSFET is important to alleviate driver requirements, such that a turn on time of less than 30 nanoseconds (ns) is achievable with less than 30 V of gate voltage and 10's of amps of gate current.

PHASE I: Develop a conceptual design for a MOSFET solution meeting the requirements in the Description. Include methodology and prototype performance through description and modeling that will demonstrate the proposed concept. Perform a tradespace assessment of size and performance for the proposed solutions.
Phase I Key Parameters:

- Voltage blocking from drain to source (V_DS) for a single semiconductor wafer of greater than 10 kV
- Low drain to source resistance (R_DS_on) of 50 milli-ohms or less up to 150 degrees Celsius
- Pulsed current discharge greater than 1 kA for a period of 500 ns forward (drain to source) and 500 ns reverse (source to drain).
    - Paralleling of devices is acceptable to reach current densities
    - External body diode is acceptable for high reverse currents
    - Pulsed current design is more critical than a continuous current rating
- Turn on time less than 50 ns with a driving gate voltage of equal or less than 30 volts
- Propose methods to provide a low inductance packaging of less than 20 nH per device for drain, source, gate, and kelvin pin. Total package size anticipated to be less than 6 x 6 x 3 inches
    - Thermal dissipation through drain is acceptable
    - Propose methods to minimize partial discharge voltage degradation

    - Propose methods for component layout showing size density
    - Propose methods for improved thermal management through packaging with thermal dissipation to switch up to 1 joule of capacitive energy per discharge at a 5 kHz discharge rate for up to 10 seconds
- Propose methods for operational lifetime evaluation
    - Describe degradation from peak operating conditions
    - Describe methods for electromagnetic interference (EMI) mitigation

PHASE II: Develop and deliver to the government (Quantity 10) optimized MOSFETs for integration with solid-state pulse generator prototypes developed by the government that meet or exceed the key performance requirements listed below. Topic proposals may propose the use of commercial or Federal facilities to satisfy the performance requirements of this topic, provided the performing SBC satisfies the performance thresholds set out in 15 U.S.C. sec. 638 and as implemented in the SBIR/STTR Policy Directive. Deliver a technical data package (TDP) detailing the design and construction of the packaged MOSFET solution as well as a preliminary datasheet on performance. Support integration and testing activities performed by the DOW.

Phase II Key Parameters:
- Voltage blocking from drain to source (V_DS) for a single semiconductor wafer of greater than 10 kV, when evaluated in package in open air (ideal) or liquid dielectric (threshold)
- Pulsed current discharge greater than 5 kA (10 kA ideal) for a period of 500 ns forward (drain to source) and 500 ns reverse (source to drain)
    - The MOSFET will be evaluated in short circuit conditions in a low-impedance, capacitive, and low-inductive (RLC) circuit
    - Paralleling of devices is acceptable to reach current densities
    - External body diode is acceptable and anticipated for high reverse currents
    - Pulsed current design is more critical than a continuous current rating
- Provide a low inductance packaging of less than 10 nH per device for drain, source, gate, and kelvin pin. Total package size anticipated to be less than 3 x 3 x 1 inches
    - Evaluate the package for partial discharge voltage degradation
    - Implement methods for EMI mitigation
- Turn on time less than 30 ns with a driving gate voltage of equal or less than 30 volts
- Low drain to source resistance (R_DS_on) of 10 milli-ohms or less up to 150 degrees Celsius
- Packaging with thermal dissipation to switch up to 2 joules of capacitive energy per discharge at a 50 kHz discharge rate for up to 10 seconds
- Implement methods for operation lifetime evaluation

PHASE III DUAL USE APPLICATIONS: Fast rising edge, high voltage MOSFETs enables the generation of higher power HPM systems. A future looking Phase III award shall deliver a complete refinement of parameters to meet requirements and develop manufacturing methods to reduce production time and cost in a commercialization path. For the delivery of MOSFETs as an enabling technology for solid-state HPM systems in such applications as counter-electronics, ultra-wideband radar, and high-power jammers. Other non-HPM applications include alternative energy (such as solar and wind inverters), power distribution, and automotive and transportation.

REFERENCES:
1. Xiao, Q.; Tan, Y.; Wu, X'; Ren, N. and Sheng, K. "A 10KV/200A SiC MOSFET Module with series-parallel hybrid connection of 1200V/50A dies." 2015 IEEE 27th International Symposium on Power Semiconductor Devices & IC's (ISPSD), Hong Kong, China, 2015, pp. 349-352. doi: 10.1109/ISPSD.2015.7123461

2. Kumar, A. et al. “Effect of capacitive current on reverse body diode of 10kV SiC MOSFETs and external 10kV SiC JBS diodes.” 2017 IEEE 5th Workshop on Wide-Bandgap Power Devices and Applications (WiPDA), Albuquerque, NM, USA, 2017, pp. 208-212. doi: 10.1109/WiPDA.2017.8170548

KEYWORDS: Semiconductor; Metal-Oxide Semiconductor Field-Effect Transistor; MOSFET; body diode; high voltage; high current; Directed Energy; DE; High Power Radio Frequency; HPRF; High Power Microwave; HPM; Pulse repetition frequency; PRF; Pulse repetition rate; PRR

DEPARTMENT OF THE AIR FORCE
SMALL BUSINESS TECHNOLOGY TRANSFER (STTR)
DoW 2026 STTR BAA Release 1
PROPOSAL SUBMISSION INSTRUCTIONS

The Air Force intends these Phase I proposal submission instructions to clarify the Department of War (DoW) Broad Agency Announcement (BAA) as it applies to the topics solicited herein. **Offerors must ensure proposals meet all requirements of the FY26 STTR BAA posted on the Defense SBIR/STTR Innovation Portal (DSIP) at the proposal submission deadline date/time.**

Applicants are encouraged to thoroughly review the DoW Program BAA and register for the DSIP Listserv to remain apprised of important programmatic and contractual changes.

- Full component-specific instructions and topic descriptions are available on DSIP at https://www.dodsbirsttr.mil/submissions/solicitation-documents/active-solicitations. Be sure to select the tab for the appropriate BAA cycle.

Please ensure all e-mail addresses listed in the proposal are current and accurate. The DAF is not responsible for ensuring notifications are received by firms changing mailing address/e-mail address/company points of contact after proposal submission without proper notification to the AF. **If changes occur to the company mail or email addresses or points of contact after proposal submission, the information must be provided to the AF SBIR/STTR One Help Desk.** The message shall include the subject line, "FY26 STTR BAA Address Change".

Points of Contact:

- General information related to the DAF SBIR/STTR program and proposal preparation instructions, contact the DAF SBIR/STTR One Help Desk at usaf.team@afsbirsttr.us. All applicants have ample opportunity to request clarifying information. **The DAF encourages applicants to request clarifying information as early as possible, as delays in such requests constrain the DAF's ability to provide satisfactory resolution to applicant concerns.**
- Questions regarding the DSIP electronic submission system, contact the DoW SBIR/STTR Help Desk at dodsbirsupport@reisystems.com.
- For technical questions about the topics during the pre-announcement and open period, please reference the FY26 STTR BAA.
- Air Force SBIR/STTR Contracting Officer (CO):
    - Mr. Daniel J. Brewer, Daniel.Brewer.13@us.af.mil

General information related to the DAF Small Business Program can be found at the DAF Small Business website, http://www.airforcesmallbiz.af.mil/. The site contains information related to contracting opportunities within the DAF, as well as business information and upcoming outreach events. Other informative sites include those for the Small Business Administration (SBA), www.sba.gov, and the Procurement Technical Assistance Centers (PTACs), http://www.aptacus.us.org. These centers provide Government contracting assistance and guidance to small businesses, generally at no cost.

PHASE I PROPOSAL SUBMISSION: The FY26 STTR BAA, https://www.dodsbirsttr.mil/submissions/login, includes all program requirements. Phase I efforts should address the feasibility of a solution to the selected topic's requirements.

PHASE I PROPOSAL FORMAT

Complete proposals must include all of the following:

Volume 1: Cover Sheet

Note: If selected for funding, the proposal's technical abstract and discussion of anticipated benefits will be publicly released. Therefore, do not include proprietary information in this section.

Volume 2: Technical Volume
Volume 3: Cost Volume
Volume 4: Company Commercialization Report
Volume 5: Supporting Documents
Volume 6: Fraud, Waste, and Abuse Training
Volume 7: Disclosures of Foreign Affiliations or Relationships to Foreign Countries

COVER SHEET (VOLUME 1)

Complete the proposal Cover Sheet in accordance with the instructions provided via DSIP. The technical abstract should include a brief description of the program objective(s), a description of the effort, anticipated benefits and commercial applications of the proposed research, and a list of keywords/terms. The technical abstract of each successful proposal will be submitted to the Office of the Secretary of War (OSW) for publication and, therefore, must not contain proprietary or classified information.

TECHNICAL VOLUME (VOLUME 2):

The Technical Volume should include all graphics and attachments but should not include the Cover Sheet, which is completed separately as Volume 1. The Phase I technical volume (uploaded in Volume 2) shall contain the required elements found below. Ensure that all graphics are distinguishable in black and white.

The Phase I Technical Volume page/slide limits identified for the topics do not include the Cover Sheet, Cost Volume, Cost Volume Itemized Listing (a-h). The Technical Volume must be no smaller than 10-point on standard 8-1/2" x 11" paper with one-inch margins. Only the Technical Volume and any enclosures or attachments count toward the page limit. In the interest of equity, pages/slides in excess of the stated limits will not be reviewed. The documents required for upload into Volume 5, "Other", do not count toward the specified limits.

These instructions supplement the FY26 STTR BAA. In addition to the requirements found in the FY26 STTR BAA applicants are required to provide the following information in Volume 2:

Key Personnel: Identify in the Technical Volume all key personnel who will be involved in this project; include information on directly related education, experience, and citizenship.

- A technical resume of the principal investigator, including a list of publications, if any, must be included. Only one principal investigator/project manager can be designated to a proposal at any given time.
- Concise technical resumes for subcontractors and consultants, if any, are also useful.
- Identify all U.S. permanent residents to be involved in the project as direct employees, subcontractors, or consultants.
- Identify all non-U.S. citizens expected to be involved in the project as direct employees, subcontractors, or consultants. For all non-U.S. citizens, in addition to technical resumes, please provide countries of origin, the type of visa or work permit under which they are performing and an explanation of their anticipated level of involvement on this project, as appropriate. Additional information may be requested during negotiations in order to verify the foreign citizen's

eligibility to participate on a contract issued as a result of this announcement. **Note:** Do not upload information such as Permanent Resident Cards (Green Cards), birth certificates, Social Security Numbers, or other PII to the DSIP system.

Phase I Statement of Work Outline

NOTE: The DAF uses the work plan outline as the initial draft of the Phase I Statement of Work (SOW). Therefore, **do not include proprietary information in the work plan outline**. To do so will necessitate a request for revision, if selected, and may delay award.

Include a work plan outline in the following format:
Scope: List the effort's major requirements and specifications.
Task Outline: Provide a brief outline of the work to be accomplished during the Phase I effort.
Milestone Schedule
Deliverables
Progress reports
Final report with SF 298

COST VOLUME (VOLUME 3)

Cost information should be provided by completing the Cost Volume in DSIP and including the Cost Volume Itemized Listing specified below. The Cost Volume detail must be adequate to enable Air Force personnel to determine the purpose, necessity and reasonability of each cost element. Provide sufficient information (a.-g. below) regarding funds use. The DSIP Cost Volume and Itemized Cost Volume Information will not count against the specified page limit. The itemized listing also may be submitted in Volume 5 under the "Other" dropdown option.

a. **Direct Cost Materials**: Justify costs for materials, parts, and supplies with an itemized list containing types, quantities, prices and where appropriate, purpose. Material costs may include the costs of such items as raw materials, parts, subassemblies, components, and manufacturing supplies.

b. **Other Direct Costs**: This category includes, but is not limited to, specialized services such as machining, milling, special testing or analysis, and costs incurred in temporarily using specialized equipment. Proposals including leased hardware must include an adequate lease v. purchase justification.

c. **Direct Labor**: Identify key personnel by name, if possible, or by labor category, if not. Direct labor hours, labor overhead and/or fringe benefits, and actual hourly rates for each individual are also necessary for the CO to determine whether these hours, fringe rates, and hourly rates are fair and reasonable.

d. **Travel**: Travel costs must relate to project needs. Break out travel costs by trip, number of travelers, airfare, per diem, lodging, etc. The number of trips required, as well as the destination and purpose of each, should be reflected. Recommend budgeting at least one trip to the Air Force location managing the contract.

e. **Subcontracts**: Involvement of consultant in the project's planning and/or research stages may be appropriate. If so, describe in detail and include information in the Cost Volume. A minimum of 40% of each STTR project must be conducted by the SBC and a minimum of 30% of the effort performed by the single partnering Research Institution. Deviations from these performance of work requirements are not permitted. The STTR funded work percentage calculation considers both direct and indirect costs after removal of the SBC's proposed profit. Support subcontract costs with copies of executed agreements. The documents must adequately describe the work to be performed. At a minimum, include a Statement of Work (SOW) with a corresponding detailed Cost Volume for each planned subcontract. Additionally, see DoW STTR FY26 BAA for more information regarding the required Allocation of Rights Agreement.

f. **Special Tooling, Special Test Equipment, and Material**: The inclusion of equipment and materials will be carefully reviewed relative to need and appropriateness to the work proposed. Special tooling and special test equipment purchases must, in the CO's opinion, be advantageous to the Government and relate directly to the effort. These toolings or equipment should not be of a type that an offeror would otherwise possess in the normal course of business. These may include items such as innovative instrumentation and/or automatic test equipment.

g. **Consultants**: Provide a separate agreement letter for each consultant. The letter should briefly state what service or assistance will be provided, the number of hours required, and the hourly rate.

NOTE: If no exceptions are taken to an offeror's proposal, the Government may award a contract without negotiations. . Therefore, the offeror's initial proposal should contain the offeror's best terms from a cost or price and technical standpoint. If there are questions regarding the award document, contact the Phase I CO identified on the cover page. The Government reserves the right to reopen negotiations later if the CO determines doing so to be necessary.

COMPANY COMMERCIALIZATION REPORT (VOLUME 4)

Completion of the CCR as Volume 4 of the proposal submission in DSIP is required. Please refer to the FY26 STTR BAA for full details on this requirement. Information contained in the CCR will not be considered by the Air Force during proposal evaluations.

SUPPORTING DOCUMENTS VOLUME (VOLUME 5)

The following documents may be required if applicable to your proposal:

1. DD Form 2345: For proposals submitted under export-controlled topics, either International Traffic in Arms or Export Administration Regulations (ITAR/EAR), a copy of the certified DD Form 2345, Militarily Critical Technical Data Agreement, or evidence of application submission must be included. The form, instructions, and FAQs may be found at the United States/Canada Joint Certification Program website, http://www.dla.mil/HQ/InformationOperations/Offers/Products/LogisticsApplications/JCP/DD2345Ins tructions.aspx. DD Form 2345 approval will be required if proposal if selected for award.
2. Verification of Eligibility of Small Business Joint Ventures (Attachment 3 to the FY26 STTR BAA)
3. Technical Data Rights Assertions (if asserting data rights restrictions)

FRAUD, WASTE AND ABUSE TRAINING (VOLUME 6)

Fraud, Waste and Abuse training material can be found in the Volume 6 section of the proposal submission module in DSIP and must be thoroughly reviewed once per year to proceed with proposal submission.

DISCLOSURES OF FOREIGN AFFILIATIONS OR RELATIONSHIPS TO FOREIGN COUNTRIES (VOLUME 7)

Small business concerns must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission. Please be aware that the Disclosures of Foreign Affiliations or Relationships to Foreign Countries WILL NOT be accepted as a PDF Supporting Document in Volume 5 of the DSIP proposal submission. Do not upload any previous versions of this form to Volume 5. For additional details, please refer to the STTR Program BAA.

DISCRETIONARY TECHNICAL AND BUSINESS ASSISTANCE (TABA)

The Air Force does not participate in the Discretionary Technical and Business Assistance (TABA) Program. Proposals submitted in response to DAF topics shall not include TABA.

AIR FORCE PROPOSAL EVALUATIONS

Proposals will be evaluated for overall merit in accordance with the criteria discussed in the FY26 STTR BAA. DAF is seeking varying technical/scientific approaches and/or varying and new technologies that would be responsive to the problem statement(s) and area(s) of interest in the topic. Multiple procurements are planned and anticipated to be awarded as a result of the topic, each proposal is considered a separate procurement and will be evaluated on its own merit, and that the Government may award all, some, or none of the proposals. Any per-award or per-topic funding caps are budgetary estimates only, and more or less funding may become available. Funding decisions are made with complete disregard to the other awards under the same topic.

In accordance with 15 USC 638(vv) (Section 4 of the SBIR and STTR Extension Act of 2022), and the Deputy Secretary of Defense Memorandum; Subject: Defense Small Business Innovation Research and Small Business Technology Transfer Due Diligence Program dated May 13, 2024, and the AF61-105_DAFGM 2025-1 "Department of the Air Force Guidance Memorandum to AFI 61-102, on DAF Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) Due Diligence Program", the DAF will review all proposals submitted in response to this BAA to assess security risks presented by small business concerns seeking a Federally funded award. The DAF will use information provided by the small business concern in response to the Disclosure of Foreign Affiliations or Relationships to Foreign Countries and the proposal to conduct a risk-based due diligence review on the cybersecurity practices, patent analysis, employee analysis, and foreign ownership of a small business concern, including the small business concern and employees of the small business concern to a foreign country, foreign person, foreign affiliation, or foreign entity. The DAF will also assess proposals utilizing open-source analysis and analytical tools, for the nondisclosures of the information set forth in 15 U.S.C. 638(g)(13). If DAF assesses that a small business concern has security risk(s), DAF will review the proposal, the evaluation, and the security risks and may decide not to select the proposal for award based upon a totality of the review.

MAJORITY OWNERSHIP IN PART BY MULTIPLE VENTURE CAPITAL, HEDGE FUND, AND PRIVATE EQUITY FIRMS

Small business concerns that are owned in majority part by multiple venture capital operating companies (VCOCs), hedge funds, or private equity funds are not eligible to submit applications or receive awards for Department of Air Force Topics.

PERFORMANCE OF WORK REQUIREMENTS AND LOCATION OF WORK

For both Phase I and Phase II, a minimum of 40% of each STTR award must be conducted by the awardee and a minimum of 30% of the effort must be performed by the single partnering Research Institution. Note, applicants and awardees may partner with multiple entities that separately meet the definition of a "Research Institution" as indicated in the STTR BAA. Applicants may use only one partnering Research Institution to meet STTR eligibility requirements. The DAF will not consider requests for deviations to these performance of work requirements.

All R/R&D work must be performed in the United States. Based on a rare and unique circumstance, the DAF may approve a particular portion of the R/R&D work to be performed or obtained in a country outside of the United States. The awarding Funding Agreement officer must approve each specific condition in writing. Applicants seeking this approval must make such a request with their initial proposal submission. The DAF will not consider these requests prior to proposal submission.

DAF USE OF SUPPORT CONTRACTORS

Restrictive notices notwithstanding, proposals may be handled for administrative purposes only, by support contractors. These support contractors may include, but are not limited to TEC Solutions, Inc., APEX, Oasis Systems, Riverside Research, Peerless Technologies, HPC-COM, Mile Two, Montech, Wright Brothers Institute, and MacB (an Alion Company). In addition, only Government employees and technical personnel from Federally Funded Research and Development Centers (FFRDCs) MITRE and Aerospace Corporations working under contract to provide technical support to AF Life Cycle Management Center and Space and Missiles Centers may evaluate proposals. All support contractors are bound by appropriate non-disclosure agreements. Contact the DAF SBIR/STTR CO Daniel J. Brewer (Daniel.Brewer.13@us.af.mil) with concerns regarding the use of support contractors.

PROPOSAL STATUS AND FEEDBACK

The Principal Investigator (PI) and Corporate Official (CO) indicated on the Proposal Cover Sheet will be notified by e-mail regarding proposal selection or non-selection. Small Businesses will receive a notification for each proposal submitted. Please read each notification carefully and note the Proposal Number and Topic Number referenced.

Automated feedback will be provided for Phase I proposals designated Not Selected. Additional feedback may be provided at the sole discretion of the DAF.

IMPORTANT: Proposals submitted to the DAF are received and evaluated by different organizations, handled by topic. Each organization operates within its own schedule for proposal evaluation and selection. Updates and notification timeframes will vary. If contacted regarding a proposal submission, it is not necessary to request information regarding additional submissions. Separate notifications are provided for each proposal.

The Air Force anticipates that all proposals will be evaluated and selections finalized within approximately 90 calendar days of solicitation close. Refrain from contacting the BAA CO for proposal status before that time.

Refer to the FY26 STTR BAA for procedures to protest the Announcement.
As further prescribed in FAR 33.106(b), FAR 52.233-3, Protests after Award should be submitted to: Air Force SBIR/STTR Contracting Officer Daniel J. Brewer, Daniel.Brewer.13@us.af.mil.

AIR FORCE SUBMISSION OF FINAL REPORTS

All Final Reports will be submitted to the awarding DAF organization in accordance with Contract instructions. Companies will not submit Final Reports directly to the Defense Technical Information Center (DTIC).

PHASE II PROPOSAL SUBMISSIONS

DAF organizations may request Phase II proposals while Phase I technical performance is ongoing or at any time after the conclusion of the period of performance. This decision will be based on the awardee's technical progress, as determined by an DAF Technical Point of Contact review using the Phase II review criteria outlined above.

Phase II is the demonstration of the technology found feasible in Phase I. Only Phase I awardees are eligible to submit a Phase II proposal. All Phase I awardees will be sent a notification with the Phase II proposal submittal date and detailed Phase II proposal preparation instructions. If the physical or email addresses or firm points of contact have changed since submission of the Phase I proposal, correct

information shall be sent to the AF SBIR/STTR One Help Desk. Phase II dollar values, performance periods, and proposal content will be specified in the Phase II request for proposal.

NOTE: The DAF primarily makes STTR Phase I and II awards as Firm-Fixed-Price contracts. However, awardees are strongly urged to work toward a Defense Contract Audit Agency (DCAA)-approved accounting system. If the company intends to continue work with the DoD, an approved accounting system will allow for competition in a broader array of acquisition opportunities, including award of Cost-Reimbursement types of contracts. Please address questions to the Phase II CO, if selected for award.

All proposals must be submitted electronically via DSIP by the date indicated in the Phase II proposal instructions. Note: Only ONE Phase II proposal may be submitted for each Phase I award.

AIR FORCE SBIR/STTR PROGRAM MANAGEMENT IMPROVEMENTS

The DAF reserves the right to modify the Phase II submission requirements. Should the requirements change, all Phase I awardees will be notified. The DAF also reserves the right to change any administrative procedures that will improve management of the DAF SBIR/STTR Program at any time.

TOPIC-SPECIFIC INFORMATION

Awards made under this topic will have a maximum value of $275,000.00. Proposals that exceed this maximum value will not be considered for evaluations or award. Awards made under this topic will have a maximum period of performance of six (6) months. Proposals that exceed this duration will not be considered for evaluations or award. Technical volumes submitted under this topic have a page/slide limit of 20 pages. Pages/slides in excess of this amount will not be considered by the DAF during evaluations.

DAF STTR BAA Topic Index
Release 1

DAF26TZ01-NV004 Low-Cost, Phased Array Antennas for Collaborative Jamming in sUAS Swarms

DAF26TZ01-NV004 TITLE: Low-Cost, Phased Array Antennas for Collaborative Jamming in sUAS Swarms

COMPONENT TECHNOLOGY PRIORITY AREA(S): Microelectronics

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop a low-cost, phased array antenna system and associated signal processing techniques for collaborative jamming applications using small to medium-sized sUAS swarms (3-7 units).

DESCRIPTION: This topic addresses the need for affordable and scalable jamming capabilities leveraging sUAS swarms. Instead of focusing on individual, high-power jammers, this STTR seeks to develop a collaborative jamming approach using multiple sUAS equipped with low-cost phased array antennas.
The key innovation is the development of a low-cost phased array antenna system that can be precisely controlled to focus jamming energy on specific targets. By coordinating the signals from multiple sUAS in a swarm, the effective jamming power can be significantly increased. The focus on low to medium-sized swarms (3-7 units) allows for manageable coordination and control strategies.
This approach offers several advantages over traditional jamming techniques, including:

- Increased jamming effectiveness through beamforming.
- Improved resilience through redundancy.
- Reduced risk of detection and counter-attack.
- Lower cost compared to high-power jamming systems.

PHASE I: Develop a concept for a low-cost phased array antenna system suitable for sUAS-based collaborative jamming. The proposal should demonstrate the feasibility of achieving beamforming, low cost, and performance suitable for disrupting target communication systems. Documentation should include preliminary designs, performance simulations, and component selection rationale.

PHASE II: Develop, integrate, and demonstrate a prototype collaborative jamming system using a swarm of 3-7 sUAS equipped with the developed phased array antennas. Specific goals include:

- Demonstrate effective beamforming and jamming of target communication signals.
- Develop a software interface for controlling and coordinating the phased array antennas within the swarm.
- Utilize open-source software and off-the-shelf hardware components where possible.

PHASE III DUAL USE APPLICATIONS: Successful development of low-cost phased array antennas and collaborative jamming techniques will have significant applications in both military and commercial sectors. Potential applications include:

- Disrupting illegal drone activity.
- Protecting critical infrastructure from cyber-attacks.
- Enhancing communication security in contested environments

REFERENCES:

1. Department of Defense, Unmanned Systems Integrated Roadmap, FY2013-2038; https://apps.dtic.mil/sti/tr/pdf/ADA592015.pdf
2. 2021 U.S. Department of Defense Counter-Small Unmanned Aircraft Systems Strategy; https://media.defense.gov/2021/Jan/07/2002561080/-1/-1/1/DEPARTMENT-OF-DEFENSE-COUNTER-SMALL-UNMANNED-AIRCRAFT-SYSTEMS-STRATEGY.PDF
3. Department of the Air Force Operational Imperatives; https://www.af.mil/Portals/1/documents/2023SAF/OPERATIONAL_IMPARITIVES_INFOGRAPHIC.pdf

KEYWORDS: Phased Array Antenna; Collaborative Jamming; sUAS Swarm; Beamforming; Electronic Warfare; Low-Cost; Agile EW

Defense Advanced Research Projects Agency (DARPA)
DoW 2026 Small Business Technology Transfer (STTR)
Annual Broad Agency Announcement (BAA)
Proposal Submission Instructions

INTRODUCTION

DARPA's mission is to create and prevent technological surprise for our national security. As part of this mission, DARPA makes high-risk, high-reward investments in science and technology that have the potential to disrupt current understandings and/or approaches. The topics that follow focus on technical domains important to DARPA's mission, pursuing innovative research concepts that fall within one of its technical offices. More information about DARPA's technical domains and research topics of interest can be found at https://www.darpa.mil/about/offices. DARPA offers free resources through DARPAConnect to help potential performers navigate DARPA, including "Tips for DARPA Proposal Success." Join DARPAConnect at www.DARPAConnect.us to leverage on-demand learning and networking resources.

The pace of discovery in both science and technology is accelerating worldwide, resulting in new fields of study and the identification of scientific areas ripe for small business utilization through the SBIR and Small Business Technology Transfer (STTR) programs. Small businesses are critical for developing technology to support national security. Proposers are encouraged to consider whether the Research/Research and Development (R/R&D) being proposed to Department of War (DoW) Components also has private sector potential, either for the proposed application or as a base for other applications.

Proposers responding to a topic in this BAA must follow all general instructions provided in the DoW STTR Program BAA. DARPA requirements, in addition to/or deviating from the DoW SBIR Program BAA, are provided in the instructions below. All DARPA STTR proposals must be submitted electronically through the Defense SBIR/STTR Innovation Portal (DSIP) as described in the Proposal Preparation and Submission sections of these instructions. It is recommended that organizations register as soon as possible upon identification of a proposal opportunity to avoid delays in the proposal submission process. Proposers are encouraged to submit proposals as early as possible to avoid unexpected delays due to a high volume of traffic during the final hours before a BAA closes. ***DARPA will not accept any late proposals.***

Proposers are encouraged to thoroughly review the DoD Program BAA and register for the Defense SBIR/STTR Innovation Portal (DSIP) Listserv to remain apprised of important programmatic and contractual changes.

- The DoD Program BAA is located at: https://www.dodsbirsttr.mil/submissions/solicitation-documents/active-solicitations. Please select the tab for the appropriate BAA cycle.
- Register for the DSIP Listserv at: https://secure.campaigner.com/CSB/Public/Form.aspx?fid=667492&ac=g9gk.

Specific questions on the administration of the DARPA Program and these proposal preparation instructions should be directed to DARPA Small Business Programs Office at SBIR_BAA@darpa.mil. DSIP Topic Q&A will NOT be available for these DARPA topics. Technical questions related to improving the understanding of a topic's requirements must be submitted NLT seven days

PHASE I PROPOSAL GUIDELINES

The Defense SBIR/STTR Innovation Portal (DSIP) is the official portal for DoW SBIR/STTR proposal submission. Proposers are required to submit proposals via DSIP; proposals submitted by any other means will be disregarded. Detailed instructions regarding registration and proposal submission via DSIP are provided in Appendix A.

Current Release Award Structure by Topic

Topic Number	Phase I			
	Technical Volume		Award Amount	Period of Performance (PoP)
	White Paper	Slide Deck		
DPA26TZ01-NV001	10 pages	5 pages	$300,000	6 months

Technical Volume (Volume 2) – White Paper & Slide Deck Format
The white paper shall not exceed 10 pages, and the slide deck shall not exceed five (5) pages. For information on the content of each of these elements of the Technical Volume and the commercialization strategy, please see Appendix A: DARPA PHASE I INSTRUCTIONS.

Cost Volume (Volume 3)
Please see the chart above for award amounts listed by topic. Proposers are required to use the applicable cost proposal template as instructed in Appendix A.

Content of the Cost Volume
Proposers should refer to the DARPA Phase I Proposal Instructions, provided in Appendix A.

Please review the Percentage of Work (POW) calculation details included in the DoW Program BAA. DARPA cannot accept deviations from the POW requirements for STTR proposals.

Company Commercialization Report (CCR) (Volume 4)
Completion of the CCR as Volume 4 of the proposal submission in DSIP is required. Please refer to the DoW STTR Program BAA for full details on this requirement. Information contained in the CCR will not be considered by DARPA during proposal evaluations.

Supporting Documents (Volume 5)
In addition to the documents required by DoW, small businesses may also submit additional documentation to support the Technical Volume (Volume 2) and the Cost Volume (Volume 3) in Volume 5. See Appendix A for required certifications that must be included in Volume 5. For additional information, see the STTR 26.D Annual Program BAA at https://www.dodsbirsttr.mil/submissions/solicitation-documents/active-solicitations.

Fraud, Waste and Abuse Training (Volume 6)
Fraud, Waste and Abuse training material can be found in the Volume 6 section of the proposal submission module in DSIP and must be thoroughly reviewed once per year to proceed with proposal submission.

Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)
Small business concerns must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission.

PHASE II PROPOSAL GUIDELINES
Phase II proposals may only be submitted by Phase I awardees. Should DARPA have funding available and decide to proceed with a Phase II, proposers awarded a Phase I contract will be eligible to submit a proposal for Phase II and will be contacted by the DARPA Small Business Programs Office at the appropriate time during their Phase I period of performance. Phase II proposals will be evaluated in accordance with the evaluation criteria listed in the applicable DoW STTR Program BAA. Phase II selection(s) are at the sole discretion of the Government and are subject to funding availability and Phase I performance. Phase II Instructions are available at https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-topics.

Current Release Award Structure by Topic

Topic Number	Phase II				
	Technical Volume	Award Amount	Period of Performance (PoP)	Option Amount	Option PoP
DPA26TZ01-NV001	25 pages	$1,500,000	18 months	$500,000	6 months

Technical Volume (Volume 2)
The Technical Volume shall not exceed 25 pages. The Phase II commercialization strategy shall not exceed 5 pages. This should be the last section of the Technical Volume and is included in the 25-page total. Any pages in the Technical Volume over 25 pages will not be considered in proposal evaluations.

DISCRETIONARY TECHNICAL AND BUSINESS ASSISTANCE (TABA)
DARPA does not offer TABA funding.

EVALUATION AND SELECTION
All proposals will be evaluated in accordance with the evaluation criteria listed in the Dow STTR Program BAA. DARPA will conduct an evaluation of each conforming proposal. Proposals that do not comply with the requirements detailed in this BAA and the research objective(s) of the corresponding topic are considered non-conforming and, therefore, will not be evaluated nor considered for award.

Using the evaluation criteria, the Government will evaluate each proposal in its entirety, documenting the strengths and weaknesses relative to each evaluation criterion. Based on these identified strengths and weaknesses, the Government will determine the proposal's overall selectability for funding. Proposals will not be evaluated against each other during the evaluation process but rather evaluated on their own individual merit to determine how well the proposal meets the criteria stated in this BAA and the corresponding topic.

Awards will be made to proposers whose proposals are determined to be the most advantageous to the Government, consistent with instructions and evaluation criteria specified in the DoW STTR Program BAA and availability of funding.

For the purposes of this proposal evaluation process, a selectable proposal is defined as follows:

Selectable: A selectable proposal is a proposal that the Government has evaluated against the evaluation criteria listed in the DoW STTR Program BAA and DARPA topic, and the strengths of the overall proposal outweigh its weaknesses. Additionally, there are no accumulated weaknesses that would require extensive negotiations and/or a resubmitted proposal.

For the purposes of this proposal evaluation process, a non-selectable proposal is defined as follows:

Non-Selectable: A proposal is considered non-selectable when the Government has evaluated the proposal against the evaluation criteria listed in the DoW STTR Program BAA and DARPA topic, and the strengths of the overall proposal do not outweigh its weaknesses.

Proposing firms will be notified of selection or non-selection status for a Phase I award within 90 calendar days of the closing date of the BAA. The Corporate Official (CO) indicated on the Proposal Cover Sheet will be notified via e-mail regarding proposal selection or non-selection. In accordance with the Small Business Administration (SBA) SBIR/STTR Policy Directive, Appendix I, paragraph 4, Method of Selection and Evaluation Criteria, subparagraph (d) Release of Proposal Review Information, DARPA will provide a technical evaluation narrative to the proposer for each proposal submitted in response to a topic. An informal feedback session may be requested by the proposing firm via e-mail at sbir@darpa.mil. The informal feedback is provided at the sole discretion of DARPA.

It is the policy of DARPA to treat all proposals as source selection information and to disclose their contents only for the purpose of evaluation. Restrictive notices notwithstanding, during the evaluation process, submissions may be handled by support contractors for administrative purposes and/or to assist with technical evaluation. All DARPA support contractors are expressly prohibited from performing DARPA-sponsored technical research and are bound by appropriate non-disclosure agreements. Input on technical aspects of the proposals may be solicited by DARPA from other Government and/or non-Government consultants/experts who are strictly bound by the appropriate non-disclosure requirements. No submissions in response to the BAA will be returned. Upon completion of the evaluation and selection process, an electronic copy of each proposal received will be retained at DARPA.

Proposal titles, abstracts, anticipated benefits, and keywords of proposals that are selected for contract award will undergo a DARPA Policy and Security Review. Proposal titles, abstracts, anticipated benefits, and keywords are subject to revision and/or redaction by DARPA. Final approved versions of proposal titles, abstracts, anticipated benefits, and keywords may appear on the DoW SBIR/STTR awards website and/or the SBA's SBIR/STTR award website (https://www.sbir.gov/awards).

Refer to the DoW SBIR Program BAA for procedures to protest this BAA. As further prescribed in the Federal Acquisition Regulation (FAR 33.106(b), FAR 52.233-3), protests regarding the selection decision should be submitted to:

DARPA
Contracts Management Office (CMO)
675 N. Randolph Street
Arlington, VA 22203
E-mail: CMO_SBIRProtests@darpa.mil and sbir@darpa.mil

AWARD AND CONTRACT INFORMATION

1. General Award Information

Multiple awards are anticipated. DARPA may award FAR-based Government contracts (Firm-Fixed Price or Cost-Plus Reimbursement) or Other Transactions (OT) for Prototypes agreement (under the authority of 10 U.S.C. § 4022) subject to approval of the Contracting Officer or Agreements Officer, respectively. The resources made available under each topic issued under this BAA will depend on the quality of the proposals received and the availability of funds.

The Government reserves the right to select for negotiation all, some, one, or none of the proposals received in response to this BAA and to make awards with or without communications with proposers. Additionally, the Government reserves the right to award all, some, one, or none of the options on the contract(s)/agreement(s) based on available funding and the performer's technical performance. If warranted, portions of resulting awards may be segregated into pre-priced options. Additionally, DARPA reserves the right to accept proposals in their entirety or to select only portions of proposals for award. In the event that DARPA desires to award only portions of a proposal, negotiations may be opened with that proposer. The Government reserves the right to fund proposals in phases with options for continued work, as applicable.

The Government reserves the right to request any additional, necessary documentation once it makes the award instrument determination. The Government reserves the right to remove a proposal from award consideration should the parties fail to reach an agreement on award terms, conditions, and price within a reasonable time, and/or the proposer fails to provide requested additional information within three (3) business days.

It is DARPA's goal to award SBIRs and STTRs as expeditiously as possible. In an effort to do so, please review and ensure the DARPA SBIR/STTR Pre-Award Checklist located on the DARPA Small Business website at https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-overview/ is completed prior to being selected for award.

In all cases, the Government Contracting Officer reserves the right to select award instrument type, regardless of the instrument type proposed, and to negotiate all instrument terms and conditions with selectees. DARPA will apply publication or other restrictions, as necessary, if it determines that the research resulting from the proposed effort will present a high likelihood of disclosing performance characteristics of military systems or manufacturing technologies that are unique and critical to defense. Any award resulting from such a determination will include a requirement for DARPA permission before publishing any information or results on the program. For more information on publication restrictions, see the DoW STTR Program BAA.

All proposers that wish to consider an OT award should carefully read the following information:

The flexibility of the OT award instrument is beneficial to the program because the performer will be able to apply its commercial best practices as required to carry out the research project that may be outside of the FAR process-driven requirements. Streamlined practices will be used, such as milestone-driven performance, intended to reduce time and effort on award administration tasks and permit performers to focus on the research effort and rapid prototyping. Because of this ability, OTs provide the Agreements Officer the flexibility to create an award instrument that contains terms and conditions that promote commercial transition, reduce some administratively burdensome acquisition regulations, and meet SBIR/STTR program goals.

Please see https://acquisitioninnovation.darpa.mil/what-are-ots for more information on OTs.

2. **Cybersecurity Maturity Model Certification (CMMC)**

Firms engaging in Controlled Unclassified Information (CUI), Export Controlled (EXPT), or International Traffic in Arms Regulations (ITAR) work for DARPA must have CMMC Level 2 self-assessment certification. Please see https://www.sprs.csd.disa.mil/nistsp.htm for more information. Many resources exist to assist companies with meeting this criterion, one of which is Project Spectrum (http://www.projectspectrum.io). Users can create accounts and gain access to useful tools, including educational videos and material, as well as a cyber self-assessment.

3. **Transition and Commercialization Support Program (TCSP)**

DARPA will provide services to Phase II awardees upon contract execution through TCSP at no cost to awardees. The TCSP goal is to maximize the potential for SBIR/STTR companies to move their technology beyond Phase II and into other research and development programs for further maturity or into solutions or products for DoW acquisition programs, other Federal programs, and/or the commercial market. Please visit https://www.darpa.mil/work-with-us/communities/small-business/transition-commercialization for more information on DARPA TCSP.

4. **Embedded Entrepreneurship Initiative (Phase II only)**

Awardees of Phase II STTR funding pursuant to this BAA may be eligible to participate in the DARPA Embedded Entrepreneurship Initiative (EEI). An invitation to participate in EEI is at the sole discretion of the Government based on evaluation of technical and commercial factors and subject to program balance and the availability of funding. EEI is a limited scope program offered by DARPA, at the Agency's discretion, to a small subset of awardees. The goal of DARPA's EEI is to increase the likelihood that DARPA-funded technologies take root in the U.S. and provide new capabilities for national defense. EEI supports DARPA's mission by accelerating the transition of innovations out of the lab and into new capabilities for the DoW. EEI investment supports development of a robust and deliberate Go-to-Market strategy for DARPA-funded advanced technology, into high-value products and capabilities for the government and commercial markets, and positions DARPA awardees to attract U.S. private investment. The following is for informational and planning purposes only and does not constitute solicitation of proposals to EEI.

There are three elements to DARPA's EEI: (1) A Senior Commercialization Advisor (SCA) from DARPA who works with the Program Manager (PM) to examine the business case for the awardee's technology and uses commercial methodologies to identify steps toward achieving a successful transition of technology to the government and commercial markets; (2) Connections to potential U.S. industry and private investor partners via EEI's Investor Working Groups; and (3) Additional funding to hire an embedded entrepreneur to achieve specific milestones in a Go-to-Market strategy for transitioning the technology into products that serve both defense and commercial markets. This embedded entrepreneur's qualifications should include business experience within the target industries of interest, experience in commercializing early-stage technology, and the ability to communicate and interact with technical and non-technical stakeholders, and customers. Funding for EEI is typically no more than $310,000 per awardee over the duration of the award. An awardee will attend one commercialization workshop and also may apportion EEI funding to hire more than one embedded entrepreneur, if achieving the milestones requires a unique expertise that can be obtained without exceeding the awardee's total EEI funding.

EEI Application Process:

After receiving an SBIR/STTR Phase II award, awardees interested in being considered for EEI should notify their DARPA PM during the period of performance. If the DARPA PM determines that EEI could be of benefit to transition the technology to product(s) the Government needs, the PM will refer the performer to the DARPA Commercial Strategy Team. The SCA will then contact the performer, assess fitness for EEI and determine, in consultation with the PM and Commercial Strategy Team, whether to invite the performer to participate in the EEI. Factors that are considered in determining fitness for EEI

include DoW/Government need for the technology; competitive approaches to enable a similar capability or product; risks and impact of the Government's being unable to access the technology from a sustainable source; Government and commercial markets for the technology; cost and affordability; manufacturability and scalability; supply chain requirements and barriers; regulatory requirements and timelines; intellectual property and Government use rights, and available funding.

After SCA review, the Commercial Strategy Team may request the SBIR/STTR awardee to submit additional tasks for review.

EEI awards are at the sole discretion of DARPA and are subject to program balance and the availability of funding. For more information, please refer to the EEI website https://www.darpa.mil/about/offices/commercial-strategy.

ADDITIONAL INFORMATION

DARPA intends to use electronic mail for all correspondence regarding these topics. Questions related to the technical aspect of the research objectives and awards specifically related to a topic should be e-mailed to SBIR_BAA@darpa.mil. Please reference the topic number in the subject line. All questions must be in English and must include the name, e-mail address, and the telephone number of a point of contact.

DARPA will attempt to answer questions in a timely manner; however, questions submitted within seven (7) calendar days of the proposal due date listed herein may not be answered. DARPA will post a consolidated Frequently Asked Questions (FAQ) document. To access the posting please visit: https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-topics Under the topic number summary, there will be a link to the FAQ. The FAQ will be updated on an ongoing basis until one week prior to the proposal due date.

Technical support for the Defense SBIR/STTR Innovation Portal (DSIP) is available Monday through Friday, 9:00 a.m. – 5:00 p.m. ET. Requests for technical support must be e-mailed to DoDSBIRSupport@reisystems.com with a copy to SBIR_BAA@darpa.mil.

DARPA
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DPA26TZ01-NV001 TITLE: MEDICAL SWARM ROBOTICS FOR EXTRACTION AND LIFE-SAVING INTERVENTIONS

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COMPONENT TECHNOLOGY PRIORITY AREA(S): Biotechnology, Human-Machine Interfaces, Trusted AI and Autonomy

PROJECTED CMMC LEVEL REQUIREMENT: Level 1

OBJECTIVE: Develop and demonstrate small-robotic swarms capable of autonomous battlefield medical assistance, including short casualty movement, hemorrhage control, fracture stabilization, and medication delivery.

DESCRIPTION: This topic addresses a critical battlefield medical need through the development of innovative swarm-based small-robotic systems capable of autonomous medical assistance to incapacitated and difficult to reach casualties. Future Large Scale Combat Operations (LSCO) predict massive casualty incidents, delayed evacuation, and insufficient capacity of the medical system, especially from the point-of-injury to Role 1 medical care [1]. With a delayed medical response casualties have a high chance of dying due to lack of hemorrhage control which is the leading cause of potentially survivable death in both battlefield and civilian trauma cases prehospital [2,3]. Autonomous medical care may be essential for saving lives in these future contexts, especially for casualties who are unable to treat themselves, have no buddy aide in proximity, or are in inaccessible areas for human medical response. This topic calls for a solution of an autonomous, self-deploying, wound assessing, swarm-capable, self-linking, mobile robotic solution to assist reaching and moving casualties and perform life-saving interventions (LSIs) at the point-of-need. Small swarm robotics provide advantages in their ability to access difficult to reach casualties obscured by rubble and terrain, can adapt by conjoining or detaching to meet the need of a detected casualty, and are ideal for limited space during unmanned evacuation assistance. We believe this is achievable due to the recent advancements in swarm, self-assembling, and mobile robotics, as well as robotics for medical applications [4–7]. For this topic, the novel robotic system should demonstrate at least two of the following four essential capabilities, one form each category:

Extraction:

1) Movement of a casualty a short distance (10m) and/or onto a SKED or litter
2) Stabilization of a fractured limb through the entanglement of rigid structures

Treatment:

1) Manage massive extremity or junctional hemorrhage control
2) Delivery of medications through intramuscular injection or placement of intraosseous needle

First, a solution can demonstrate the ability to leverage a swarm architecture to coordinate the capability to drag/move a casualty a short distance and or position the casualty onto an extraction SKED or litter. Coordinated swarm maneuvers would allow smaller robotics capable of navigating tight spaces lift or drag a casualty together when a single unit may not have the dragging capacity otherwise.

Second, a solution can demonstrate protective limb stabilization by entangling multiple robotic units around or along a limb. By interlocking systems, a solution should provide protective bracing around an injured body-part to prevent further injury during casualty movement.

Third, a solution can demonstrate the ability to self-arrange and reassemble into shapes to provide massive hemorrhage control. The goal will be to create a “smart tourniquet” capable of autonomously clamping around injured limbs to stop arterial blood flow as well as apply sufficient pressure and coverage over a junctional wound. The solution will need the necessary sensing and intelligence to identify and locate the hemorrhage injury and advanced capabilities and swarm architecture to reassemble into a hemostatic tool.

Fourth, a solution can demonstrate the ability to deliver medications to a casualty either through intramuscular injection or establishing an intraosseous infusion. Due to the intended smaller size, the medical robotic solution could be an ideal assistant in the tight quarters of unmanned evacuation vehicles. The ability to provide medications and fluids enables higher qualities of care by autonomous and unmanned systems.

The proposed design for a robotic solution should aim to accomplish two of the four tasks with at least one from each category of extraction and treatment while being as small, light, and portable as possible. Proposals that can accomplish more tasks will be reviewed more favorably. The topic is not prescribing a design choice, and proposers are welcome to propose any form factor of robotic system that provides mobility, a swarm architecture, and self-assembly and deployment. The desired utilization of these robots is for individual/self-aid in a frontline environment with the aim to fit into an Individual First Aid Kid (IFAK) or lightweight enough to be deployed via drone-swarm. By leveraging a modular, interconnected, swarm-capable architecture the design should allow for mobility across dynamic environments, adaptability to various anatomies and injuries, and low-cost manufacturing.

PHASE I: Proposals are required to have selected design targets prior to Phase I acceptance, i.e., Phase I is not an exploratory phase into multiple designs. Phase I will include development of subcomponents of the design to demonstrate feasibility into the overall Phase II proposed solution. While not necessary to be integrated into a fully realized prototype system, Phase I will require the demonstration into feasibility of the following system subcomponents: 1) a modular swarm architecture and communication capability, 2) sensing capability to identify injury and anatomy, 3) mobility of the units across dynamic terrain and over/across the human body, 4) design feasibility into achieving treatment capabilities, 5) design feasibility of interlocking units, shape-changing, & rigid stability. Demonstrations can be conducted in lab environments and on hemorrhage control trainers / patient manikins. Deliverables in Phase I include reports and video demonstrations when milestones are made.
Milestones

- Month 1: Design, build, test, and learn plan
- Month 3: Mid-point subcomponent design review
- Month 6: Subcomponent feasibility demonstrations

Key Deliverables

- Month 1: Design, build, test, and learn plan
- Month 3: Mid-point design review report
- Month 6: A comprehensive subcomponent feasibility report and demonstrations (videos) of functional subcomponents:

1. Modular swarm architecture and communication capability
2. Sensing capability to identify injury and anatomy
3. Mobility of units across dynamic terrain and over/across the human body
4. Feasibility into design to achieve treatment capabilities
5. Feasibility into design of interlocking units, shape-changing & rigid stability

PHASE II: Phase II requires proposers to develop a fully realized physical prototype of the swarm-capable small robotic system and demonstrate its ability to perform the four listed medical interventions on high-fidelity phantoms in operationally relevant environments. All developed subcomponents developed under Phase I will be advanced and integrated into features of a realized system. Advancements must be made to the components throughout Phase II to enable more effective capability in LSIs, and ruggedization towards fielding the technology. Sensing capabilities will need to be added to understand if proper tourniquet application has been made across varying limb sizes. Proposers must also develop sensing capabilities to ensure proper pressure is applied to wounds and monitoring if hemostasis is achieved. Additionally advanced interlocking strategies to stabilize fractures must be developed.

At the conclusion of Phase II a physical demonstration must be conducted of the proposed solution autonomously accomplishing two of the four medical interventions by leveraging its swarm architecture and self-reassembling capabilities. Performers must physically demonstrate one procedure from each of the following categories:
Extraction:
1) Movement of a casualty a short distance (10m) and/or onto a SKED or litter
2) Stabilization of a fractured limb through the entanglement of rigid structures

Treatment:
1) Manage massive extremity or junctional hemorrhage control
2) Delivery of medications through intramuscular injection or placement of intraosseous needle

Demonstrations must be conducted in operationally relevant environments on either perfused cadavers, animal models, or high-fidelity medical training phantoms. The system by the end of Phase II must be ruggedized to operate outdoors and in unideal weather conditions.
Phase II will also require commercialization and transition planning along with technology development. Throughout the phase, the proposers must collaborate with military end-users to refine operational requirements and deployment scenarios of their developed solution. Manufacturing and scaling plans for production must also be developed before the end of the Phase including designing packaging and deployment systems for rapid battlefield deployment. Since this is a medical device intended to interact with people, an FDA acceptance plan must be developed in Phase II involving a regulatory pathway and protocols for safe operation. The final report must also include technology transfer documents outlining planned opportunities for commercial and military applications.
The deliverables for Phase II will include the ruggedized final prototype system consisting of multiple modular units with integrated enhanced capabilities, a comprehensive testing and validation report outlining the final demonstration, the technology transfer plan with commercialization plans, FDA regulatory pathway, and manufacturing and scaling strategy.

Milestones
- Month 1: Design, build, test, and learn plan
- Month 6: Fabrication of all necessary subcomponents
- Month 9: Physical testing of subcomponents and subcomponent capabilities
- Month 12: Integrated prototype
- Month 15: Successful demonstration of Extraction task with integrated prototype
- Month 18: Successful demonstration of Treatment task with integrated prototype

Key Deliverables
- Month 1: Design, build, test, and learn plan
- Month 6: Report of subcomponent fabrication
- Month 9: Video demonstration of subcomponent testing
- Month 12: Report and design documentation of integrated prototype
- Month 15: Video demonstration of Extraction task with integrated prototype
- Month 18: Video demonstration of Treatment task with integrated prototype.

Comprehensive report including: testing and validation reports, technology transition plan with commercialization pathway with description of integration specifications for existing military medical systems, and discussion of manufacturing and logistics framework for production scaling.

PHASE III DUAL USE APPLICATIONS: Phase III offers the opportunity for the proposers to apply secured outside funding, not from the SBIR program, to advance and mature the technology further for commercial and Government use cases. Many commercial applications could be applied for an advanced

swarm-like medical response system specifically in disaster response. Collapsed buildings, fires, and hazardous chemicals can make reaching civilian casualties impossible outside of the means of robotic and autonomous systems. Initial LSIs can provide the necessary time for stabilization until medical response teams arrive and continue care. For the military there are expectations that LSCO environments will put a strain on the current doctrine of casualty response time and alternative and advanced solutions must be applied to manage massive hemorrhage. Possible military transition partners could include Project Manager Soldier Medical Devices (PMSMD) or Program Executive Office Operational Medicine (PEO OpMED).

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KEYWORDS: Swarm robotics, battlefield medicine, autonomous systems, hemorrhage control, smart tourniquet, medical robotics, entanglement dynamics, medical automation, emergency response, combat casualty care.

Appendix A: DARPA PHASE I PROPOSAL INSTRUCTIONS

I. Introduction

A complete proposal submission consists of:

Volume 1: Proposal Cover Sheet
Volume 2: Technical Volume
Volume 3: Cost Volume
Volume 4: Company Commercialization Report
Volume 5: Supporting Documents
- a. Verification of Eligibility of Small Business Joint Ventures (Attachment 3), if applicable
- b. Data Rights Assertions, if applicable
- c. Allocation of Rights
- d. Other supporting documentation

Volume 6: Fraud, Waste and Abuse Training
Volume 7: Disclosures of Foreign Affiliations or Relationships to Foreign Countries

The Defense SBIR/STTR Innovation Portal (DSIP) provides a structure for building the proposal volumes and submitting a consolidated proposal package. If this is your first time submitting an STTR proposal using DSIP, please review detailed training guides at https://www.dodsbirsttr.mil/submissions/learning-support/training-materials. It is the responsibility of the proposing firm to ensure that a complete proposal package is certified and submitted by the close date listed in the topic to which they are responding. ***DARPA will not accept late proposals.***

To assist in proposal development, templates for Volume 2: Technical Volume and Volume 3: Cost Volume have been provided as attachments on the DARPA Small Business website at https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-topics. Use of the DARPA Cost Proposal template is mandatory.

Proposers should ensure that they have an accurate and active entity registration on SAM.gov, and those engaging in International Traffic in Arms Regulations (ITAR) or Controlled Unclassified Information (CUI) work for DARPA must have Level 2 CMMC certification. Please see https://www.sprs.csd.disa.mil/nistsp.htm for more information. Many resources exist to assist companies with meeting this criterion, one of which is Project Spectrum (http://www.projectspectrum.io). Users can create accounts and gain access to useful tools, including educational videos and material, as well as a cyber self-assessment.

II. Proprietary Information

Proposers that include in their proposals data that they do not want disclosed to the public for any purpose, or used by the Government except for evaluation purposes, shall follow instructions in the DoD STTR 26.TZ BAA regarding marking propriety proposal information.

III. Phase I Proposal Instructions

a. Classification and Marking of Proposal Submission

1. **All proposals are required to be UNCLASSIFIED or CUI. Do not include any classified information in your proposal submission.**

2. Do not include any proprietary information (PROPIN) on the Proposal Coversheet (Volume 1), as this may be released publicly, if selected for award.

3. PROPIN or any other CUI information may be included in the Technical Volume (Volume 2), as needed. Ensure that the document is marked appropriately as CUI with the appropriate CUI Control Block on the first page of Volume 2.

4. The Cost Volume (Volume 3) should be marked CUI for PROPIN as it includes the proposer's budget/cost information.

5. Volumes 4-7 should be marked as appropriate based on content included in each volume. For additional CUI marking information refer to https://www.archives.gov/cui/registry/category-list to properly identify the categories of CUI.

b. Proposal Cover Sheet (Volume 1)

The Cover Sheet must include a brief technical abstract of no more than 3000 characters that describes the proposed research and development project with a discussion of anticipated benefits and potential commercial applications. **Do not include proprietary or classified information in the Proposal Cover Sheet**. If your proposal is selected for award, the technical abstract and discussion of anticipated benefits may be publicly released.

c. Format of the Technical Volume (Volume 2) – White Paper & Slide Deck Format

1. The Technical Volume must include two parts, PART ONE: white paper, and PART TWO: slide deck, combined as a single Portable Document Format (PDF) for upload to DSIP.

2. Type of File: The Technical Volume must be a single PDF file, including graphics. Perform a virus check before uploading the Technical Volume file. If a virus is detected, it may cause rejection of the proposal. Do not lock or encrypt the uploaded file. Do not include or embed active graphics such as videos, moving pictures, or other similar media in the document.

3. Length: The length of the white paper shall not exceed ten (10) pages, and the slide deck shall not exceed five (5) pages/slides. The Government will not consider pages in excess of the page count limitations.

4. Layout: Number all pages of your proposal consecutively. Font size should not be smaller than 10-point on standard 8-1/2" x 11" paper with one-inch margins. The header on each page of the Technical Volume should contain your company name, topic number, and proposal number assigned by DSIP when the Cover Sheet was created. The header may be included in the one-inch margin.

d. Content of the Technical Volume (Volume 2) – White Paper & Slide Deck Format

See **Section II** of the Phase I Template – Volume 2: Technical Volume at https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-topics for the slide deck and white paper template.

White Paper (not to exceed 10 pages). Provide the following information:

1. Goals and Impact: Clearly describe what is being proposed and what difference it will make (qualitatively and quantitatively), including a brief discussion on how this directly relates to the topic.

2. Technical Plan: Provide an explicit, detailed description of the Phase I approach. The Statement of Work should indicate what tasks are planned, how and where the work will be conducted, a schedule of major events, and the final product(s) to be delivered. The Phase I effort should attempt to determine the technical feasibility of the proposed concept. The methods planned to achieve each objective or task should be discussed explicitly and in detail.

3. Management and Capabilities: Designate key personnel who will be involved in the Phase I effort. Provide a brief summary of expertise of the team, including subcontractors and key personnel. Describe the organizational experience in this technology area, previous work not directly related to the proposed effort but similar, existing intellectual property required to complete the project, and any specialized facilities to be used as part of the project. List Government-furnished materials or data assumed to be available. Describe any specialized facilities to be used as part of the project, the extent of access to these facilities, and any biological containment, biosafety, and certification requirements.

4. Transition and Commercialization Plan (not to exceed five (5) pages of the 10-page white paper):
    a) Describe the commercial product or DoW system to be developed.
    b) Discuss the potential end users – DoW, Federal, and/or private sector customers. Discuss your business model for this technology (i.e., how do you anticipate generating revenue with this technology?). Who are you selling to directly or indirectly, a supplier, an integrator, or an end user?
    c) Describe your company's funding history. Discuss how much additional funding above this proposed effort (include additional required technology development, staffing requirements, infrastructure requirements, intellectual property (IP) strategy costs, etc.) will be required to bring this technology to market and how you anticipate going about getting that funding (e.g., Government S&T contracts, investment).
    d) Describe the timeline to maturity for sales or transition to an end user. Describe your IP strategy.
    e) Describe the technology, market, team and business risks associated with this proposed effort and your plan to mitigate these risks.

Slide Deck (not to exceed five (5) slides). Provide the following information (convert the completed deck to a PDF and attach it to the white paper):

1. What are you trying to do and how does this directly relate to the topic?

2. Technology and commercial product: Specifically, what are you proposing to produce – software, system, application? Be specific on what your proposed technology development is targeting as an end state.

3. How is the technology approached today? Who is doing the research, development and delivering products/services? What are the current limitations in the technology and commercial marketplaces?

4. Management: Overview of team, facilities and qualifications.

5. Technical summary quad chart: Use template provided at https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-topics.

NOTE: All letters of recommendation, CVs, and Data Rights Assertions can be loaded in Volume 5: Supporting Documents.

Advocacy Letters (OPTIONAL)* Feedback received from potential Commercial and/or DoW customers and other end-users regarding their interest in the technology to support their capability gaps. Advocacy letters that are faxed or e-mailed separately will NOT be accepted.

Letters of Intent/Commitment (OPTIONAL)* Relationships established, feedback received, support and commitment for the technology with one or more of the following: Commercial customer, DoW Program Management, Program Executive Office (PEO), a Defense Prime, or vendor/supplier to the Primes and/or other vendors/suppliers identified as having a potential role in the integration of the technology into fielded systems/products or those under development. Letters of Intent/Commitment that are faxed or e-mailed separately will NOT be accepted.

*Advocacy Letters and Letters of Intent/Commitment are optional and should ONLY be submitted to substantiate any transition or commercialization claims made in the commercialization strategy. Please DO NOT submit these letters just for the sake of including them in your proposal. These letters DO NOT count against any page limit.

In accordance with section 3-102d of DoD Directive 5500.07, Joint Ethics Regulation, May 15, 2024, letters from Government personnel will NOT be considered during the evaluation process.

e. Format of Cost Volume (Volume 3)

Proposers are required to use the Phase I – Volume 3: Cost Proposal Template (Excel Spreadsheet) provided at https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-topics.

f. Content of the Cost Volume (Volume 3)

Some items in the Cost Breakdown Guidance below may not apply to the proposed project. If such is the case, there is no need to provide information on every item.

For Phase I proposals, proposers should NOT provide documentation to substantiate how all proposed costs were derived. However, proposers should be prepared to provide such documentation should the Contracting Officer request this documentation. If any substantiating documentation is requested by the Contracting Officer, it is important to respond as quickly as possible to the request as to not delay contract negotiation.

Examples of substantiating documentation are as follows: If you proposed travel cost to attend a project-related meeting or conference, and used a travel website to compare flight costs, include a screen shot of the comparison. Similarly, if you proposed to purchase materials or equipment, and used the internet to search for the best source, include your market research for

those items. You do not necessarily have to propose the cheapest item or supplier, but you should be able to explain your decision to choose one item or supplier over another. It's important to provide enough information to allow contracting personnel to understand how the proposer plans to use the requested funds.

Cost Breakdown Guidance:

- List all key personnel by name as well as by number of hours dedicated to the project as direct labor.
- Special tooling and test equipment and material cost may be included. The inclusion of equipment and material will be carefully reviewed relative to need and appropriateness for the work proposed. The purchase of special tooling and test equipment must, in the opinion of the Contracting Officer, be advantageous to the Government and should be related directly to the specific topic. These may include such items as innovative instrumentation and/or automatic test equipment. Title to property furnished by the Government or acquired with Government funds will be vested with DARPA; unless it is determined that transfer of title to the contractor would be more cost effective than recovery of the equipment by the DARPA.
- Cost sharing is permitted for proposals under this BAA; however, cost sharing is not required, nor will it be an evaluation factor in the consideration of a proposal.
- If Subcontractors will be performing Fundamental Research under the effort, please incorporate the following into proposal: 1) a separate statement of work (SOW) outlining the specific work that the proposer finds to qualify as Fundamental Research; OR 2) Within Prime contractor SOW identify which tasks are to be performed that are fundamental research.
- Proposers should complete both tabs within the Cost Proposal Spreadsheet (Cost Model & Milestone Chart)

For more information about cost proposals and accounting standards associated with contract awards, see the Defense Contract Audit Agency (DCAA) publication titled "Audit Process Overview – Information for Contractors" at http://www.dcaa.mil.

Please note, a separate, more detailed cost proposal spreadsheet will be provided for any Phase II Proposals.

g. Company Commercialization Report (Volume 4)

The Company Commercialization Report (CCR) allows companies to report funding outcomes resulting from prior SBIR and STTR awards. The Company Commercialization Report (CCR) is required for Phase I and Direct to Phase II proposals. Please refer to the DoW STTR Program BAA for full details on this requirement. Information contained in the CCR will not be considered by DARPA during proposal evaluations.

h. Supporting Documents (Volume 5)

In addition to required DoW documentation and certifications, small businesses may also submit additional documentation to support the Technical Volume (Volume 2) and the Cost Volume (Volume 3) in Volume 5. See Introduction for **required** certifications that must be included in Volume 5. The template for Data Rights Assertions is included in the Phase I Template – Volume 2: Technical Volume at https://www.darpa.mil/work-with-us/communities/small-business/sbir-sttr-topics. For additional information, see the STTR

26.D Annual Program Broad Agency Announcement (BAA) at https://www.dodsbirsttr.mil/submissions/solicitation-documents/active-solicitations.

i. Fraud, Waste and Abuse (Volume 6)

The Fraud, Waste and Abuse (FWA) training is required for Phase I and Direct to Phase II proposals. FWA training provides information on what represents FWA in the SBIR/STTR program, the most common mistakes that lead to FWA, as well as the penalties and ways to prevent FWA in your firm. This training material must be thoroughly reviewed once per year. Plan ahead and leave ample time to complete this training based on the proposal submission deadline. Knowingly and willfully making any false, fictitious, or fraudulent statements or representations may be a felony under the Federal Criminal False Statement Act (18 U.S.C. §1001), punishable by a fine of up to $10,000, up to five years in prison, or both. Understanding the indicators and types of fraud, waste and abuse that can occur is critical for the SBIR/STTR awardees' role in preventing the loss of research dollars.

j. Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7)

In accordance with Section 4 of the SBIR and STTR Extension Act of 2022 and the SBA SBIR/STTR Policy Directive, the DoW will review all proposals submitted in response to this BAA to assess security risks presented by small business concerns seeking a Federally funded award. Small business concerns must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission (NOTE: PDF uploads will no longer be accepted). Full proposal submissions cannot be certified and submitted by the Corporate Official until Volume 7 is fully completed and the webform is submitted.

For additional details, please refer to the DoW STTR Program BAA.

Appendix A
TECHNICAL PROPOSAL TEMPLATE (VOLUME 2)

INSTRUCTIONS

These instructions and template apply to DoW SBIR/STTR Phase I topics and provide general guidelines for completing the Phase I Technical Volume. Information provided in the Service/Component-specific instructions for the topic of interest take precedence over any instructions listed below.

The template (beginning on the following page) is the format model that may be used to prepare the Phase I Technical Volume. Do not include the instructions provided on this page or any bracketed [ ] guidance in the template.

Disclosure

Offerors that include in their proposals data which they do not want disclosed to the public for any purpose, or used by the U.S. Government except for evaluation purposes, must:

(1) Mark the first page of each Volume of the Submission with the following legend:
"This proposal includes data that must not be disclosed outside the Government and must not be duplicated, used, or disclosed-in whole or in part-for any purpose other than to evaluate this proposal. If, however, a contract is awarded to this offeror as a result of-or in connection with-the submission of this data, the Government has the right to duplicate, use, or disclose the data to the extent provided in the resulting contract. This restriction does not limit the Government's right to use information contained in this data if it is obtained from another source without restriction. The data subject to this restriction are contained in pages [insert numbers or other identification of sheets]";

(2) Mark each sheet of data it wishes to restrict with the following legend:
"Use or disclosure of data contained on this page is subject to the restriction on the first page of this volume."

Format

The Technical Volume shall meet the following requirements:

- Please refer to Service/Component-specific topic instructions for the page limit and how a technical volume is handled if the stated page count is exceeded. It is the proposing firm's responsibility to verify that the Technical Volume does not exceed the page limit after upload to DSIP. Unless otherwise noted, all content in the Technical Volume will count toward the limit.
- Single column format single-spaced typed lines.
- Standard 8 ½" x 11" paper format.
- Page margins one inch on all sides. A header and footer may be included in the one-inch margin.
- **The header on each page of the Technical Volume should contain your company name, topic number, and DSIP-assigned proposal number.**
- No font smaller than 10-point. For headers, footers, imbedded tables, figures, images, or graphics that include text, a font size of smaller than 10-point is allowable, though proposers are cautioned that the text may be unreadable by evaluators.

Do not lock or encrypt the uploaded file. Do not include or embed active graphics such as videos, moving pictures, or other similar media in the document.

Delete this instruction page and begin the Technical Volume starting with the following page.

[Title]

Volume 2: Technical Volume

[Note: Remove the disclosure statement below if not applicable to your proposal. Refer to Instructions.]

This proposal includes data that must not be disclosed outside the Government and must not be duplicated, used, or disclosed – in whole or in part – for any purpose other than to evaluate this proposal. If, however, a contract is awarded to this offeror as a result of – or in connection with – the submission of this data, the Government has the right to duplicate, use, or disclose the data to the extent provided in the resulting contract. This restriction does not limit the Government's right to use information contained in this data if it is obtained from another source without restriction. The data subject to this restriction are contained in pages <insert numbers or other identification of sheets>.

1. Identification and Significance of the Problem or Opportunity.

[Define the specific technical problem or opportunity addressed and its importance.]

2. Phase I Technical Objectives.

[Enumerate the specific objectives of the Phase I work, including the questions the research and development effort will try to answer to determine the feasibility of the proposed approach.]

3. Phase I Statement of Work (include Subcontractors and/or Research Institutions).

(a) [Provide an explicit, detailed description of the Phase I approach. If a Phase I option is required or allowed by the Component (refer to Component-specific instructions for topic of interest), describe appropriate research activities which would commence at the end of Phase I base period should the Component elect to exercise the option. The Statement of Work should indicate what tasks are planned, how and where the work will be conducted, a schedule of major events, and the final product(s) to be delivered. The Phase I effort should attempt to determine the technical feasibility of the proposed concept. The methods planned to achieve each objective or task should be discussed explicitly and in detail. This section should be a substantial portion of the Technical Volume.

(b) The solicitation may contain topics that have been identified by the Program Manager as research or activities involving Human/Animal Subjects and/or Recombinant DNA. If Phase I performance includes performance of these kinds of research or activities, please identify the applicable protocols and how those protocols will be followed during Phase I. Please note that funds cannot be released or used on any portion of the project involving human/animal subjects or recombinant DNA research or activities until all of the proper approvals have been obtained. **SBCs proposing research involving human and/or animal use are encouraged to separate these tasks in the technical proposal and cost proposal in order to avoid potential delay of contract award.**]

4. Related Work.

[Describe significant activities directly related to the proposed effort, including any conducted by the principal investigator, the proposing firm, consultants, or others. Describe how these activities interface with the proposed project and discuss any planned coordination with outside sources. The Technical Volume must persuade evaluators of the proposer's awareness of the state of the art in the topic. Describe any previous work not directly related but similar to the proposed effort. Provide the following: (1) a short description, (2) the client for which work was performed (including the Government Point of Contact to be contacted including e-mail address and phone number), and (3) date of performance including project completion.]

5. <u>Relationship with Future Research or Research and Development</u>.

(a) [State the anticipated results of the proposed approach if the project is successful.

(b) Discuss the significance of the Phase I effort in providing a foundation for a Phase II research or research and development effort.

(c) Identify the applicable clearances, certifications and approvals required to conduct Phase II testing. Outline the plan for ensuring timely completion of stated authorizations in support of a Phase II research or research and development effort.]

6. <u>Commercialization Strategy.</u>

[Describe in approximately one page the SBC's strategy for commercializing this technology in DoW, other Federal Agencies, and/or private sector markets. Provide specific information on the market need the technology will address and the size of the market. Also include a schedule showing the quantitative commercialization results from the project that your company expects to achieve.]

7. <u>Key Personnel</u>.

[Identify key personnel who will be involved in the Phase I effort including information on directly related education and experience. A concise technical resume of the principal investigator, including a list of relevant publications (if any), must be included (Please do not include Privacy Act Information). All resumes will count toward the page limit for Volume 2, as specified in the Component-specific instructions.]

[Principal Investigator Name]
[School, Degree, Year]

Relevant Experience
[A concise description of the principal investigator's relevant technical experience and its application to this topic.]

Relevant Awards or Patents
[List any awards received or patents granted or applications submitted for work related to this topic.]

Relevant Publications
[List any publications relevant to this topic.]

[Repeat this format as necessary to address the qualifications of all key personnel.]

8. <u>Foreign Citizens</u>.

[Identify any foreign citizens or individuals holding dual citizenship expected to be involved on this project as a direct employee, subcontractor, or consultant. For these individuals, please specify their country of origin, the type of visa or work permit under which they are performing and an explanation of their anticipated level of involvement on this project. The proposal may be deemed nonresponsive if the requested information is not provided. You may be asked to provide additional information during negotiations in order to verify the foreign citizen's eligibility to participate on a SBIR/STTR contract. Supplemental information provided in response to this paragraph will be protected in accordance with the Privacy Act (5 U.S.C. 552a), if applicable, and the Freedom of Information Act (5 U.S.C. 552(b)(6)). Note: If no foreign nationals will be involved in proposed work, the word "None" can be substituted for the table.]

Name [include direct employees, subcontractors, and consultants]	Foreign National (Yes/No)	Country of Origin	Type of Visa or Work Permit	Level of Involvement (Role)

9. Facilities/Equipment.

[Describe available instrumentation and physical facilities necessary to carry out the Phase I effort. Justify equipment purchases in this section and include detailed pricing information in the Cost Volume. State whether or not the facilities where the proposed work will be performed meet environmental laws and regulations of federal, state (name), and local Governments for, but not limited to, the following groupings: airborne emissions, waterborne effluents, external radiation levels, outdoor noise, solid and bulk waste disposal practices, and handling and storage of toxic and hazardous materials.]

10. Subcontractors/Consultants.

[Propose efforts as applicable to either SBIR or STTR as follows:

SBIR. Involvement of a university or other subcontractors or consultants in the project may be appropriate. A minimum of two-thirds of the research and/or analytical work in Phase I, as measured by direct and indirect costs, must be carried out by the proposing small business firm, unless otherwise approved in writing by the Contracting Officer. SBIR efforts may include subcontracts with Federal Laboratories and Federally Funded Research and Development Centers (FFRDCs). A waiver is not required for the use of Federal Laboratories and FFRDCs; however, proposers must certify their use of such facilities on the proposal cover sheet. Subcontracts with other Federal organizations are not permitted. Note that universities cannot publicly release information related to Export Controlled/ITAR restricted topics. (Refer to the DoW SBIR/STTR solicitation for detailed eligibility requirements as it pertains to the use of subcontractors/consultants.)

STTR. Involvement of a Research Institution in the project is required. A minimum of 40 percent of the research and/or analytical work in Phase I, as measured by direct and indirect costs, must be conducted by the proposing small business firm, and a minimum of 30 percent of the research and/or tasks in Phase I, as measured by direct and indirect costs, must be conducted by a single Research Institution. STTR efforts may include subcontracts with Federally Funded Research and Development Centers (FFRDCs). A waiver is not required for the use of Federal Laboratories, but they do not qualify as a Research Partner; proposers may only subcontract to Federal Laboratories within the remaining 30 percent and must certify their use of such facilities on the proposal cover sheet. Subcontracts with other Federal organizations are not permitted. Note that universities cannot publicly release information related to Export Controlled/ITAR restricted topics. (Refer to the DoW SBIR/STTR solicitation for detailed eligibility requirements as it pertains to the use of subcontractors/consultants.]

11. Prior, Current or Pending Support of Similar Proposals or Awards.

[If a proposal submitted in response to this solicitation is substantially the same as another proposal that was funded, is now being funded, or is pending with another Federal Agency, another or the same DoW Service/Component, you must disclose this on the proposal cover sheet and provide the following information:

a) Name and address of the Federal Agency(s) or DoW Component to which a proposal was submitted, will be submitted, or from which an award is expected or has been received.

b) Date of proposal submission or date of award.

c) Title of proposal.

d) Name and title of principal investigator for each proposal submitted or award received.

e) Title, number, and date of solicitation(s) under which the proposal was submitted, will be submitted, or under which award is expected or has been received.

f) If award was received, provide contract number.

g) Specify the applicable topics for each proposal submitted or award received.

Note: If this does not apply, state in the proposal "No prior, current, or pending support has been provided for proposed work."]

12. <u>Identification and Assertion of Restrictions on the Government's Use, Release, or Disclosure of Technical Data or Computer Software.</u>

The Offeror asserts for itself, or the persons identified below, that the Government's rights to use, release, or disclose the following technical data or computer software should be restricted:

Technical Data or Computer Software to be Furnished with Restrictions	Basis for Assertion	Asserted Rights Category	Name of Person or Organization Asserting Restrictions
[(LIST)]	[(LIST)]	[(LIST)]	[(LIST)]

[Completion of this table and submission of the proposal constitutes signature for the information listed in the table above.]

[**ADDITIONAL INFORMATION/INSTRUCTION:** Rights in technical data, including software, developed under the terms of any contract resulting from proposals submitted in response to this solicitation generally remain with the contractor, except that the Government obtains a royalty-free license to use such technical data only for Government purposes during the period commencing with contract award and ending twenty years after completion of the project under which the data were generated. This data <u>must</u> be marked with the restrictive legend specified in DFARS 252.227-7018 (JAN 2025). Upon expiration of the twenty-year restrictive license, the Government has unlimited rights in the SBIR data. During the license period, the Government may not release or disclose SBIR data to any person other than its support services contractors except: (1) For evaluation purposes; (2) As expressly permitted by the contractor; or (3) A use, release, or disclosure that is necessary for emergency repair or overhaul of items operated by the Government. See DFARS clause 252.227-7018 (JAN 2025).

If a proposer plans to submit assertions in accordance with DFARS 252.227-7017 (JAN 2025), those assertions must be identified and assertion of use, release, or disclosure restriction MUST be included with your proposal submission. The contract cannot be awarded until assertions have been approved. Please note that only the table is included in the page limitation; any supporting data concerning the contract/grant number and awarding agency, as well as planned use or need of the data asserted, can be provided in Volume 5, Supporting Documents.

The following instructions apply to the fields in the table above (Identification and Assertion of Restrictions on the Government's Use, Release, or Disclosure of Technical Data or Computer Software).

1) For technical data (other than computer software documentation) pertaining to items, components, or processes developed at private expense, identify both the deliverable technical data and each such item, component, or process. For computer software or computer software documentation identify the software or documentation.

2) Generally, development at private expense, either exclusively or partially, is the only basis for asserting restrictions. For technical data, other than computer software documentation, development refers to development of the item, component, or process to which the data pertain. The Government's rights in computer software documentation generally may not be restricted. For computer software, development refers to the software. Indicate whether development was accomplished exclusively or partially at private expense. If development was not accomplished at private expense, or for computer software documentation, enter the specific basis for asserting restrictions.

3) Enter asserted rights category (e.g., Government purpose license rights from a prior contract, rights in SBIR/STTR data generated under another contract, limited, restricted, or government purpose rights under this or a prior contract, or specially negotiated licenses).

4) Corporation, individual, or other person, as appropriate.

Enter “none” when all data or software will be submitted without restrictions.]

Appendix B
DEFINITIONS

The following definitions from the SBA SBIR/STTR Policy Directive, the Federal Acquisition Regulation (FAR) and other cited regulations apply to this solicitation.

Commercialization

The process of developing products, processes, technologies, or services, and the production and delivery (whether by the originating party or others) of the products, processes, technologies, or services for Federal Government or commercial markets purchase or use.

Cooperative Research and Development

An SBC and a research institution jointly conduct R&D. For purposes of the STTR Program, the SBC performs 40 percent of the work, and the single research institution performs not less than 30 percent of the work. For purposes of the SBIR Program, this refers to work a research institution conducts as the SBC's subcontractor. The proposing SBC must conduct at least two-thirds of the research and/or analytical work in Phase I.

Covered Individual

An individual who contributes in a substantive, meaningful way to the scientific development or execution of a R&D project proposed to be carried out with a DoW-funded award. DoW has further designated covered individuals as including all proposed key personnel.

Essentially Equivalent Work

Work that is substantially the same research, which is proposed for funding in more than one contract proposal or grant application submitted to the same federal agency or submitted to two or more different federal agencies for review and funding consideration; or work where a specific research objective and the research design for accomplishing the objective are the same or closely related to another proposal or award, regardless of the funding source.

Export Control

The International Traffic in Arms Regulations (ITAR), 22 CFR Parts 120 through 130, and the Export Administration Regulations (EAR), 15 CFR Parts 730 through 799, apply to all projects with military or dual-use applications that develop beyond fundamental research, which is basic and applied research ordinarily published and shared broadly within the scientific community. Details are available here.

NOTE: Export control compliance statements found in the individual Service/Component-specific proposal instructions are not meant to be all inclusive. They do not remove any liability from the submitter to comply with applicable ITAR or EAR export control restrictions or from informing the U.S. Government of any potential export restriction as fundamental R&D efforts proceed.

Federal Laboratory

In 15 U.S.C. §3703, it means any laboratory, any federally funded R&D center (FFRDC), or any center established under 15 U.S.C. §§ 3705 & 3707 that a federal agency owns, leases, or otherwise uses and the Federal Government funds, whether the U.S. Government or the contractor operates.

Federally Funded Award

A Phase I, Phase II (including Direct to Phase II, sequential Phase II/subsequent Phase II and cross-agency Phase II), or Phase III SBIR or STTR award made using a funding agreement.

Foreign Affiliation

Under 15 U.S.C. § 638(e)(16), foreign affiliation means a funded or unfunded academic, professional, or institutional appointment or position with a foreign government or government-owned entity, whether full-time, part-time, or voluntary (including adjunct, visiting, or honorary). This includes appointments or positions deemed adjunct, visiting, or honorary with research institutions located in a foreign country of concern.

Foreign Country of Concern

In 15 U.S.C. § 638(e)(17), foreign country of concern means the People's Republic of China, the Democratic People's Republic of Korea, the Russian Federation, the Islamic Republic of Iran, or any other country the Secretary of State determines as a country of concern.

Foreign Entity

Foreign entity means any branch, partnership, group or sub-group, association, estate, trust, corporation or division of a corporation, non-profit, academic institution, research center, or organization that foreign owners, foreign investors, foreign management, or a foreign government establish, direct, or control.

Foreign Government

Foreign government means any government or governmental body, organization, or instrumentality, including government owned-corporations, other than the U.S. Government or U.S. state, territorial, tribal, or jurisdictional governments or governmental bodies. The term includes, but is not limited to, non-U.S. national and subnational governments, including their respective departments, agencies, and instrumentalities.

Foreign National

Foreign National (also known as Foreign Person) under 22 CFR 120.16 mean any natural person who is not a lawful permanent resident as defined by 8 U.S.C. § 1101(a)(20) or who is not a protected individual as defined by 8 U.S.C. § 1324b(a)(3). It also means any foreign corporation, business association, partnership, trust, society, or any other entity or group that is not incorporated or organized to do business in the United States, as well as international organizations, foreign governments and any agency or subdivision of foreign governments (e.g., diplomatic missions).

"Lawfully admitted for permanent residence" means the status of having been lawfully accorded the privilege of residing permanently in the United States as an immigrant in accordance with the immigration laws, such status not having changed.

"Protected individual'' means an individual who (A) is a citizen or national of the United States, or (B) is an alien who is lawfully admitted for permanent residence, is granted the status of an alien lawfully admitted for temporary residence under 8 U.S.C. § 1160(a) or 8 U.S.C. § 1255a(a)(1), is admitted as a refugee under 8 U.S.C. § 1157, or is granted asylum under Section 8 U.S.C. § 1158; but does not include

(i) an alien who fails to apply for naturalization within six months of the date the alien first becomes eligible (by virtue of period of lawful permanent residence) to apply for naturalization or, if later, within six months after November 6, 1986, and (ii) an alien who has applied on a timely basis, but has not been naturalized as a citizen within two years after the date of the application, unless the alien can establish that the alien is actively pursuing naturalization, except that time consumed in the Service's processing the application shall not be counted toward the two-year period.

Fraud, Waste and Abuse

a. Fraud includes any false
b. about a material fact or any intentional deception designed to deprive the United States unlawfully of something of value or to secure from the United States a benefit, privilege, allowance, or consideration to which an individual or business is not entitled.
c. Waste includes extravagant, careless or needless expenditure of government funds, or the consumption of government property, that results from deficient practices, systems, controls, or decisions.
d. Abuse includes any intentional or improper use of government resources, such as misuse of rank, position, or authority or resources.
e. The SBIR/STTR Program training related to Fraud, Waste and Abuse is available here. See Section 1.13 for reporting fraud, waste, and abuse.

Funding Agreement

Any contract, grant, or cooperative agreement entered between any Federal Agency and any SBC for the performance of experimental, developmental, or research work, including products or services, Federal Government-funded in whole or in part. DoW Services/Components will only use contracts and other transaction authority (OTA) agreements for all SBIR awards.

Historically Black Colleges and Universities, and Minority-Serving Institutions

Department of Education list for historically Black colleges and universities and minority-serving institutions.

HUBZone Certified Small Business Concern

An SBC with SBA certification under the Historically Underutilized Business Zones (HUBZone) Program (13 C.F.R. § 126) as a HUBZone firm listed in the dynamic small business search (DSBS).

Innovation

As defined and as used in SBA Policy Directive 3(t), the term innovation means-
Something new or improved, having marketable potential, that includes the development of new technology, the refinement of existing technology, or the development of new applications for existing technology.

Malign Foreign Talent Recruitment Program

As defined in 42 U.S.C § 19237, the term “malign foreign talent recruitment program” means-

(A) any program, position, or activity that includes compensation in the form of cash, in-kind compensation, including research funding, promised future compensation, complimentary foreign travel, things of non de minimis value, honorific titles, career advancement opportunities, or other

types of remuneration or consideration directly provided by a foreign country at any level (national, provincial, or local) or their designee, or an entity based in, funded by, or affiliated with a foreign country, whether or not directly sponsored by the foreign country, to the targeted individual, whether directly or indirectly stated in the arrangement, contract, or other documentation at issue, in exchange for the individual-

(i) engaging in the unauthorized transfer of intellectual property, materials, data products, or other nonpublic information owned by a United States entity or developed with a Federal R&D award to the government of a foreign country or an entity based in, funded by, or affiliated with a foreign country regardless of whether that government or entity provided support for the development of the IP, materials, or data products;

(ii) being required to recruit trainees or researchers to enroll in such program, position, or activity;

(iii) establishing a laboratory or company, accepting a faculty position, or undertaking any other employment or appointment in a foreign country or with an entity based in, funded by, or affiliated with a foreign country if such activities are in violation of the standard terms and conditions of a Federal R&D award;

(iv) being unable to terminate the foreign talent recruitment program contract or agreement except in extraordinary circumstances;

(v) through funding or effort related to the foreign talent recruitment program, being limited in the capacity to carry out a R&D award or required to engage in work that would result in substantial overlap or duplication with a Federal R&D award;

(vi) being required to apply for and successfully receive funding from the sponsoring foreign government's funding agencies with the sponsoring foreign organization as the recipient;

(vii) being required to omit acknowledgment of the recipient institution with which the individual is affiliated, or the Federal research agency sponsoring the R&D award, contrary to the institutional policies or standard terms and conditions of the Federal R&D award;

(viii) being required to not disclose to the Federal research agency or employing institution the participation of such individual in such program, position, or activity; or

(ix) having a conflict of interest or conflict of commitment contrary to the standard terms and conditions of the Federal R&D award; and

(B) a program that is sponsored by-

(i) a foreign country of concern or an entity based in a foreign country of concern, whether or not directly sponsored by the foreign country of concern;

(ii) an academic institution on the list developed under section 1286(c)(8) of the John S. McCain National Defense Authorization Act for Fiscal Year 2019 (10 U.S.C. 2358 note; 1 Public Law 115–232) ; or

(iii) a foreign talent recruitment program on the list developed under section 1286(c)(9) of the John S. McCain National Defense Authorization Act for Fiscal Year 2019 (10 U.S.C. 2358 note; 1 Public Law 115–232).

Performance Benchmark Requirements

SBCs with multiple SBIR/STTR awards must meet minimum performance requirements to be eligible to apply for a new Phase I or Direct-to-Phase II award. The purpose of these requirements is to ensure that Phase I SBCs with multiple prior SBIR/STTR awards are making progress towards commercializing the work done under those awards. The Phase I to Phase II transition rate addresses the extent to which an awardee progresses a project from Phase I to Phase II. The commercialization benchmark addresses the extent to which an awardee has moved past Phase II work towards commercialization.

The SBIR and STTR Extension Act of 2022 (Pub. L. 117-183) amended the benchmarks' applications for more experienced firms. Detailed information on benchmark calculations and increased performance standards for more experienced firms can be found here.

Personal Conflict of Interest

A situation in which an individual has a financial interest, personal activity, or relationship that could impair the employee's ability to act impartially and in the best interest of the government when performing under the contract. (A de minimis interest that would not "impair the employee's ability to act impartially and in the best interest of the government" is not covered under this definition.)

Among the sources of personal conflicts of interest are:

a. The covered employee's, their close family members' or other members of their household's financial interests;
b. Other employment or financial relationships (including seeking or negotiating for prospective employment or business); and
c. Gifts, including travel.

Financial interests referred to in this definition's first paragraph may arise from:

a. Compensation, including wages, salaries, commissions, professional fees, or fees for business referrals;
b. Consulting relationships (including commercial and professional consulting and service arrangements, scientific and technical advisory board memberships, or serving as an expert witness in litigation);
c. Services provided in exchange for honorariums or travel expense reimbursements;
d. Research funding or other forms of research support;
e. Investment in the form of stock or bond ownership or partnership interest (excluding diversified mutual fund investments);
f. Real estate investments;
g. Patents, copyrights, and other IP interests; or
h. Business ownership and investment interests.

Principal Investigator/Program Manager

The principal investigator/project manager is the proposing SBC-designated individual who provides the scientific and technical direction to a funding agreement-supported project.

Proprietary Information

Proprietary information is any information that an SBC considers to be non-public information the SBC owns and is marked accordingly.

Research Institution

Any organization located in the United States that is:

a. A university.
b. A nonprofit institution as defined in Section 4(5) of the Stevenson-Wydler Technology Innovation Act of 1980.
c. A contractor-operated federally funded R&D center, as identified by the National Science Foundation in accordance with the government-wide FAR issued in accordance with the Office of Federal Procurement Policy Act Section 35(c)(1). A list of eligible FFRDCs is here.

Research or Research and Development

Any activity that is:

a. A systematic, intensive study directed toward greater knowledge or understanding of the subject studied.
b. A systematic study directed specifically toward applying new knowledge to meet a recognized need; or
c. A systematic knowledge application toward the production of useful materials, devices, systems, or methods, including design, development, and improvement of prototypes and new processes to meet specific requirements.

Research Involving Animal Subjects

All activities involving animal subjects shall be conducted in accordance with DoDI 3216.01 "Use of Animals in DoD Programs," 9 C.F.R. parts 1-4 "Animal Welfare Regulations," National Academy of Sciences Publication "Guide for the Care & Use of Laboratory Animals," as amended, and the Department of Agriculture rules implementing the Animal Welfare Act (7 U.S.C. §§ 2131-2159), as well as other applicable federal and state law and regulation and DoW instructions.

"Animal use" protocols apply to all activities that meet any of the following criteria:

a. Any research, development, test, evaluation or training, (including experimentation) involving an animal or animals.
b. An animal is defined as any living or dead, vertebrate organism (non-human) that is being used or is intended for use in research, development, test, evaluation or training.
c. A vertebrate is a member of the subphylum Vertebrata (within the phylum Chordata), including birds and cold-blooded animals.

See DoDI 3216.01 for definitions of these terms and more information about the applicability of DoDI 3216.01 to work involving animals.

Research Involving Human Subjects

All research involving human subjects shall be conducted in accordance with 32 C.F.R. § 219 "The Common Rule," 10 U.S.C. § 980 "Limitation on Use of Humans as Experimental Subjects," and DoDI 3216.02 "Protection of Human Subjects and Adherence to Ethical Standards in DoD-Supported Research," as well as other applicable federal and state law and regulations, and DoW Services/Component guidance. Proposing SBCs must be cognizant of and abide by the additional restrictions and limitations imposed on the DoW regarding research involving human subjects, specifically as they regard vulnerable populations (DoDI 3216.02), recruitment of military research subjects (DoDI 3216.02), and informed consent and surrogate consent (10 U.S.C. § 980) and chemical and biological agent research (DoDI 3216.02). Food and Drug Administration regulation and policies may also apply.

"Human use" protocols apply to all research that meets any of the following criteria:

a. Any research involving an intervention or an interaction with a living person that would not be occurring or would be occurring in some other fashion but for this research.
b. Any research involving identifiable private information. This may include data/information/specimens collected originally from living individuals (broadcast video, web-use logs, tissue, blood, medical or personnel records, health data repositories, etc.) in which the

identity of the subject is known, or the identity may be readily ascertained by the investigator or associated with the data/information/specimens.

See DoDI 3216.02 for definitions of these terms and more information about the applicability of DoDI 3216.02 to research involving human subjects.

Research Involving Recombinant DNA Molecules

Any recipient performing research involving recombinant DNA molecules and/or organisms and viruses containing recombinant DNA molecules shall comply with the National Institutes of Health Guidelines for Research Involving Recombinant DNA Molecules, dated January 2011, as amended. The guidelines can be found at: https://osp.od.nih.gov/wp-content/uploads/2016/05/NIH_Guidelines.pdf. Recombinant DNA is defined as (i) molecules that are constructed outside living cells by joining natural or synthetic DNA segments to DNA molecules that can replicate in living cells or (ii) molecules that result from the replication of those described in (i) above.

Service-Disabled Veteran-Owned Small Business (SDVOSB)

A service-disabled veteran or service-disabled veterans-owned and controlled SBC defined in Small Business Act 15 USC § 632(q)(2) and SBA's implementing SDVOSB regulations (13 CFR 125).

Small Business Concern (SBC)

A concern that meets the requirements set forth in 13 C.F.R. § 121.702 (available here).

An SBC must satisfy the following conditions on the date of award:

a. Is organized for profit, with a place of business located in the United States, which operates primarily within the United States, or which makes a significant contribution to the United States economy through payment of taxes or use of American products, materials or labor;
b. Is in the legal form of an individual proprietorship, partnership, limited liability company, corporation, joint venture, association, trust or cooperative, except that if the concern is a joint venture, each entity to the venture must meet the requirements set forth in paragraph (c) below;
c. Is more than 50 percent directly owned and controlled by one or more individuals (who are citizens or permanent resident aliens of the United States), other SBCs (each of which is more than 50 percent directly owned and controlled by individuals who are citizens or permanent resident aliens of the United States), or any combination of these; and
d. Has, including its affiliates, not more than 500 employees. (See here for definition of an affiliate.)

Subcontract

A subcontract is any agreement, other than one involving an employer-employee relationship, including consultants, the funding agreement awardee enters calling for supplies or services for the performance of the original funding agreement.

Subcontractor

Subcontractor means any supplier, distributor, vendor, firm, academic institution, research center, or other person or entity that furnishes supplies or services pursuant to a subcontract, at any tier.

United States

United States means the fifty states, the territories and possessions of the Federal Government, the Commonwealth of Puerto Rico, the Republic of the Marshall Islands, the Federated States of Micronesia, the Republic of Palau, and the District of Columbia.

Women-Owned Small Business Concern

An SBC where one or more women own at least 51 percent, or in the case of any publicly owned business, women own at least 51 percent of the stock, and women control the management and daily business operations.

Appendix C
POTENTIAL APPLICABLE FEDERAL ACQUISITION REGULATION, DEFENSE FEDERAL ACQUISITION REGULATION SUPPLEMENT CLAUSES

Note: An independent review and determination of required clauses should be complete for each award. Green cells are potential required Federal Acquisition Regulation (FAR) and Defense Federal Acquisition Regulation Supplement (DFARS) clauses. Blue cells are potential required FAR and DFARS clauses, when applicable.

Clause Number	Title	Date	When Applicable
52.203-17	Contractor Employee Whistleblower Rights	Nov-23	
52.203-19	Prohibition on Requiring Certain Internal Confidentiality Agreements or Statements	Jan-17	
52.204-10	Reporting Executive Compensation and First-Tier Subcontract Awards	Jun-20	
52.204-13	System for Award Management Maintenance	Oct-18	
52.204-18	Commercial and Government Entity Code Maintenance	Aug-20	
52.204-19	Incorporation by Reference of Representations and Certifications	Dec-14	
52.204-21	Basic Safeguarding of Covered Contractor Information Systems	Nov-21	
52.204-23	Prohibition on Contracting for Hardware, Software, and Services Developed or Provided by Kaspersky Lab and Other Covered Entities	Dec-23	
52.204-25	Prohibition on Contracting for Certain Telecommunications and Video Surveillance Services or Equipment	Nov-21	
52.204-27	Prohibition on a ByteDance Covered Application	Jun-23	
52.204-30	Federal Acquisition Supply Chain Security Act Orders—Prohibition	Dec-23	
52.209-06	Protecting the Government's Interest When Subcontracting with Contractors Debarred, Suspended, or Proposed for Debarment	Jan-25	
52.209-10	Prohibition on Contracting with Inverted Domestic Corporations	Nov-15	
52.219-06	Notice of Total Small Business Set-Aside	Nov-20	
52.219-08	Utilization of Small Business Concerns	Jan-25	
52.219-28	Post-Award Small Business Program Representation	Jan-25	
52.222-03	Convict Labor	Jun-03	
52.222-19	Child Labor-Cooperation with Authorities and Remedies	Jan-25	
52.222-21	Prohibition of Segregated Facilities	Apr-15	
52.222-25	Affirmative Action Compliance	Apr-84	
52.222-26	Equal Opportunity	Sep-16	
52.222-36	Equal Opportunity for Workers with Disabilities	Jun-20	

Clause Number	Title	Date	When Applicable
52.222-50	Combating Trafficking in Persons	Nov-21	
52.225-01	Buy American-Supplies	Oct-22	
52.225-13	Restrictions on Certain Foreign Purchases	Feb-21	
52.226-07	Drug-Free Workplace	May-24	
52.226-08	Encouraging Contractor Policies to Ban Text Messaging While Driving	May-24	
52.227-01 Alt I	Authorization and Consent - Alternate I (Apr-84)	Jun-20	
52.227-02	Notice and Assistance Regarding Patent and Copyright Infringement	Jun-20	
52.227-11	Patent Rights-Ownership by the Contractor	May-14	
52.227-20	Rights in Data-SBIR Program	May-14	
52.232-01	Payments	Apr-84	
52.232-11	Extras	Apr-84	
52.232-23	Assignment of Claims	May-14	
52.232-25	Prompt Payment	Jan-17	
52.232-33	Payment by Electronic Funds Transfer - System for Award Management	Oct-18	
52.232-39	Unenforceability of Unauthorized Obligations	Jun-13	
52.232-40	Providing Accelerated Payments to Small Business Subcontractors	Mar-23	
52.233-01	Disputes	May-14	
52.233-04	Applicable Law for Breach of Contract Claim	Oct-04	
52.242-15	Stop-Work Order	Aug-89	
52.243-01 Alt V	Changes-Fixed-Price Alternate V (Apr-84)	Aug-87	
52.244-06	Subcontracts for Commercial Products and Commercial Services	Jan-25	
52.246-09	Inspection of Research and Development (Short Form)	Apr-84	
52.252-02	Clauses Incorporated by Reference	Feb-98	
52.252-06	Authorized Deviations in Clauses	Nov-20	
52.253-01	Computer Generated Forms	Jan-91	
252.203-7000	Requirements Relating to Compensation of Former DoD Officials	Sep-11	
252.203-7002	Requirement to Inform Employees of Whistleblower Rights	Dec-22	
252.204-7000	Disclosure of Information	Oct-16	
252.204-7003	Control of Government Personnel Work Product	Apr-92	
252.204-7008	Compliance with Safeguarding Covered Defense Information Controls	Oct-16	
252.204-7009	Limitations on the Use or Disclosure of Third-Party Contractor Reported Cyber Incident Information	Jan-23	
252.204-7012	Safeguarding Covered Defense Information and Cyber Incident Reporting (DEVIATION 2024-O0013)	May-24	

Clause Number	Title	Date	When Applicable
252.204-7016	Covered Defense Telecommunications Equipment or Services—Representation	Dec-19	
252.204-7017	Prohibition on the Acquisition of Covered Defense Telecommunications Equipment or Services—Representation	May-21	
252.204-7018	Prohibition on the Acquisition of Covered Defense Telecommunications Equipment or Services	Jan-23	
252.204-7019	Notice of NISTSP 800-171 DoD Assessment Requirements	Nov-23	
252.204-7020	NIST SP 800-171 DoD Assessment Requirements	Nov-23	
252.204-7022	Expediting Contract Closeout	May-21	
252.204-7024	Notice on the use of the Supplier Performance Risk System	Mar-23	
252.227-7016	Rights in Bid or Proposal Information	Jan-23	
252.227-7018	Rights in Other Than Commercial Technical Data and Computer Software—Small Business Innovation Research Program and Small Business Technology Transfer Program	Jan-25	
252.227-7019	Validation of Asserted Restrictions--Computer Software	Jan-25	
252.227-7025	Limitations on the Use or Disclosure of Government-Furnished Information Marked with Restrictive Legends	Jan-25	
252.227-7030	Technical Data--Withholding of Payment	Mar-00	
252.227-7037	Validation of Restrictive Markings on Technical Data	Jan-25	
252.227-7039	Patents--Reporting of Subject Inventions	Apr-90	
252.232-7003	Electronic Submission of Payment Requests and Receiving Reports	Dec-18	
252.232-7006	Wide Area WorkFlow Payment Instructions	Jan-23	
252.232-7010	Levies on Contract Payments	Dec-06	
252.235-7010	Acknowledgment of Support and Disclaimer	May-95	
252.235-7011	Final Scientific or Technical Report	Dec-19	
252.243-7001	Pricing of Contract Modifications	Dec-91	
252.244-7000	Subcontracts for Commercial Items	Nov-23	
52.203-03	Gratuities	Apr-84	Exceeding the simplified acquisition threshold.
52.203-05	Covenant Against Contingent Fees	May-14	Exceeding the simplified acquisition threshold.
52.203-06 or Alt I	Restrictions on Subcontractor Sales to the Government or ALT I	Nov-21	Exceeding the simplified acquisition threshold, Alt I commercial products or commercial services.
52.203-07	Anti-Kickback Procedures	Jun-20	Contracts exceeding $150,000.
52.203-08	Cancellation, Rescission, and Recovery of Funds for Illegal or Improper Activity	May-14	Exceeding the simplified acquisition threshold.
52.203-10	Price or Fee Adjustment for Illegal or Improper Activity	May-14	Exceeding the simplified acquisition threshold.
52.203-12	Limitation on Payments to Influence Certain Federal Transactions	Jun-20	Contracts exceeding $150,000.

Clause Number	Title	Date	When Applicable
52.204-02	Security Requirements	Mar-21	May require access to classified information; cost contract (see 16.302) for research and development with an educational institution is contemplated.
52.212-04	Contract Terms and Conditions—Commercial Products and Commercial Services	Nov-23	Commercial products or commercial services.
52.212-05	Contract Terms and Conditions Required to Implement Statutes or Executive Orders-Commercial Items	May-24	Commercial products or commercial services.
52.219-14	Limitations on Subcontracting	Oct-22	Set aside for small business and the contract amount is expected to exceed the simplified acquisition threshold.
52.222-35	Equal Opportunity for Veterans	Jun-20	Contracts exceed $150,000.
52.222-37	Employment Reports on Veterans	Jun-20	If contract contains 52.222-35.
52.222-40	Notification of Employee Rights Under the National Labor Relations Act	Dec-10	Exceeding the simplified acquisition threshold.
52.222-54	Employment Eligibility Verification	Jan-25	Contracts exceed $150,000.
52.223-03	Hazardous Material Identification and Material Safety Data	Feb-21	Requires the delivery of hazardous materials.
52.229-03	Federal, State, and Local Taxes	Feb-13	A fixed-price contract is contemplated; and the contract is expected to exceed the simplified acquisition threshold.
52.232-02	Payment under Fixed-Price Research and Development Contracts	Apr-84	Fixed Price R&D.
52.233-03	Protest After Award	Aug-96	Exceed the simplified acquisition threshold.
52.242-13	Bankruptcy	Jul-95	Exceed the simplified acquisition threshold.
52.242-17	Government Delay of Work	Apr-84	Supplies other than commercial or modified-commercial products.
52.245-01	Government Property	Sep-21	When property is expected to be furnished.
52.245-09	Use and Charges	Apr-12	When the clause at 52.245-1 is included.
52.246-04	Inspection of Services-Fixed Price	Aug-96	Services, or supplies that involve the furnishing of services, when a fixed-price contract is contemplated, and the contract amount is expected to exceed the simplified acquisition threshold.
52.246-16	Responsibility for Supplies	Apr-84	Supplies, services involving the furnishing of supplies, or research and development, when a fixed-price contract is contemplated, and the contract amount is expected to exceed the simplified acquisition threshold.
52.246-23	Limitation of Liability	Feb-97	Exceed the simplified acquisition threshold.
52.247-34	F.o.b. Destination	Nov-91	When the delivery term is f.o.b. destination.
252.204-7015	Notice of Authorized Disclosure of Information for Litigation Support	Jan-23	Commercial products and commercial services.
252.209-7004	Subcontracting with Firms that are Owned or Controlled by the Government of a Country that is a State Sponsor of Terrorism	May-19	Contracts value of $150,000 or more.
252.211-7003	Item Identification and Valuation	Jan-23	For supplies, and for services involving the furnishing of supplies.
252.223-7001	Hazard Warning Labels	Dec-91	Requires submission of hazardous material data sheets.
252.223-7008	Prohibition of Hexavalent Chromium	Jan-23	For supplies, maintenance and repair services, or construction.

Clause Number	Title	Date	When Applicable
252.225-7001	Buy American and Balance of Payments Program	Feb-24	Acquisition of commercial products and commercial services.
252.225-7002	Qualifying Country Sources as Subcontractors	Mar-22	(i) 252.225-7001, Buy American and Balance of Payments Program. Use if one or more is included: (ii) 252.225-7021, Trade Agreements. (iii) 252.225-7036, Buy American - Free Trade Agreements - Balance of Payments Program.
252.225-7012	Preference for Certain Domestic Commodities	Apr-22	Commercial products and commercial services.
252.225-7052	Restriction on Acquisition of Certain Magnets, Tantalum, and Tungsten	May-24	Products and commercial services, that exceed the simplified acquisition threshold.
252.225-7056	Prohibition Regarding Business Operations with the Maduro Regime	Jan-23	Commercial products and commercial services.
252.225-7060	Prohibition on Certain Procurements from the Xinjiang Uyghur Autonomous Region	Jun-23	Products utilizing funds appropriated or otherwise made available for any fiscal year.
252.225-7972	Prohibition on the Procurement of Foreign-Made Unmanned Aircraft Systems (DEVIATION 2024-O0014)	Aug-24	Commercial products and commercial services.
252.225-7967	Prohibition Regarding Russian Fossil Fuel Business Operations (DEVIATION 2024-O0006, Revision 1)	Feb-24	Exceeding the simplified acquisition threshold and for commercial products and commercial services.
252.228-7001	Ground and Flight Risk	Mar-23	Acquisition, development, production, modification, maintenance, repair, flight, or overhaul of aircraft owned by or to be delivered to the Government.
252.228-7005	Mishap Reporting and Investigation Involving Aircraft, Missiles, and Space Launch Vehicles	Nov-19	Acquisition, development, production, modification, maintenance, repair, flight, or overhaul of aircraft owned by or to be delivered to the Government.
252.235-7002	Animal Welfare	Dec-14	Uses live vertebrate animals.
252.235-7004	Protection of Human Subjects	Jul-09	Involving human subjects.
252.243-7002	Requests for Equitable Adjustment	Dec-22	Exceeding the simplified acquisition threshold.
252.245-7003	Contractor Property Management System Administration	Jan-25	Containing the clause at FAR 52.245-1, Government Property.
252.245-7005	Management and Reporting of Government Property	Jan-24	Containing the clause at FAR 52.245-1, Government Property.
252.247-7023	Transportation of Supplies by Sea	Oct-24	Except - those with an anticipated value at or below the simplified acquisition threshold.